

THE MEASURE OF ALL THINGS

Desk Volume Two

*The formal statements, the complete proofs, and the problems
belonging to the audio season of the same name.*

Narrated by Jeeves

Written for and with P. Milovanov, Esq.

Issued chapter by chapter alongside the episodes.
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How to Use This Volume

This volume is the desk's half of the second season, and the partnership's terms are unchanged from Volume One: the episodes carry motivation, history, definitions in words, and the *strategy* of proofs; this volume carries precise notation, formal statements, complete proofs, and the problems. Listen first, then work. Nothing appears here that the corresponding episode has not motivated, and nothing is claimed aloud as proved unless the proof appears here — or the narration says plainly where the full argument lives, or that it lies beyond the course.

Standing conventions, carried over. The natural numbers begin at one. Facts used on credit are flagged where they occur, entered in the running ledger, and paid in a named later chapter or declared honestly beyond the course. Each chapter opens with an unnumbered Episode Digest and a Notation Ledger; problems come in three tiers (Calisthenics; The Real Thing; For the Ambitious, starred); hints are spoiler-safe; solutions are complete and terse for Tier I, provided for selected Tier II problems, and withheld for Tier III. The bell of completeness (⊗) counts marquee deployments only, and its count *continues from Volume One*: fourteen rings stand on the register as this volume opens.

What this season assumes. Volume One entire — in particular: suprema and infima and their approximation properties; countability and the diagonal; sequences, series, and the huddle; open, closed, compact, and Baire's theorem; continuity and uniform continuity; the derivative; the Riemann integral, its fundamental theorem, and its discontents. Where a Volume One item is load-bearing, the chapters cite it by chapter and number.

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Chapter 1

How Long Is a Set?

*Companion to Episode One of the second season. The episode wrote the contract any honest length must keep, watched the hinge question — additivity across how many pieces? — select the countable clause as the season’s engine, saw Jordan’s finite machinery declare the rationals unmeasurable absurdly, built Lebesgue’s outer measure and rang the bell in the new theatre, measured the rationals at zero and made “measure zero” lawful, stated the catch, executed the honest retreat to sigma-algebras and the Borel estate, forged the oscillation gauge and the continuity-set anatomy, heard the rationals refuse to be a G-delta at Baire’s second and final discharge, and paid debt nine with ceremony: no function on the line is continuous exactly at the rationals. This chapter is the paperwork. Register notes: the bell rings **once** this chapter — the fifteenth ring, the first of the volume, marked (⊗) at Definition 1.7; the count carries over from Volume One’s fourteen. Debt nine is **paid** at Theorem 1.29; debt eleven pays its first instalment at Definition 1.13; the catch is **stated on credit** at Theorem 1.15, trial scheduled for Chapter 3 (Vitali presiding); the epsilon-over-two-to-the-k gambit is christened at Theorem 1.11.*

Episode Digest

The argument of the hour. Length, made honest, selects its own jurisdiction. The contract: nonnegativity, nought for the empty set, intervals at $b - a$, translation invariance, and additivity across *countably* many pieces — finite additivity goes mute before limits, uncountable additivity contradicts the schoolroom, and the countable clause is exactly the engine analysis needs. Jordan’s finite covers fail the rationals absurdly (inner content zero, outer one). Lebesgue’s countable covers succeed: the outer measure — infimum over countable interval covers, completeness delivering the number (ring fifteen) — assigns every set a value, honours the contract’s first four clauses, and measures every countable set at zero: “measure zero” becomes lawful, the prophecy’s language redeemed. But countable additivity cannot hold on all sets (the catch, on credit until Chapter 3); the honest retreat defines sigma-algebras, and the intersection-from-above construction yields the Borel estate with its first ladder rungs, F-sigma and G-delta. The oscillation gauge proves every continuity set is a G-delta; Baire’s theorem proves the rationals are not; and the two instruments cross in a two-line ceremony: no function is continuous exactly on the rationals. The census closes exactly — continuity sets are precisely the G-delta sets (converse starred) — and the season’s method stands demonstrated: functions ask, sets answer.

The episode, beat by beat. A New Sign Above the Door. The establishment reopens; terms renewed (ear/desk, honesty clause, tiers); Season One’s debts carried at face value; the bell’s

count stands at fourteen as the season opens. Credit line read into the record: Borel (1898) proposed countable operations and measured his estate; Lebesgue's thesis (1902) finishes the house; master and student, credit divided exactly.

The Length of the Obvious. The contract's four clauses (Definition 1.1): nought for the empty set; intervals at $b - a$; translation invariance; additivity of disjoint pieces — with the hinge question attached: how many pieces?

How Many Pieces? Uncountable additivity dies in a sentence (the unit interval is uncountably many weightless points; Remark 1.2). Finite additivity is safe and useless: limits are countable processes, and length must pass through them. The engine turned over once: countable additivity yields continuity along swelling chains (Proposition 1.3, by disjointification and the telescope), with the shrinking-chain caveat exhibited (half-lines $[n, \infty)$: infinite measures, empty intersection). The converse — continuity recovers countable additivity — is Problem II.2, solved in full.

Jordan's Honest Attempt. Inner and outer content by finite packings and covers (Definition 1.4); the kinship made exact: contents are the Darboux ledgers of the indicator (Proposition 1.5) — the rationals' Jordan failure and the static's Riemann defeat are one failure. The verdict: inner content of $\mathbb{Q} \cap [0, 1]$ is zero (no interval fits), outer is one (finite covers close up to endpoints; Proposition 1.6); unmeasurable, absurdly. Diagnosis in one word: finiteness.

The Civil Servant's Idea. Why 1902 and not 1802: the permission to complete an infinite act — Cantor's jurisdiction, Episode Three of the first season — had to exist before the schoolroom tools could be used countably at once. The definition (Definition 1.7, $(*)$ ring fifteen): infimum over countable interval covers of the total length. Properties with their strategies: intervals measure $b - a$ (Theorem 1.10 — compactness converts the countable conspiracy to a finite committee; Volume One's coronation cashed); monotonicity (Proposition 1.9); countable subadditivity by the epsilon-over-two-to-the-k gambit (Theorem 1.11, the season's technique-character christened); translation invariance (Problem II.3, solved in full); the extended nonnegative arithmetic with its one prohibition (Remark 1.8). Scope: every set receives a number.

The Rationals Measured. Every countable set is null (Theorem 1.12: enumerate, house the k -th point under $\varepsilon/2^k$, geometric series pays). "Measure zero" defined (Definition 1.13) — debt eleven's first instalment; the prophecy's exhibits re-read. The null family is a sigma-ideal: hereditary, closed under countable unions (Proposition 1.14). The dust's number is withheld — next chapter's set piece.

The Scorecard. Four clauses banked (nonnegativity, nought, intervals, translation); the engine clause half-paid: subadditivity holds, additivity awaits a jurisdiction — next chapter's criterion.

The Catch. Stated on credit (Theorem 1.15): no assignment on *all* subsets keeps the full contract; trial at Chapter 3, Vitali presiding, Choice in the dock.

The Honest Retreat. Sigma-algebras (Definition 1.16): complement and countable union — the traffic limits require; countable intersections free by De Morgan. Specimens: the tribe of two, the all-subsets tribe (barred by the catch), the countable-or-co-countable tribe (Problem II.1). The strategy: restrict the universal instrument to a tribe on which the contract can be proved.

The Borel Estate. The intersection-from-above construction (Theorem 1.18): the smallest sigma-algebra containing a given collection exists — unanimity across all containing tribes. The Borel sets (Definition 1.19); robustness: intervals, open sets, rational-endpoint intervals all generate the

same estate (Theorem 1.20, via Volume One's structure theorem, Proposition 5.3 — every open set a countable string of beads). The ladder's first rungs: F-sigma and G-delta (Definition 1.21); the rationals are F-sigma, the irrationals G-delta; open sets are F-sigma too (Proposition 1.22, the fattened-endpoints trick).

Where a Function Is Continuous. The oscillation gauge (Definition 1.23): infimum over windows of the swing; continuity at a point is oscillation zero (Lemma 1.24); calibrations — the static at one everywhere, Thomae reading each fraction's loudness, the step reading its jump, the cornered ramp reading zero (Remark 1.25). The n -quiet sets are open (Lemma 1.26, quietness inherited within the witness window); the continuity set is the countable intersection of the n -quiet sets: a G-delta, for every function whatsoever (Theorem 1.27).

The Rationals Refuse. $\mathbb{Q}$ is not a G-delta (Theorem 1.28): were $\mathbb{Q} = \bigcap G_n$, the G_n and the punctured lines $\mathbb{R} \setminus \{q\}$ would form a countable family of open dense sets with empty intersection, contradicting Baire (Volume One, Theorem 5.28) — his second and final engagement, discharged.

The Ceremony. Debt nine paid (Theorem 1.29, boxed): continuity sets are G-delta; the rationals are not; no function is continuous exactly on the rationals — two lines after nine episodes. The census closes exactly: every G-delta *is* a continuity set (Problem III.1, starred) — the taxonomy was the theorem. The season's method demonstrated: functions ask, sets answer.

The Season Before Us. The itinerary in eight sittings: the criterion and the dust's number; Vitali and Banach–Tarski; integration rebuilt and the static tamed; the convergence theorems (the crown jewels); Littlewood's principles; Fubini; Kolmogorov at the listener's own door.

The toolkit acquired. The contract and its hinge; the epsilon-over-two-to-the- k gambit; the fattening trick (open covers at negligible cost); the intersection-from-above construction; the ladder's first rungs; the oscillation gauge and quietness-inheritance; the punctured-lines assembly for Baire; and the two-instrument ceremony pattern — anatomy plus refusal.

Dates worth keeping. Jordan's content in its mature form, 1893; Borel's countably additive measure on his estate, 1898; Lebesgue's thesis — *Intégrale, longueur, aire* — 1902; Vitali's catch, 1905 (trial: Chapter 3); Baire's theorem, 1899 (Volume One's eleventh ring, spent here for the second and last time).

1.1 Notation Ledger

Spoken in the episode	Written here	Remarks
“the outer measure”	$m^*(A) = \inf\{\sum_k I_k : A \subseteq \bigcup_k I_k\}$	(*) Def. 1.7
“an interval’s length”	$ I = b - a$	Def. 1.1
“measure zero”; “null”	$m^*(A) = 0$	Def. 1.13
“a sigma-algebra”; “the tribe”	$\mathcal{A}: \emptyset \in \mathcal{A}; A^c \in \mathcal{A}; \bigcup_k A_k \in \mathcal{A}$	Def. 1.16
“the generated sigma-algebra”	$\sigma(\mathcal{G})$	Thm. 1.18
“the Borel estate”	$\mathcal{B} = \sigma(\text{open sets})$	Def. 1.19
“F-sigma; G-delta”	$F_\sigma: \bigcup_n F_n \text{ (} F_n \text{ closed)}; G_\delta: \bigcap_n U_n \text{ (} U_n \text{ open)}$	Def. 1.21
“the oscillation of f at x ”	$\omega_f(x) = \inf_{\delta>0} \omega_f((x - \delta, x + \delta))$	Def. 1.23
“the swing on a window”	$\omega_f(W) = \sup_W f - \inf_W f$	Def. 1.23
“the continuity set”	$C_f = \{x : f \text{ continuous at } x\}$	Thm. 1.27

On the extended nonnegative reals (the ledger’s fine print). — Outer measure takes values in $[0, \infty]$; the symbol ∞ is admitted as an honest verdict, with the arithmetic $a + \infty = \infty$ and $a \leq \infty$ for all a , and one standing prohibition — $\infty - \infty$ is never written, and no argument of this season will need it (Remark 1.8). Series of nonnegative terms converge or diverge to ∞ with no third possibility (Volume One, the monotone license, Chapter 4); every sum below is of this kind.

1.2 The Contract

Definition 1.1 (The contract for a length). A candidate notion of length is an assignment μ of a value in $[0, \infty]$ to sets of reals (all sets, or a declared tribe of them), demanded to satisfy:

- (i) $\mu(\emptyset) = 0$;
- (ii) $\mu(I) = b - a$ for every interval I with endpoints $a \leq b$, open, closed, or half-open;
- (iii) (translation invariance) $\mu(A + t) = \mu(A)$ for every $t \in \mathbb{R}$, where $A + t = \{a + t : a \in A\}$;
- (iv) (countable additivity) for every sequence $A_1, A_2, \dots$ of pairwise disjoint sets in the domain whose union is in the domain, $\mu(\bigcup_k A_k) = \sum_k \mu(A_k)$.

Finite additivity is the weakening of (iv) to finitely many pieces; it follows from (iv) by padding with empty sets, clause (i) absorbing the padding.

Remark 1.2 (Additivity across arbitrary families is not on the table). $[0, 1]$ is the disjoint union of its singletons. Each singleton has length zero under any candidate honouring (ii) and monotone common sense: $\{x\} \subseteq (x - \varepsilon, x + \varepsilon)$ for every ε . An additivity clause surviving uncountable disassembly would price $[0, 1]$ as a “sum” of uncountably many zeros; no convention for such sums yields one (any reasonable definition — say, the supremum of finite subsums — yields zero). Clause (iv) stops at countable by necessity, not modesty.

Proposition 1.3 (Continuity of measure — the engine’s crankshaft). *Let $\mathcal{A}$ be a sigma-algebra and μ a countably additive assignment on $\mathcal{A}$ with values in $[0, \infty]$ and $\mu(\emptyset) = 0$. (This proposition is stated for the tribes to come; from Chapter 2 onward it is used constantly.)*

- (i) (Swelling chains) If $A_1 \subseteq A_2 \subseteq \cdots$, all in $\mathcal{A}$, then $\mu(\bigcup_n A_n) = \lim_n \mu(A_n)$.
- (ii) (Shrinking chains) If $A_1 \supseteq A_2 \supseteq \cdots$, all in $\mathcal{A}$, and $\mu(A_1) < \infty$, then $\mu(\bigcap_n A_n) = \lim_n \mu(A_n)$.
The finiteness hypothesis is not removable: $A_n = [n, \infty)$ shrinks to the empty set with every $\mu(A_n)$ infinite (once Chapter 2 delivers the measure, this is a lawful exhibit; the Archimedean property of Volume One, Theorem 2.12, empties the intersection).

Proof. (i) Disjointify into rings of growth: $B_1 = A_1$ and $B_k = A_k \setminus A_{k-1}$ for $k \geq 2$. Each B_k lies in $\mathcal{A}$ (complement and intersection, via De Morgan); the B_k are pairwise disjoint; $\bigcup_{k \leq n} B_k = A_n$; and $\bigcup_k B_k = \bigcup_n A_n$. Countable additivity prices the union as $\sum_k \mu(B_k)$, whose partial sums are, by finite additivity, $\mu(A_n)$ — the telescope, Volume One's oldest cancellation. A series of nonnegative terms is the limit of its partial sums: $\mu(\bigcup_n A_n) = \lim \mu(A_n)$.

(ii) Pass to complements within A_1 : the sets $A_1 \setminus A_n$ swell to $A_1 \setminus \bigcap_n A_n$. Apply (i), then subtract from $\mu(A_1)$ — lawful precisely because $\mu(A_1) < \infty$ keeps every quantity finite and the prohibited $\infty - \infty$ off the page (finite additivity gives $\mu(A_1) = \mu(A_n) + \mu(A_1 \setminus A_n)$, so $\mu(A_1 \setminus A_n) = \mu(A_1) - \mu(A_n)$, and likewise for the intersection). The identity rearranges to the claim. The converse direction — (i) plus finite additivity recovers countable additivity — is Problem II.2, solved in full: the engine and the clause are one mechanism. $\square$

1.3 Jordan's Content

Definition 1.4 (Inner and outer content). For bounded $A \subseteq \mathbb{R}$: the *inner content* $\underline{c}(A)$ is the supremum of $\sum_{j=1}^N |I_j|$ over all *finite* families of pairwise disjoint intervals with $\bigcup_j I_j \subseteq A$; the *outer content* $\bar{c}(A)$ is the infimum of $\sum_{j=1}^N |I_j|$ over all finite families with $A \subseteq \bigcup_j I_j$. Always $\underline{c}(A) \leq \bar{c}(A)$; when equal, A is *Jordan measurable* with content the common value.

Proposition 1.5 (The kinship: contents are Darboux ledgers). Let $A \subseteq [a, b]$ and let $\mathbf{1}_A$ be its indicator. Then

$$\underline{c}(A) = \int_a^b \mathbf{1}_A \quad \text{and} \quad \bar{c}(A) = \int_a^b \mathbf{1}_A.$$

In particular A is Jordan measurable exactly when $\mathbf{1}_A$ is Riemann integrable, with content equal to integral. The rationals' failure below and the static's defeat (Volume One, Proposition 6.25 and Chapter 8) are one failure.

Proof. Lower ledgers. A partition P of $[a, b]$ has $L(\mathbf{1}_A, P) = \sum' \Delta x_i$, the primed sum over slices lying *entirely* inside A (the floor is one exactly there) — a finite disjoint family of intervals inside A , so $L(\mathbf{1}_A, P) \leq \underline{c}(A)$. Conversely a finite disjoint packing of A by intervals of total length ℓ yields, after listing all endpoints as a partition and shaving ε at the joins (closed slices must sit inside A ; open packing intervals lose two points' worth, which cost nothing after shaving), a lower sum exceeding $\ell - \varepsilon$. Suprema agree. The upper ledgers mirror: slices meeting A versus finite covers, with the same endpoint bookkeeping. The desk leaves the mirrored half as drill within Problem I.4's ambit. $\square$

Proposition 1.6 (Jordan's verdict on the rationals). For $A = \mathbb{Q} \cap [0, 1]$: $\underline{c}(A) = 0$ and $\bar{c}(A) = 1$. The set is not Jordan measurable.

Proof. Inner: an interval of positive length inside A would consist of rationals only, impossible — the irrationals are dense (Volume One, Corollary 2.16). Only degenerate packings exist; the supremum is zero.

Outer: let $I_1, \dots, I_N$ be a finite cover of A and let $F = \overline{I_1} \cup \dots \cup \overline{I_N}$, the union of closures — a closed set of total length $\sum_j |I_j|$ at most (closing an interval adds endpoints, which carry no length). F contains A , and F is closed, so F contains $\overline{A} = [0, 1]$: the rationals are dense in the unit interval (Volume One, Theorem 2.15), and a closed superset of a set contains its closure. A finite union of intervals containing $[0, 1]$ has total length at least 1 (order the pieces meeting $[0, 1]$ and walk from 0 to 1: each step is paid for by disjoint stretches of the pieces; the desk notes this is Jordan's own counting, or Volume One's Heine–Borel walk in miniature). Hence every finite cover totals at least 1; the interval $[0, 1]$ itself achieves 1; the infimum is 1. $\square$

1.4 The Outer Measure

Definition 1.7 ((*) Lebesgue outer measure — ring fifteen). For any $A \subseteq \mathbb{R}$,

$$m^*(A) = \inf \left\{ \sum_{k=1}^{\infty} |I_k| : A \subseteq \bigcup_{k=1}^{\infty} I_k, I_k \text{ intervals} \right\},$$

the infimum over all countable interval covers of the total length (a series of nonnegative terms: convergent or honestly infinite). Covers exist for every A ($\mathbb{R}$ itself is a countable union of intervals), so the set of totals is a nonempty subset of $[0, \infty]$; if any total is finite, the set of finite totals is nonempty and bounded below by 0, and *the bell rings — the fifteenth, the first of the volume*: the completeness axiom (Volume One, Axiom 2.6), and nothing else, delivers the greatest lower bound. (If every total is infinite, $m^*(A) = \infty$, an honest verdict.) The remaining suprema and infima of this chapter are clerical work upon this one and carry the standing annotation: completeness — small change, uncounted.

Remark 1.8 (Extended arithmetic; open covers at negligible cost). Values live in $[0, \infty]$ with $a + \infty = \infty$; $\infty - \infty$ is never written. Two service facts, used silently hereafter. *Fattening*: any cover $\{I_k\}$ may be upgraded to *open* intervals $J_k \supseteq I_k$ with $|J_k| \leq |I_k| + \varepsilon/2^k$ — total cost at most ε , the geometric series footing the bill — so the infimum may be computed over open covers alone. *Degenerate intervals* (points) are permitted in covers and change nothing.

Proposition 1.9 (Monotonicity). $A \subseteq B$ implies $m^*(A) \leq m^*(B)$.

Proof. Every cover of B covers A , so the infimum for A runs over a larger family of candidate totals and cannot exceed the infimum for B . (With clause (i) of the contract: $m^*(\emptyset) = 0$, the empty set hiding under covers of arbitrarily small total.) $\square$

Theorem 1.10 (The interval clause is kept). For every interval I with endpoints $a \leq b$: $m^*(I) = b - a$.

Proof. ($\leq$) $I \subseteq (a - \varepsilon, b + \varepsilon)$, a one-interval cover of total $b - a + 2\varepsilon$, for every $\varepsilon > 0$; the infimum dips below each, hence to $b - a$ at most.

($\geq$), the deep direction; it suffices to treat $I = [a, b]$ closed (any interval contains closed subintervals of length as near its own as desired, and monotonicity transfers the bound). Let $\{I_k\}$ be any countable cover of $[a, b]$; fatten to open J_k at cost ε (Remark 1.8). The open J_k cover the compact $[a, b]$; Heine–Borel (Volume One, Theorem 5.18) extracts a finite committee $J_{k_1}, \dots, J_{k_N}$

still covering. Audit the committee by the walk: some member (α_1, β_1) contains a ; if $\beta_1 > b$ stop; otherwise $\beta_1 \in [a, b]$ is covered by some member (α_2, β_2) with $\alpha_2 < \beta_1 < \beta_2$; continue. The walk visits distinct members with strictly increasing right endpoints, so it stops within N steps, at some $\beta_M > b$. Then

$$\sum_{i=1}^M |J_{(i)}| \geq (\beta_1 - \alpha_1) + (\beta_2 - \beta_1) + \cdots + (\beta_M - \beta_{M-1}) = \beta_M - \alpha_1 > b - a,$$

each step's length at least the fresh ground gained (since $\alpha_{i+1} < \beta_i$). Hence $\sum_k |I_k| \geq \sum_k |J_k| - \varepsilon \geq b - a - \varepsilon$, for every cover and every ε ; the infimum is at least $b - a$. — The clause cost compactness, exactly as narrated: the countable conspiracy collapses to an auditable committee, and Jordan's own counting convicts the committee. $\square$

Theorem 1.11 (Countable subadditivity — the gambit's debut). *For any sets $A_1, A_2, A_3, \dots$*

$$m^*\left(\bigcup_k A_k\right) \leq \sum_k m^*(A_k).$$

Proof. If any $m^*(A_k) = \infty$ the claim is vacuous. Otherwise let $\varepsilon > 0$. For each k , the infimum's approximation property (Volume One's oldest reflex) supplies a cover $\{I_{k,j}\}_j$ of A_k with $\sum_j |I_{k,j}| \leq m^*(A_k) + \varepsilon/2^k$. Pool the covers: $\{I_{k,j}\}_{k,j}$ is a countable family (a countable union of countable families — Volume One, Chapter 3's census) covering $\bigcup_k A_k$, of total

$$\sum_k \sum_j |I_{k,j}| \leq \sum_k \left(m^*(A_k) + \frac{\varepsilon}{2^k}\right) = \sum_k m^*(A_k) + \varepsilon,$$

the geometric series picking up the bill. One cover within ε of the target, for every ε : the infimum obeys. *The epsilon-over-two-to-the-k gambit, christened* — a fixed tolerance spent across countably many claimants; it recurs until the finale. $\square$

Theorem 1.12 (Every countable set is null). *If $A \subseteq \mathbb{R}$ is countable, then $m^*(A) = 0$.*

Proof. Enumerate $A = \{a_1, a_2, \dots\}$ (finite sets: pad the argument or stop early). Given $\varepsilon > 0$, house a_k in an interval of length $\varepsilon/2^k$ centred there. The family covers A with total ε exactly. So $m^*(A) \leq \varepsilon$ for every $\varepsilon > 0$, and $m^*(A) \geq 0$ always: $m^*(A) = 0$. In particular $m^*(\mathbb{Q} \cap [0, 1]) = 0$ and $m^*(\mathbb{Q}) = 0$: the set that defeated Jordan is dispatched, and the entire difference is the word *countable*. $\square$

Definition 1.13 (Measure zero). $A \subseteq \mathbb{R}$ has *measure zero* (is *null*) when $m^*(A) = 0$: for every $\varepsilon > 0$ there are countably many intervals covering A of total length below ε . This is the phrase Volume One's finale quoted as prophecy (Volume One, Chapter 8, the Lebesgue criterion, still on credit — proof at Chapter 4 of this volume); debt eleven pays its first instalment here.

Proposition 1.14 (The null family is a sigma-ideal). (i) *If $B \subseteq A$ and A is null, B is null.* (ii) *If each of $A_1, A_2, \dots$ is null, so is $\bigcup_k A_k$.*

Proof. (i) is monotonicity (Proposition 1.9). (ii) is countable subadditivity (Theorem 1.11): $m^*(\bigcup_k A_k) \leq \sum_k 0 = 0$. Hereditary downward, closed under countable assembly: negligibility with working papers. When “almost everywhere” arrives in force, these two laws discharge exceptional sets by the countable armful. $\square$

Theorem 1.15 (The catch — stated on credit). *There is no assignment μ defined on all subsets of $\mathbb{R}$ satisfying the full contract of Definition 1.1 — nonnegativity with $\mu(\emptyset) = 0$, intervals to length, translation invariance, and countable additivity. In particular m^* , which keeps the first three clauses on all sets, is not countably additive on all sets.*

On credit. The proof is the Vitali construction, Chapter 3's centrepiece, with the Axiom of Choice named in the indictment; nothing in this volume uses the catch before its trial. Until then it stands as the season's motivation: some sets must be turned away, and the honest question is which stay.

1.5 Sigma-Algebras and the Borel Estate

Definition 1.16 (Algebra; sigma-algebra). A collection $\mathcal{A}$ of subsets of $\mathbb{R}$ (or of any ambient set) is a *sigma-algebra* when: (i) $\emptyset \in \mathcal{A}$; (ii) $A \in \mathcal{A}$ implies $A^c \in \mathcal{A}$; (iii) $A_1, A_2, \dots \in \mathcal{A}$ implies $\bigcup_k A_k \in \mathcal{A}$. Weakening (iii) to finite unions defines an *algebra*. Consequences, by De Morgan (Volume One, Chapter 1's toolkit): the ambient set belongs ($\emptyset^c$); countable intersections belong ($\bigcap A_k = (\bigcup A_k^c)^c$); differences belong. The walls are drawn exactly where the traffic runs: complements and countable assemblies are what limiting arguments perform.

Remark 1.17 (Three specimens). The tribe of two, $\{\emptyset, \mathbb{R}\}$: closed under everything, useless for everything. The all-subsets tribe $\mathcal{P}(\mathbb{R})$: maximal hospitality, and the catch (Theorem 1.15) bars it from carrying the contract. The countable-or-co-countable tribe $\{A : A \text{ countable or } A^c \text{ countable}\}$: a genuine sigma-algebra containing every singleton yet not $[0, \frac{1}{2}]$ — dissected in Problem II.1. Membership is a distinction.

Theorem 1.18 (Generated sigma-algebras exist — from above). *Let $\mathcal{G}$ be any collection of subsets. Among all sigma-algebras containing $\mathcal{G}$ there is a smallest, written $\sigma(\mathcal{G})$: it is contained in every sigma-algebra containing $\mathcal{G}$.*

Proof. The family of sigma-algebras containing $\mathcal{G}$ is nonempty: $\mathcal{P}(\mathbb{R})$ qualifies. Let $\sigma(\mathcal{G})$ be their intersection — the sets belonging to *every* sigma-algebra containing $\mathcal{G}$, the unanimous sets. The three closure checks, each one line: $\emptyset$ belongs to every chamber, hence unanimously. If A is unanimous then, chamber by chamber, A^c belongs (each chamber is a sigma-algebra), hence A^c is unanimous. If $A_1, A_2, \dots$ are unanimous then each chamber contains them all and therefore their union: the union is unanimous. So $\sigma(\mathcal{G})$ is a sigma-algebra; it contains $\mathcal{G}$ (every chamber does); and it is contained in every containing sigma-algebra, being their intersection — unanimity cannot outvote a chamber. Floor-by-floor construction from below is not available at any countable stage (the Notes gesture at the transfinite abyss); the intersection from above is the whole trick. $\square$

Definition 1.19 (The Borel estate). $\mathcal{B} = \sigma(\{\text{open subsets of } \mathbb{R}\})$: the *Borel sets*. Every set this course has ever named in anger — intervals, opens, closedes, $\mathbb{Q}$, the irrationals, the dust, every continuity set — is Borel; producing a non-Borel set requires the ghost-machinery of Chapter 3 or deliberate transfinite bureaucracy (Notes).

Theorem 1.20 (The estate does not depend on the founder's taste). $\sigma(\text{open sets}) = \sigma(\text{all intervals}) = \sigma(\text{open intervals}) = \sigma(\text{open intervals with rational endpoints})$.

Proof. Write $\mathcal{B}, \mathcal{I}, \mathcal{I}_o, \mathcal{I}_Q$ for the four generated tribes; we chain inclusions both ways, using twice the master principle: if every member of $\mathcal{G}_1$ lies in $\sigma(\mathcal{G}_2)$, then $\sigma(\mathcal{G}_1) \subseteq \sigma(\mathcal{G}_2) - \sigma(\mathcal{G}_2)$ is a sigma-algebra containing $\mathcal{G}_1$, and the smallest such is contained in it.

$\mathcal{I}_Q \subseteq \mathcal{I}_o \subseteq \mathcal{I}$: generators nest. $\mathcal{I} \subseteq \mathcal{B}$: every interval is Borel — open intervals are open; $[a, b] = \bigcap_n (a - \frac{1}{n}, b + \frac{1}{n})$, a countable intersection of opens; half-open likewise, one side at a time. $\mathcal{B} \subseteq \mathcal{I}_o$: every open set is a countable disjoint union of open beads — Volume One's structure theorem (Proposition 5.3) — hence belongs to $\sigma(\text{open intervals})$, care of closure clause (iii); unbounded beads are countable unions of bounded ones. $\mathcal{I}_o \subseteq \mathcal{I}_Q$: every open interval is a countable union of rational-endpoint ones — pick rational endpoints marching out to the true ones, density (Volume One, Theorem 2.15) supplying the recruits. The chain closes: all four tribes are one estate. The rational-endpoint generating set is *countable* — a fact of consequence for the season's bookkeeping and for Chapter 8's dictionary. $\square$

Definition 1.21 (The ladder's first rungs). A set is F_σ when it is a countable union of closed sets; G_δ when a countable intersection of open sets. (F for *fermé*, sigma for sum; G for *Gebiet*, delta for *Durchschnitt*: the subject's bilingual fossils.) Complementation exchanges the two rungs, by De Morgan.

Proposition 1.22 (Rung calibrations). (i) $\mathbb{Q}$ is F_σ ; the irrationals are G_δ . (ii) Every closed set is F_σ and every open set G_δ , trivially; moreover every open set is F_σ and every closed set G_δ .

Proof. (i) Singletons are closed; $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ is a countable union of them. Complement and De Morgan deliver the irrationals as G_δ . (ii) For open U : write U as a countable union of open beads (Volume One, Proposition 5.3) and each bounded bead (a, b) as $\bigcup_n [a + \frac{1}{n}, b - \frac{1}{n}]$ (fattened endpoints, run inward; unbounded beads as $\bigcup_n [a + \frac{1}{n}, n]$ and kin) — a countable union of closed sets. A countable union of countable unions is countable (Volume One, Chapter 3). Closed sets as G_δ : complements. $\square$

1.6 The Continuity-Set Anatomy

Definition 1.23 (Oscillation). For $f : \mathbb{R} \rightarrow \mathbb{R}$ (or on an interval), the *swing* on a set W where f is bounded is $\omega_f(W) = \sup_W f - \inf_W f$ (unbounded: $\omega_f(W) = \infty$). The *oscillation at a point* is

$$\omega_f(x) = \inf_{\delta > 0} \omega_f((x - \delta, x + \delta)),$$

the swings settling downward as windows shrink (monotone in δ ; the infimum is small change, uncounted).

Lemma 1.24 (Continuity is oscillation zero). f is continuous at x if and only if $\omega_f(x) = 0$.

Proof. ($\Rightarrow$) Given $\varepsilon > 0$, continuity yields δ with $|f(y) - f(x)| < \varepsilon/2$ on the window, so the swing there is at most ε ; the infimum dips below every ε . ($\Leftarrow$) If $\omega_f(x) = 0$: given ε , some window swings below ε ; on it, $|f(y) - f(x)| \leq \omega_f(W) < \varepsilon$ — both values live in the window's range. That is continuity's definition (Volume One, Chapter 6). $\square$

Remark 1.25 (Calibrations). The static (Volume One, Definition 6.24): every window contains rationals and irrationals (Volume One, 2.15–2.16), so every swing is 1: $\omega_D \equiv 1$. Thomae (Volume One, Definition 6.26): at irrational x , oscillation zero (continuity, Volume One, Theorem 6.27); at

rational p/q in lowest terms, oscillation exactly $1/q$ — the value there is $1/q$, while nearby values are small (the counting lemma inside Volume One's Theorem 6.27 forces large denominators nearby) and irrationals nearby read zero, so the swing settles to the fraction's own loudness. The step function: oscillation equal to the jump's height at the jump, zero elsewhere. The cornered ramp (Volume One, Chapter 8): oscillation zero even at the corner — the gauge detects broken continuity, not broken smoothness.

Lemma 1.26 (The quiet sets are open). *For every $c > 0$, the set $V_c = \{x : \omega_f(x) < c\}$ is open.*

Proof. Let $x \in V_c$. The infimum's approximation property produces a witness window $W = (x - \delta, x + \delta)$ with $\omega_f(W) < c$. Every resident $y \in W$ owns a sub-window $W' \subseteq W$ centred at y (openness of the interval), and swings only shrink on subsets: $\omega_f(W') \leq \omega_f(W) < c$, so $\omega_f(y) < c$. Thus $W \subseteq V_c$: quietness certified by one window is inherited by everyone living in it, and V_c contains a neighbourhood of each of its points — open, by the very definition (Volume One, Chapter 5). $\square$

Theorem 1.27 (Every continuity set is a G_δ). *For every f , the set C_f of points of continuity satisfies*

$$C_f = \bigcap_{n=1}^{\infty} \left\{ x : \omega_f(x) < \frac{1}{n} \right\},$$

a countable intersection of open sets.

Proof. By Lemma 1.24, $x \in C_f$ iff $\omega_f(x) = 0$ iff $\omega_f(x) < 1/n$ for every n — a nonnegative quantity below every threshold is zero, the Archimedean gate (Volume One, Theorem 2.12). Each threshold set is open by Lemma 1.26. No hypothesis on f was used: the theorem binds every function whatsoever. $\square$

1.7 The Refusal and the Ceremony

Theorem 1.28 (The rationals are not a G_δ). *There do not exist open sets $G_1, G_2, \dots$ with $\mathbb{Q} = \bigcap_n G_n$.*

Proof. Suppose otherwise. Each $G_n \supseteq \mathbb{Q}$ is open, and dense — it houses the dense tenant $\mathbb{Q}$ (Volume One, Theorem 2.15). For each rational q , the punctured line $\mathbb{R} \setminus \{q\}$ is open (a point is closed) and dense (no interval fits in a point). The family $\{G_n\}_n \cup \{\mathbb{R} \setminus \{q\}\}_{q \in \mathbb{Q}}$ is a countable family of open dense sets (countable plus countable, Volume One, Chapter 3). Its intersection is empty: a surviving point would be rational (unanimity of the G_n) yet avoid every rational (the punctures). But Baire's theorem (Volume One, Theorem 5.28, with its dense-intersection reading) makes any countable intersection of open dense subsets of $\mathbb{R}$ dense, hence nonempty. Contradiction; the supposition dies. Monsieur Baire's second and final engagement, discharged. $\square$

Theorem 1.29 (The ceremony: debt nine paid). *There is no function $f : \mathbb{R} \rightarrow \mathbb{R}$ whose set of continuity points is exactly $\mathbb{Q}$.*

Proof. C_f is a G_δ (Theorem 1.27); $\mathbb{Q}$ is not (Theorem 1.28). Two lines, nine episodes in the arranging. $\square$

Remark 1.30 (The census closes exactly). Thomae's function realizes the irrationals as a continuity set (lawfully: they are G_δ); the rationals are forbidden; and the characterization is complete in both directions: *every* G_δ subset of $\mathbb{R}$ is the continuity set of some function — Problem III.1, starred, the two-sign construction staged in hints. The continuity sets are precisely the G_δ sets; nothing else about a candidate set — cardinality, density, order — is relevant. Volume One's Episode Six inventory (everything; nothing; exactly the irrationals; never exactly the rationals) stands fully explained.

1.8 Problems

Tier I — Calisthenics

- I.1 Directly from Definition 1.7: every finite set has outer measure zero. (Do not cite Theorem 1.12; build the cover.)
- I.2 For the displaced interval $[a, b] + t$: exhibit covers witnessing $m^*([a, b] + t) \leq b - a$, and cite the theorem that forces equality once translation invariance (Problem II.3) is in hand.
- I.3 Nested pairs, worked: from the definitions alone, verify $m^*({0}) \leq m^*([0, \frac{1}{2}]) \leq m^*([0, 1])$, then replace the middle term by its exact value.
- I.4 The two-set rehearsal for the gambit: given $m^*(A), m^*(B) < \infty$ and $\varepsilon > 0$, produce a single cover of $A \cup B$ of total below $m^*(A) + m^*(B) + \varepsilon$, splitting the tolerance in half. Then complete the mirrored half of Proposition 1.5 (upper ledgers), which uses the same endpoint bookkeeping.
- I.5 De Morgan drills: from Definition 1.16's three clauses, derive membership of the ambient set, of countable intersections, and of set differences, each in one line.
- I.6 Membership audits in the countable-or-co-countable tribe: the even integers; the rationals; $[0, 1]$; the irrationals; a set containing exactly one point of each interval $[n, n + 1)$. Verdicts with one-line warrants.
- I.7 Certify $[a, b)$ as both F_σ and G_δ , with explicit sequences of closed and open sets.
- I.8 Oscillation readings, with one-line justifications: the step function at and off its jump; the parabola everywhere; the static everywhere.

Tier II — The Real Thing

- II.1 (*The middle specimen.*) Prove the countable-or-co-countable collection is a sigma-algebra (the union clause splits by cases on whether some member is co-countable); show it contains every singleton but not $[0, \frac{1}{2}]$; and identify it as $\sigma(\{\text{singletons}\})$ — the estate the points alone can afford.
- II.2 (*The engine's two faces.*) Let $\mu \geq 0$ on a sigma-algebra satisfy $\mu(\emptyset) = 0$ and finite additivity. Prove: μ is countably additive if and only if μ is continuous along swelling chains (Proposition 1.3(i)). Solved in full below, completing the crankshaft argument.

- II.3 (*Clause three for the new instrument.*) Prove $m^*(A + t) = m^*(A)$ for every A and t : covers slide rigidly with their lengths in their luggage. Solved in full below.
- II.4 (*The refusal, generalized.*) Prove there is no countable *dense* G_δ subset of $\mathbb{R}$: Baire refires under your hand. Conclude anew that $\mathbb{Q}$ is not G_δ , and locate exactly where density entered.
- II.5 (*The gauge's grain.*) Prove that for every $c > 0$, the set $\{x : \omega_f(x) \geq c\}$ is closed; conclude the discontinuity set of any function is an F_σ . (The dual reading of Lemma 1.26; the discontinuity set's anatomy, dual to the continuity set's.)
- II.6 (*Thomae, certified twice.*) Certify that Thomae's continuity set — the irrationals — is G_δ two ways: by the anatomy (Theorem 1.27 applied to Thomae), and by direct exhibition of open sets intersecting to the irrationals. Verify the two exhibitions differ.

Tier III — For the Ambitious ★

- III.1 ★ (*The census closed: every G_δ is a continuity set.*) Let $G = \bigcap_n U_n$ with U_n open; arrange $U_1 \supseteq U_2 \supseteq \cdots$ (intersect initial segments). For $x \notin G$ let $m(x)$ be the least n with $x \notin U_n$ (well-ordering, Volume One, Chapter 1). Define $f = 0$ on G ; $f(x) = 1/m(x)$ for $x \notin G$ rational; $f(x) = -1/m(x)$ for $x \notin G$ irrational. Prove $C_f = G$. Hints below; this is the two-sign construction the ceremony's closing remark invoked.
- III.2 ★ (*The ladder ascends.*) Prove the rationals are F_σ but not G_δ , and the irrationals G_δ but not F_σ ; then show $A = (\mathbb{Q} \cap (-\infty, 0)) \cup ((\mathbb{R} \setminus \mathbb{Q}) \cap (0, \infty))$ is neither F_σ nor G_δ — localize Baire on each half-line. References in the Notes for the ladder's transfinite continuation.
- III.3 ★ (*Jordan anatomized.*) Prove: a bounded set is Jordan measurable if and only if its topological boundary ∂A has outer content zero — equivalently (for bounded sets, via Theorem 1.10's methods) measure zero. Test the criterion on the rationals of the unit interval ($\partial = [0, 1]$) and on the dust (next chapter completes the verdict).

1.9 Hints

II.1. For the union clause: if every A_k is countable, the union is countable (Volume One, Chapter 3); if some A_{k_0} is co-countable, the union's complement sits inside $A_{k_0}^c$. For $\sigma(\{\text{singletons}\})$: show the tribe is a sigma-algebra containing the singletons, and that any such sigma-algebra contains every countable set and every co-countable set.

II.4. Suppose $D = \{d_1, d_2, \dots\}$ countable, dense, and $= \bigcap_n G_n$ with G_n open (dense, since D is). The punctured lines $\mathbb{R} \setminus \{d_k\}$ are open; dense because a puncture cannot contain an interval. Intersect everything and apply Baire. Density of D entered twice: making the G_n dense, and making the contradiction bite ($\mathbb{Q}$ qualifies; a countable *closed* set need not).

II.5. The complement of $\{\omega_f \geq c\}$ is $\bigcup_{c' < c} V_{c'}$ — or directly: V_c open for every c (Lemma 1.26) and $\{\omega_f \geq c\} = \mathbb{R} \setminus V_c$. For the F_σ conclusion: the discontinuity set is $\bigcup_n \{\omega_f \geq 1/n\}$.

II.6. Direct exhibition: the irrationals are $\bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\})$, a countable intersection of opens. The anatomy's exhibition uses the quiet sets $V_{1/n}$ of Thomae's oscillation — compute which points each $V_{1/n}$ admits (rationals with denominator above n , and all irrationals), and observe the two families differ set by set though their intersections agree.

III.1. Continuity on G : for $x \in G$ and N given, $W = \bigcap_{n \leq N} U_n$ is an open neighbourhood of x ; any $y \in W$ has either $f(y) = 0$ ($y \in G$) or $m(y) > N$, whence $|f(y)| < 1/N$. Discontinuity at $x \notin G$ with $m(x) = k$: every window at x contains points of the opposite rationality class (Volume One, 2.15–2.16); such a point y gives either $f(y) = 0$ (if $y \in G$: gap $1/k$) or $f(y)$ of opposite sign (gap exceeding $1/k$). Either way $\omega_f(x) \geq 1/k > 0$. Mind that m is well-defined only off G , and that the decreasing arrangement of the U_n is what makes $m(y) > N$ mean $y \in U_1, \dots, U_N$.

III.2. If the irrationals were F_σ , the rationals would be G_δ (De Morgan): Theorem 1.28 forbids. For A : were A G_δ , then $A \cap (-\infty, 0) = \mathbb{Q} \cap (-\infty, 0)$ would be G_δ in the open half-line, and Baire localizes to any complete-metric-flavoured stage — run the punctured-lines argument inside $(-\infty, 0)$; the mirrored half kills F_σ . References: Oxtoby for the honest treatment of relative Baire arguments.

III.3. Key identity: for bounded A , $\bar{c}(A) = \bar{c}(\bar{A})$ (closing a finite cover costs endpoints only) and $\underline{c}(A) = \underline{c}(A^\circ)$ (an interval inside A avoids the boundary after shaving). Then $\bar{c}(A) - \underline{c}(A) = \bar{c}(\partial A)$ via $\bar{A} = A^\circ \sqcup \partial A$ and finite additivity of content on such splittings.

1.10 Solutions

Tier I (terse). **I.1** House the j -th of the N points in an interval of length ε/N : total ε , for every ε ; the infimum is 0. **I.2** $[a, b] + t = [a + t, b + t]$, an interval of the same length; the cover $(a + t - \varepsilon, b + t + \varepsilon)$ witnesses $\leq b - a + 2\varepsilon$; Theorem 1.10 forces $= b - a$ (or II.3 plus the theorem, as the drill anticipates). **I.3** Monotonicity twice (Proposition 1.9); middle term $= \frac{1}{2}$ by Theorem 1.10. **I.4** Covers within $\varepsilon/2$ of each outer measure pool to a countable cover of the union (two countables); total below the sum plus ε . Mirrored ledgers: slices of a partition meeting A form a finite cover of A ; conversely a finite cover's endpoints, listed as a partition with ε -shaving at the joins, give an upper sum within ε of the cover's total. **I.5** Ambient set $= \emptyset^c$; intersections by $\bigcap A_k = (\bigcup A_k^c)^c$; difference $A \setminus B = A \cap B^c$. **I.6** Evens: countable, in. Rationals: countable, in. $[0, 1]$: uncountable with uncountable complement, out. Irrationals: complement countable, in. The transversal of the intervals $[n, n + 1)$: countably infinite (one point per integer), in — the tribe cares only for cardinality, not location. **I.7** $[a, b] = \bigcup_n [a, b - \frac{b-a}{n}]$ (closed) $= \bigcap_n (a - \frac{1}{n}, b)$ (open). **I.8** Step: at the jump every window swings the jump's height h , and constancy elsewhere swings nought — $\omega = h$ at the jump, 0 off it. Parabola: continuous everywhere (Volume One, Chapter 6), so $\omega \equiv 0$ (Lemma 1.24). Static: every window contains both values 0 and 1 (density both ways): $\omega \equiv 1$.

II.2 (solution in full). $(\Rightarrow)$ is Proposition 1.3(i). $(\Leftarrow)$ Let $E_1, E_2, \dots$ be pairwise disjoint members with union E . The partial unions $U_n = \bigcup_{k \leq n} E_k$ form a swelling chain in the sigma-algebra with union E , so continuity gives $\mu(E) = \lim_n \mu(U_n)$. Finite additivity prices each stage:

$\mu(U_n) = \sum_{k \leq n} \mu(E_k)$, a partial sum. The limit of the partial sums *is* the series (nonnegative terms; Volume One, Chapter 4's monotone license makes the limit exist in $[0, \infty]$): $\mu(E) = \sum_k \mu(E_k)$. Countable additivity holds; the engine and the clause are one mechanism seen from two sides, as the episode claimed.

II.3 (solution in full). Fix A and t . If $\{I_k\}$ covers A , then $\{I_k + t\}$ covers $A + t$ — $a \in A$ gives $a \in I_k$ for some k , so $a + t \in I_k + t$ — and translated intervals keep their lengths: $|I_k + t| = |I_k|$ (endpoints shift together). So every candidate total for A is a candidate total for $A + t$: $m^*(A + t) \leq m^*(A)$, the infimum running over a family at least as rich. Apply the same inequality to the set $A + t$ and the shift $-t$: $m^*(A) = m^*((A + t) + (-t)) \leq m^*(A + t)$. The two inequalities close: $m^*(A + t) = m^*(A)$. Clause three of the contract is kept by the new instrument — the clause that will sign Chapter 3's confession is, for now, in impeccable standing.

1.11 Notes

History. Jordan's content reached its mature form in the second edition of his *Cours d'analyse* (1893), building on Peano; it is the exact set-theoretic shadow of Riemann's integral, as Proposition 1.5 makes precise. Borel introduced countably additive measure on the sets his operations generate in *Leçons sur la théorie des fonctions* (1898) — the estate bears his name by right. Lebesgue's thesis, *Intégrale, longueur, aire* (Annali di Matematica, 1902), contains the outer measure, the measurable sets, and most of this season in embryo; Lebesgue was indeed teaching at the lycée of Nancy as he wrote it. Vitali's construction (1905) waits at Chapter 3. Baire's theorem is from his own thesis (1899). The characterization of continuity sets — necessity tonight, the converse in Problem III.1 — goes back to W. H. Young (1903) and is treated in Hahn's *Theorie der reellen Funktionen*; our two-sign construction is the standard modern staging.

The ladder above the first rungs. $F_{\sigma\delta}$, $G_{\delta\sigma}$, and onward: the Borel hierarchy genuinely ascends through all countable transfinite stages — no countable list of operations exhausts $\mathcal{B}$ — and its honest treatment needs ordinals. Oxtoby, *Measure and Category*, chapters 1–3, is the season's recommended companion for exactly this circle of ideas (Baire, hierarchy, the category/measure duality previewed next chapter); Kechris, *Classical Descriptive Set Theory*, for the abyss in full.

Reading. Axler, *Measure, Integration & Real Analysis* (the season's companion, per the plan's shelf: chapter 2A covers tonight's ground); Royden–Fitzpatrick, chapters 1–2; Folland, chapter 1; Stein–Shakarchi, *Real Analysis*, chapter 1. Tao's *An Introduction to Measure Theory* opens with a beautiful extended meditation on exactly tonight's contract-and-catch dialectic.

Forward pointers. The scorecard's unpaid half — countable additivity on a tribe — is Chapter 2 (Carathéodory); the catch's trial, Chapter 3 (Vitali, Choice, Banach–Tarski); the null ideal meets “almost everywhere” in force at Chapter 4, where the static is tamed and Volume One's criterion is finally proved.

Chapter 2

Lebesgue Measure

*Companion to Episode Two. The episode met Carathéodory's criterion under three metaphors and one rehearsal, admitted the first citizens (null sets; the half-line turnstile), narrated the four-step architecture of the 1914 theorem — symmetry, composition, proportion, limit — settled the contract on the tribe with uniqueness attached, drove the machine (beads, frames, the irrationals at full measure, the removals eating the interval), took the approximation manners as Littlewood's first principle in embryo, paid **debt eight's question of extent** — the dust at measure zero, act three — met the fat dust and divorced the two smallnesses, honoured Jordan's old regime inside the new, licensed the construction in every dimension, and found the Borel estate a suburb of the commonwealth: vast veil, Borel substance. This chapter is the paperwork, including the season's longest single proof (Theorem 2.5), written out in full as promised, with relish. Register notes: **the bell is silent this chapter** — the count holds at fifteen; every argument below runs on rings already paid for (the Episode Six precedent: wages to dividends), and the standing small-change annotation applies to the chapter's routine suprema. Debt eight's extent is **paid** at Theorem 2.18; the two-courts counsel is adopted as standing convention at Remark 2.21.*

Episode Digest

The argument of the hour. One strange sentence selects the tribe. A set is measurable when it splits every test set cleanly — the good citizen, the honest knife, the balanced audit — and the criterion's entire content is the reverse of subadditivity: no phantom mass at the cut. Null sets pass (completeness in embryo); half-lines pass (the turnstile to geometry); and Carathéodory's theorem builds the rest by a four-step staircase — symmetry (complements free), composition (honest knives compound), proportion (disjoint knives cut every cake in exact proportion, by induction), limit (the monotone license closes both the sigma-algebra walls and countable additivity). On the tribe, Episode One's contract holds entire — and uniquely: nothing else was ever on offer. The machine then runs: open sets by beads; the irrationals of the unit interval at measure one; the dust — uncountable, the size of the continuum — at measure **zero**, the account three episodes and two seasons old, paid in four sentences; the fat dust (Smith–Volterra–Cantor) at measure one half though nowhere dense, divorcing category's smallness from measure's; the stacked fat dusts, meagre yet of full measure, and their complement, residual yet null — never say small without naming the judge. Jordan's regime is absorbed intact; the construction is dimension-blind (area, volume: same theorems verbatim); and the commonwealth outnumbers the Borel estate two-to-the-continuum to continuum, yet every citizen is a G_δ shaved of a null set: the vastness is veil, the substance Borel.

The episode, beat by beat. The Typo. The predicament restated: outer measure for every set; four clauses kept; additivity impossible globally (the catch, on credit). Tonight's business: name the tribe, prove the sigma-algebra, prove the contract — Carathéodory 1914, the season's longest desk engagement, announced with relish.

The Good Citizen. The criterion (Definition 2.1), verbatim in words; three objections became three defences (no geometry — conduct, not appearance; content is the reverse inequality only — Remark 2.2; universal test sets — citizenship against the whole commonwealth). Three metaphors — customs, knife, audit — and a rehearsal: a half-line splits interval cakes cleanly by schoolroom arithmetic.

First Citizens. Null sets measurable (Proposition 2.3) — two lines, hereditary law plus monotonicity; completeness in embryo. Half-lines measurable (Theorem 2.4): covers split at the station, lengths add per interval, epsilon of grace accounted. The dishonest-knife preview: Vitali's defendant, next chapter. Geometry enters through one turnstile.

The Theorem of Nineteen Fourteen. Statement (Theorem 2.5); the staircase: symmetry (Lemma 2.6), composition with the four named pieces (Lemma 2.7), proportion by induction (Lemma 2.8), the two-army siege at the limit. The machinery bill audited: subadditivity, induction, the monotone license — the course's oldest instruments; no new ring. The staircase recurs at product measures (Episode Seven) and wherever abstract measures live.

Where the Typo Came From. Lebesgue's two-ledger original: inner measure by complement-within-a-frame (Definition 2.15); inner meets outer, cage meets packing — Jordan's architecture one storey up. Carathéodory's reformulation: the frame is a test cake; then all cakes, no frames. Equivalence on bounded sets: Problem II.2, solved in full.

The Scorecard Settled. Lebesgue measure (Definition 2.10); the five clauses verified on the tribe (Corollary 2.11); citizenship translates (Proposition 2.12); the crankshaft dividends legally operative. **Uniqueness** (Theorem 2.13): rivals agree on intervals, extend by beads, squeeze by manners — the contract was a description, not a specification; the stronger translation-invariant rigidity is Problem III.2. Dimension license (Remark 2.14): the construction verbatim in the plane and in space; next chapter's volumes owned, not borrowed.

Driving the Machine. Beads (the formula, drilled at Problem I.2); frames and the one arithmetic prohibition in action; the membership pipeline run once at full length (dust to citizen in five moves); the irrationals of the unit interval at measure one — almost every number is irrational, as land; transcendentals likewise; the removals eat the interval (continuity from below at the dust's complement); sigma-finiteness filed for Episode Seven.

The Old Regime, Honoured. Jordan absorbed (Proposition 2.22): every Jordan-measurable set is a citizen at its old price — the squeeze in four lines; the revolution expropriated no one. Episode Four's Riemann absorption rehearsed.

The Measure's Manners. Approximation: open from above (Theorem 2.16), closed from below, compact in bounded frames; the structural jewel — every citizen a G -delta shaved of a null set (Theorem 2.17); Littlewood's first principle proper — nearly a finite union of intervals (Problem II.4, guided); the trade's working face.

The Dust, Act Three. The ceremony (Theorem 2.18): stage covers, two-thirds to the n , geometric decay — measure zero; uncountable by the diagonal; the account's history read into the record; the wonder beat — intuition caught compounding the wrong way; pure architecture, zero

acreage; number and land divorced, decree absolute. Grace notes: the address system survives; the almost-surely wink toward the finale. The stranger return (Episode Four) stays on the payroll.

The Fat Dust. Smith–Volterra–Cantor (Proposition 2.19): removals total one half; closed, nowhere dense, measure one half. Fattening to one-minus-epsilon; never to one (open dense complements keep intervals). The stacked fat dusts: meagre, full measure (Proposition 2.20). Volterra’s stage explained: oscillation seated on positive measure is what the Riemann ledgers cannot absorb.

Two Smallnesses / What the Courts Agree On. Meagre versus null: both sigma-ideals, both convict countable sets, both humbled by starved open dense sets; the fat dust and the stacked complement disagree maximally in both directions; the partition of the unit interval into meagre plus null (Problem II.5). Field vocabulary: meagre/residual, null/full; generic maps to residual. Standing convention adopted: name the judge (Remark 2.21).

Borel and Beyond. Borel sets are citizens (Corollary 2.9); the count — Borel at continuum (on credit, references), citizens at two-to-the-continuum via the dust’s subsets and completeness; almost no citizen is Borel, yet every citizen is Borel-plus-veil (Theorem 2.17); completion (Definition 2.23, Theorem 2.24): Lebesgue is completed Borel; the trade’s events convention.

The Season’s Ledger. The bell silent at fifteen — dividends, the Episode Six precedent by name; debt eight’s extent paid; the scorecard closed; completeness banked; the two-courts counsel elevated to convention; the catch’s trial called for next week (Vitali, Choice, Banach–Tarski defused).

The toolkit acquired. The splitting criterion and its free direction; the four-step staircase; the proportion-lemma induction pattern; approximation from outside and inside; the structure theorem (Borel plus veil); the fat-dust construction and the stacking manoeuvre; the name-the-judge discipline.

Dates worth keeping. Lebesgue’s thesis, 1902 (inner-outer measurability); Carathéodory’s criterion and theorem, 1914, *Über das lineare Mass*; Smith’s fat dust, 1875, six years before Volterra’s use; Vitali’s defendant, 1905, arraigned next chapter.

2.1 Notation Ledger

Spoken in the episode	Written here	Remarks
“measurable”; “the tribe”	$E \in \mathcal{M}: \forall T, m^*(T) = m^*(T \cap E) + m^*(T \setminus E)$	Def. 2.1
“the test cake”	T (arbitrary subset of $\mathbb{R}$)	Def. 2.1
“Lebesgue measure”	$m = m^* \upharpoonright_{\mathcal{M}}$	Def. 2.10
“the inner measure” (frame $[a, b] \supseteq E$)	$m_*(E) = (b - a) - m^*([a, b] \setminus E)$	Def. 2.15
“open surplus”; “closed certificate”	$U \supseteq E$ open, $m(U \setminus E) < \varepsilon$; $F \subseteq E$ closed, $m(E \setminus F) < \varepsilon$	Thm. 2.16
“the fat dust”	S (Smith–Volterra–Cantor set)	Prop. 2.19
“meagre / residual; null / full”	the two courts’ verdicts	Rem. 2.21
“the completion”	$\overline{\mathcal{A}}$: sets $A \cup Z$, Z inside a null member	Def. 2.23

Standing conventions. The restriction bar $\upharpoonright$ reads “restricted to.” Throughout, “null” means $m^* = 0$ (Chapter 1); the extended arithmetic and its one prohibition (Chapter 1, the ledger’s fine print) remain in force — subtraction only where the minuend is finite.

2.2 The Criterion

Definition 2.1 (Carathéodory measurability). $E \subseteq \mathbb{R}$ is *measurable* when for every $T \subseteq \mathbb{R}$,

$$m^*(T) = m^*(T \cap E) + m^*(T \setminus E).$$

The collection of measurable sets is written $\mathcal{M}$.

Remark 2.2 (The free direction). $T = (T \cap E) \cup (T \setminus E)$, so countable subadditivity (Theorem 1.11, the gambit’s theorem) gives $m^*(T) \leq m^*(T \cap E) + m^*(T \setminus E)$ always, for every E and T . Membership in $\mathcal{M}$ is therefore exactly the reverse inequality

$$m^*(T \cap E) + m^*(T \setminus E) \leq m^*(T) \quad \text{for every } T,$$

and this is the only inequality ever checked below: no phantom mass at the cut. (When $m^*(T) = \infty$ the reverse inequality is automatic; all content lives on finite test cakes.)

Proposition 2.3 (Null sets are citizens). *If $m^*(N) = 0$ then $N \in \mathcal{M}$; consequently (the null ideal being hereditary, Chapter 1) every subset of a null set is measurable. This property of $\mathcal{M}$ is called completeness.*

Proof. For any T : $m^*(T \cap N) \leq m^*(N) = 0$ by monotonicity, and $m^*(T \setminus N) \leq m^*(T)$ likewise. Their sum is at most $m^*(T)$; Remark 2.2 closes the pinch. $\square$

Theorem 2.4 (The turnstile: half-lines are citizens). *For every $a \in \mathbb{R}$: $(a, \infty) \in \mathcal{M}$.*

Proof. Let T have $m^*(T) < \infty$ (Remark 2.2) and let $\varepsilon > 0$. Choose a cover $\{I_k\}$ of T with $\sum_k |I_k| \leq m^*(T) + \varepsilon$ (approximation property). The station a splits each interval cleanly: $I'_k = I_k \cap (a, \infty)$ and $I''_k = I_k \cap (-\infty, a]$ are intervals (possibly empty or degenerate) with $|I'_k| + |I''_k| = |I_k|$ — schoolroom arithmetic, case by case on where a sits relative to I_k . Now $\{I'_k\}$ covers $T \cap (a, \infty)$ and $\{I''_k\}$ covers $T \setminus (a, \infty)$, so

$$m^*(T \cap (a, \infty)) + m^*(T \setminus (a, \infty)) \leq \sum_k |I'_k| + \sum_k |I''_k| = \sum_k |I_k| \leq m^*(T) + \varepsilon.$$

As ε was arbitrary, the reverse inequality holds; the half-line balances every audit. $\square$

2.3 Carathéodory’s Theorem

Theorem 2.5 (Carathéodory, 1914). *$\mathcal{M}$ is a sigma-algebra containing every null set, and m^* restricted to $\mathcal{M}$ is countably additive: for pairwise disjoint $E_1, E_2, \dots \in \mathcal{M}$,*

$$m^*\left(\bigcup_k E_k\right) = \sum_k m^*(E_k).$$

The proof is the season’s longest, as promised, and is taken in the episode’s four steps. Throughout, T denotes an arbitrary test set with $m^*(T) < \infty$.

Lemma 2.6 (Step one: symmetry). $\emptyset, \mathbb{R} \in \mathcal{M}$, and $E \in \mathcal{M}$ implies $E^c \in \mathcal{M}$.

Proof. $T \cap \emptyset = \emptyset$ (outer measure zero) and $T \setminus \emptyset = T$: the empty set balances trivially, and $\mathbb{R}$ with it. The criterion is symmetric in E and E^c , since $T \cap E^c = T \setminus E$ and $T \setminus E^c = T \cap E$: the two slices merely exchange names, and the balanced equation is the same equation. $\square$

Lemma 2.7 (Step two: composition). If $E, F \in \mathcal{M}$ then $E \cup F \in \mathcal{M}$ (hence, with symmetry, $E \cap F$ and $E \setminus F$ as well: $\mathcal{M}$ is an algebra).

Proof. The episode's four pieces of T are: within both $(T \cap E \cap F)$; within E only; within F only; within neither $(T \setminus (E \cup F))$. The bookkeeping, tightly: cut T by E (exact, E being a citizen); then cut the outside slice $T \setminus E$ by F (exact again, the criterion holding for *every* test set — here is where the universal quantifier pays):

$$m^*(T) = m^*(T \cap E) + m^*((T \setminus E) \cap F) + m^*((T \setminus E) \setminus F).$$

Now $(T \setminus E) \setminus F = T \setminus (E \cup F)$ exactly, and the union's inside decomposes as $T \cap (E \cup F) = (T \cap E) \cup ((T \setminus E) \cap F)$ — a point of T in the union is in E , or else in F but not E . Subadditivity on that two-piece decomposition gives

$$m^*(T \cap (E \cup F)) \leq m^*(T \cap E) + m^*((T \setminus E) \cap F),$$

and substituting into the display:

$$m^*(T \cap (E \cup F)) + m^*(T \setminus (E \cup F)) \leq m^*(T).$$

Remark 2.2 closes the pinch: honest knives, used in sequence, make an honest compound cut. Finite unions follow by iteration; intersections and differences by De Morgan with Lemma 2.6. $\square$

Lemma 2.8 (Step three: proportion). Let $E_1, \dots, E_n \in \mathcal{M}$ be pairwise disjoint. Then for every test set T ,

$$m^*\left(T \cap \bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n m^*(T \cap E_k).$$

Proof. Induction on n ; the case $n = 1$ is an identity. For the step, cut the cake $T \cap \bigcup_{k \leq n} E_k$ with the knife E_n — a citizen, so the cut is exact. By disjointness, the inside slice is

$$\left(T \cap \bigcup_{k \leq n} E_k\right) \cap E_n = T \cap E_n,$$

the newest piece alone, and the outside slice is $T \cap \bigcup_{k \leq n-1} E_k$, the accumulated rest — nothing shared, nothing lost. Exactness reads

$$m^*\left(T \cap \bigcup_{k \leq n} E_k\right) = m^*(T \cap E_n) + m^*\left(T \cap \bigcup_{k \leq n-1} E_k\right),$$

and the induction hypothesis expands the second term into the remaining sum. $\square$

Proof of Theorem 2.5 (step four: the limit). Let $E_1, E_2, \dots \in \mathcal{M}$ be pairwise disjoint, with union E and partial unions $U_n = \bigcup_{k \leq n} E_k$ (citizens, by composition). Fix a test set T . For each n , cutting T by U_n exactly and applying proportion to the inside:

$$m^*(T) = \sum_{k=1}^n m^*(T \cap E_k) + m^*(T \setminus U_n) \geq \sum_{k=1}^n m^*(T \cap E_k) + m^*(T \setminus E),$$

the last step by monotonicity ($T \setminus U_n \supseteq T \setminus E$: sieging armies only grow). The left side is free of n ; let $n \rightarrow \infty$ (the monotone license: partial sums of nonnegative terms climb to the series):

$$m^*(T) \geq \sum_{k=1}^{\infty} m^*(T \cap E_k) + m^*(T \setminus E) \geq m^*(T \cap E) + m^*(T \setminus E),$$

the second inequality by countable subadditivity applied to $T \cap E = \bigcup_k (T \cap E_k)$. By Remark 2.2 the outer terms are equal, so every inequality in the chain is an equality. Reading the chain twice: first, E satisfies the criterion — $\mathcal{M}$ is closed under *disjoint* countable unions, and with the algebra operations of Lemma 2.7, under arbitrary countable unions (disjointify: $A'_k = A_k \setminus \bigcup_{j < k} A_j \in \mathcal{M}$, same union) — the walls are complete, and $\mathcal{M}$ is a sigma-algebra. Second, take $T = E$: then $T \cap E_k = E_k$ and $T \setminus E = \emptyset$, and the equality of the chain reads

$$m^*(E) = \sum_{k=1}^{\infty} m^*(E_k) :$$

countable additivity on the tribe. Null citizens were Proposition 2.3. *The machinery bill: sub-additivity, induction, the monotone license — no compactness, no Baire, no new ring of the bell. The definition carried the depth; the proof collected it.* $\square$

Corollary 2.9 (The Borel estate immigrates entire). $\mathcal{B} \subseteq \mathcal{M}$: every Borel set is measurable.

Proof. $\mathcal{M}$ is a sigma-algebra (Theorem 2.5) containing every half-line (Theorem 2.4), hence every interval ($(a, b] = (a, \infty) \setminus (b, \infty)$, and kin by the algebra operations), hence — the estate's robustness, Theorem 1.20 — all of $\mathcal{B} = \sigma(\text{intervals})$: the smallest sigma-algebra containing the intervals sits inside any sigma-algebra containing them. $\square$

2.4 Lebesgue Measure and Its Credentials

Definition 2.10 (Lebesgue measure). $m = m^* \upharpoonright_{\mathcal{M}}$: outer measure, restricted to the measurable tribe.

Corollary 2.11 (The contract, settled). On $\mathcal{M}$, all five clauses of Chapter 1's contract hold: $m \geq 0$ with $m(\emptyset) = 0$; $m(I) = b - a$ on intervals; $m(E + t) = m(E)$; and countable additivity. Moreover the crankshaft dividends operate legally: continuity along swelling chains, and along shrinking chains with a finite start (Proposition 1.3, whose hypotheses $\mathcal{M}$ and m now satisfy).

Proof. Clauses one, two, three are Chapter 1's theorems for m^* , inherited by restriction (intervals are citizens by Corollary 2.9); clause four is Theorem 2.5. The crankshaft was proved for any countably additive assignment on a sigma-algebra. $\square$

Proposition 2.12 (Citizenship translates). If $E \in \mathcal{M}$ and $t \in \mathbb{R}$, then $E + t \in \mathcal{M}$.

Proof. For any test T : $T \cap (E + t) = ((T - t) \cap E) + t$ and $T \setminus (E + t) = ((T - t) \setminus E) + t$ (shift the whole audit by $-t$, cut, shift back). Translation invariance of m^* (Chapter 1, Problem II.3) prices the shifted slices identically, and E cuts the shifted cake $T - t$ exactly. $\square$

Theorem 2.13 (Uniqueness: nothing else was on offer). *Let ν be countably additive on $\mathcal{M}$ (or on $\mathcal{B}$), with $\nu(I) = |I|$ on every interval. Then $\nu = m$ on its domain.*

Proof. On open sets: beads (Volume One, Proposition 5.3) and countable additivity force $\nu(U) = \sum |\text{beads}| = m(U)$. Within a frame $[-N, N]$, for measurable $E \subseteq [-N, N]$: given ε , Theorem 2.16 supplies open $U \supseteq E$ (intersected with an open frame-pad, still open) with $m(U) \leq m(E) + \varepsilon$; then $\nu(E) \leq \nu(U) = m(U) \leq m(E) + \varepsilon$, so $\nu(E) \leq m(E)$ — monotonicity of ν being additivity's child. Run the same estimate on the frame's remainder $[-N, N] \setminus E$ and add: both split the frame, whose ν - and m -prices agree (an interval), so the two $\leq$'s must both be equalities: $\nu(E) = m(E)$ on framed citizens. Unframed citizens are swelling unions of framed ones ($E \cap [-N, N] \uparrow E$), and both parties obey continuity from below (Corollary 2.11 for m ; the crankshaft, whose proof used only countable additivity, for ν). Frame-patching details: Problem II.6's ambit. The translation-invariant strengthening — ν shift-blind and finite on frames forces $\nu = c \cdot m$ — is Problem III.2. $\square$

Remark 2.14 (The construction is dimension-blind). Replace intervals by boxes (products of intervals) in the plane or in space, box-volume for length, and every definition and theorem of Chapter 1 and this chapter repeats verbatim: outer measure by countable box covers; the same gambit; Carathéodory's criterion and staircase unchanged; Borel sets from open sets; the contract with translation invariance in all coordinates. The desk assigns nothing — the proofs are the above with the words exchanged — but the license is now owned: Chapter 3 may speak of the volume of a ball outright, and Chapter 7 will build the plane's measure a second way (as a product) and prove the two agree. References for the transposition written out: the Notes.

2.5 Where the Typo Came From

Definition 2.15 (Lebesgue's inner measure, 1902). For bounded E contained in a frame $[a, b]$:

$$m_*(E) = (b - a) - m^*([a, b] \setminus E),$$

the frame's length less the complement's outer price (independent of the chosen frame: enlarging the frame adds equal amounts to both terms — Problem II.2's warm-up). Lebesgue's original definition: a bounded set is measurable when $m_*(E) = m^*(E)$ — cage meets packing, the two-ledger architecture of Darboux and Jordan one storey up.

The reconciliation — on bounded sets, inner-meets-outer holds if and only if Carathéodory's criterion does — is Problem II.2, solved in full. One direction is transparent (the frame is one test cake; Carathéodory quantifies over all); the other is the useful surprise: balancing the frame suffices to balance every cake, outer regularity doing the generalizing.

2.6 The Measure's Manners

Theorem 2.16 (Approximation: open surpluses, closed and compact certificates). *Let $E \in \mathcal{M}$ and $\varepsilon > 0$.*

(i) There is an open $U \supseteq E$ with $m(U \setminus E) \leq \varepsilon$.

(ii) There is a closed $F \subseteq E$ with $m(E \setminus F) \leq \varepsilon$.

(iii) If E is bounded, the F of (ii) is compact.

Proof. (i) First suppose $m(E) < \infty$. A cover within ε of $m(E)$, fattened open (Chapter 1's fattening trick, at cost folded into ε), unions to an open $U \supseteq E$ with $m(U) \leq \sum |J_k| \leq m(E) + \varepsilon$ (subadditivity); additivity on the citizen pair $(E, U \setminus E)$ converts the total into a surplus: $m(U \setminus E) = m(U) - m(E) \leq \varepsilon$ — subtraction lawful, the minuend finite. For $m(E) = \infty$: frame the line, $E_N = E \cap [-N, N]$; take open $U_N \supseteq E_N$ with surplus $\varepsilon/2^N$ (the gambit, as ever); $U = \bigcup U_N$ is open, contains E , and $U \setminus E \subseteq \bigcup (U_N \setminus E_N)$ prices below ε by subadditivity.

(ii) Complements: apply (i) to E^c (a citizen, by symmetry), obtaining open $V \supseteq E^c$ with $m(V \setminus E^c) \leq \varepsilon$; then $F = V^c$ is closed, $F \subseteq E$, and $E \setminus F = V \setminus E^c$ exactly.

(iii) A closed subset of a bounded set is closed and bounded: compact (Volume One, Theorem 5.18's characterization). The bounded case is Problem II.3, solved in full with the frame bookkeeping displayed; the display above is its skeleton. $\square$

Theorem 2.17 (The structural jewel: Borel plus veil). *$E \subseteq \mathbb{R}$ is measurable if and only if $E = G \setminus Z$ for some G_δ set G and null Z — equivalently $E = F \cup Z'$ with F an F_σ and Z' null.*

Proof. $(\Rightarrow)$ Take open $U_n \supseteq E$ with surplus $1/n$ (Theorem 2.16(i)); $G = \bigcap_n U_n$ is a G_δ containing E , and $G \setminus E \subseteq U_n \setminus E$ has measure at most $1/n$ for every n : null. Then $E = G \setminus (G \setminus E)$. The F_σ form is the mirrored (ii). $(\Leftarrow)$ G_δ sets are Borel, hence citizens (Corollary 2.9); null sets are citizens (Proposition 2.3); and $\mathcal{M}$ is closed under difference and union. — Every citizen is a Borel body wearing a null veil; the commonwealth's vastness is entirely veil. $\square$

2.7 The Dust, Act Three

Theorem 2.18 (The decree: the dust is null). *The Cantor set C (Volume One, Definition 3.20) is measurable with $m(C) = 0$ — an uncountable set (Volume One, Chapter 3) of measure zero.*

Proof. Citizenship by the pipeline: C is closed, its complement open, hence Borel, hence measurable (Corollary 2.9). The price: after stage n of the construction, C lies inside a union C_n of 2^n closed intervals of length 3^{-n} each — a finite union of citizens of total measure $(2/3)^n$ (additivity on the disjoint components). Monotonicity: $m(C) \leq (2/3)^n$ for every n , and the geometric decay (Volume One, Chapter 4) sends the bound to nought. A nonnegative number below every positive number is zero. $\square$

Debt eight, the question of extent: PAID. — Uncountably many addresses, zero acreage; number and land divorced, decree absolute. The account's history — born Episode Three, toured Episode Five, serviced Episode Eight, opened last chapter, paid here — is read into the episode's record; the stranger return (the Cantor staircase and its kin) remains contracted for Chapter 4. The complement computation — the removals swelling to full measure, continuity from below in action — is Problem I.5.

2.8 The Fat Dust and the Two Courts

Proposition 2.19 (Smith–Volterra–Cantor). *Construct $S \subseteq [0, 1]$ by removing, at stage n , from each of the 2^{n-1} surviving closed pieces its open middle interval of length 4^{-n} . Then S is closed, nowhere dense (contains no interval; Volume One, Definition 5.27), and*

$$m(S) = 1 - \sum_{n=1}^{\infty} 2^{n-1} \cdot 4^{-n} = 1 - \frac{1}{2} = \frac{1}{2}.$$

Fattening the removals to total any ε yields closed nowhere-dense sets of measure $1 - \varepsilon$; measure exactly 1 is unattainable (the complement is open, dense, and nonempty, hence contains an interval of positive measure).

Proof. The removals are pairwise disjoint open intervals; their total, the geometric series $\sum_n 2^{n-1} 4^{-n} = \frac{1}{2} \sum_n 2^{-n} = \frac{1}{2}$. S is the complement in $[0, 1]$ of their union: closed, and citizen's arithmetic (additivity, subtraction within the finite frame) prices it at $\frac{1}{2}$. Nowhere dense: the stage- n construction leaves 2^n disjoint closed pieces of total length $1 - r_n < 1$, so each piece is shorter than $2^{-n} \cdot 1$; an interval $I \subseteq S$, being connected, would lie within a single piece at every stage, forcing $|I| \leq 2^{-n} \rightarrow 0$. A closed set with empty interior is nowhere dense. $\square$

Proposition 2.20 (The stacked dusts: meagre yet full). *Let S_j be fat dusts with removals totalling 2^{-j} , and $M = \bigcup_j S_j$. Then M is meagre (a countable union of nowhere-dense sets) with $m(M) = 1$; its complement $R = [0, 1] \setminus M$ is residual in $[0, 1]$ and null. In particular $[0, 1] = M \sqcup R$ partitions the interval into a meagre set and a null set.*

Proof. Meagreness is the definition. $m(M) \geq m(S_j) = 1 - 2^{-j}$ for every j (monotonicity), and $m(M) \leq 1$: so $m(M) = 1$. Additivity within the frame prices R at nought; R is the complement of a meagre set in $[0, 1]$, which is residual by definition. Formalities of the partition claim: Problem II.5. $\square$

Remark 2.21 (The two courts; the standing convention). Category's negligible: *meagre*; its almost-everything: *residual*. Measure's: *null* and *full*. Both ideals are sigma-ideals; both convict countable sets; both acquit sets with interior; and on the closed empty-interior territory between, they rule independently — the fat dust (meagre, substantial) and the stacked complement (residual, null) disagree maximally in both directions. **Standing convention, adopted this episode:** no verdict of smallness is entered in this course without naming the judge; “almost everywhere” is henceforth measure's phrase exclusively. Volume One's *generic* maps to *residual*; whether the nowhere-differentiable monsters, generic in category's court (Volume One, Chapter 7), are also typical in measure's is answered in the affirmative by Rademacher–Kolmogorov-flavoured arguments beyond this course — references in the Notes for the ambitious.

2.9 The Old Regime, Honoured

Proposition 2.22 (Jordan absorbed). *Every Jordan-measurable bounded set A (Chapter 1, Definition 1.4) is Lebesgue measurable, and $m(A) = c(A)$, the Jordan content.*

Proof. Price first: finite covers are countable covers (pad with empties), so $m^*(A) \leq \bar{c}(A)$; and any finite disjoint packing $\bigsqcup_j I_j \subseteq A$ is a citizen of measure $\sum_j |I_j|$, so $m^*(A) \geq \underline{c}(A)$ by

monotonicity. Jordan measurability squeezes: $m^*(A) = c(A)$. Citizenship: the two contents differ by the boundary's outer content — $\bar{c}(A) - \underline{c}(A) = \bar{c}(\partial A)$, Chapter 1's Problem III.3 anatomy — so ∂A has content zero, hence outer measure zero (covers again), hence is null. Then $A = A^\circ \cup (A \cap \partial A)$: an open citizen united with a subset of a null set, measurable by completeness (Proposition 2.3) and the algebra. The old deeds transfer unaltered. $\square$

2.10 Completion, and the Size of the Commonwealth

Definition 2.23 (Completion of a measure). Given a sigma-algebra $\mathcal{A}$ with a countably additive μ , its *completion* $\bar{\mathcal{A}}$ consists of all sets $A \cup Z$ with $A \in \mathcal{A}$ and Z a subset of some μ -null member of $\mathcal{A}$; $\bar{\mu}(A \cup Z) = \mu(A)$. (Well-definedness and the sigma-algebra checks: Problem III.1, with the universal property — the smallest complete extension.)

Theorem 2.24 (Lebesgue is completed Borel). $\mathcal{M} = \bar{\mathcal{B}}$: the measurable sets are exactly the Borel sets adjusted by subsets of Borel null sets, and m is the completion of its own restriction to $\mathcal{B}$.

Proof. ($\supseteq$) Borel sets are citizens; subsets of null sets are citizens (completeness); unions of citizens are citizens. ($\subseteq$) The structural jewel (Theorem 2.17), F_σ form: $E = F \cup Z'$ with F Borel and Z' null — and Z' , being null, sits inside a Borel null set (the G_δ hull of Theorem 2.17 applied to Z'). Measures agree since $\bar{m}(F \cup Z') = m(F) = m(E)$. $\square$

Remark 2.25 (The count). The Borel estate has exactly continuum-many members — stated on credit with references (the transfinite ladder; a generated sigma-algebra from a countable seed cannot outgrow the continuum), per the honesty clause; no argument of this course uses it except here. The commonwealth: the dust is null with continuum-many points, so *every* subset of the dust is a citizen (Proposition 2.3) — two-to-the-continuum many, strictly more than the continuum (Volume One, Chapter 3, Cantor's theorem). Almost no citizen is Borel; every citizen is Borel plus veil (Theorem 2.24); and the count exhibits nothing — existence without exhibition, debt six's family trait, the patriarch arraigned next chapter.

2.11 Problems

Tier I — Calisthenics

- I.1 Verify Carathéodory's criterion for (a, ∞) directly when the test set is a single interval, by cases on where the station sits; then for a test set of two disjoint intervals. (The episode's rehearsal, in writing.)
- I.2 An open set is given as the union of the intervals from $1/(n+1)$ to $1/n$ for even n . Price it by the beads formula, and verify the answer against the complement within $(0, 1]$.
- I.3 Prove formally that the irrationals of $[0, 1]$ have measure one, naming each clause of the contract used.
- I.4 Rebuild the fat dust with stage- n middles of length 5^{-n} : total the removals, price the residue, and confirm nowhere density survives.

- I.5 The dust's complement: let r_n be the total length removed through stage n of the middle-thirds construction. Compute r_n exactly, exhibit the swelling chain, and price the limit by continuity from below (Proposition 1.3).
- I.6 Exhibit an open $U \supseteq \mathbb{Q} \cap [0, 1]$ with $m(U) < \varepsilon$, and state what its complement within $[0, 1]$ certifies (a closed subset of the irrationals of measure exceeding $1 - \varepsilon$).
- I.7 Translation drills: $C + t$ is null; the shifted fat dust $S + t$ is measurable of measure one half; and the criterion survives reflection $x \mapsto -x$ (state the reflected invariance of m^* and adapt Proposition 2.12).
- I.8 The two-courts docket: classify each of $\mathbb{Q} \cap [0, 1]$; the irrationals of $[0, 1]$; C ; S ; the stacked M ; and $R = [0, 1] \setminus M$ — meagre or residual? null or full? One line of warrant per verdict.

Tier II — The Real Thing

- II.1 (*Composition, to the letter.*) Complete Lemma 2.7's bookkeeping in the four-piece dialect of the episode: name the four pieces, perform the three exact cuts, and reconcile with the tight three-line proof given in the text.
- II.2 (*The two definitions agree.*) Prove Definition 2.15 is frame-independent; then prove that for bounded E , $m_*(E) = m^*(E)$ if and only if E satisfies Carathéodory's criterion. Solved in full below, as the provenance section promised.
- II.3 (*Compact certificates.*) For bounded measurable E and $\varepsilon > 0$, produce compact $K \subseteq E$ with $m(E \setminus K) < \varepsilon$, with the frame bookkeeping displayed. Solved in full below.
- II.4 (*Littlewood's first principle, proper.*) For measurable E with $m(E) < \infty$ and $\varepsilon > 0$, produce a finite union V of intervals with $m((E \setminus V) \cup (V \setminus E)) < \varepsilon$. (Open surplus; beads; a long enough finite prefix.)
- II.5 (*The partition.*) Formalize Proposition 2.20 into the partition statement: the unit interval is the disjoint union of a meagre set and a null set; verify each court convicts exactly one part.
- II.6 (*Frame-patching.*) Complete Theorem 2.13: show the framed agreement extends to all citizens by continuity from below, and identify precisely where countable additivity of the rival ν is used.

Tier III — For the Ambitious ★

- III.1 ★ (*Completion in the abstract.*) Prove Definition 2.23's $\overline{\mathcal{A}}$ is a sigma-algebra, $\overline{\mu}$ is well-defined and countably additive, the completed measure is complete, and the construction is the *smallest* complete extension (universal property). Then re-derive Theorem 2.24 from the abstract statement.
- III.2 ★ (*Rigidity.*) Let ν be countably additive and translation invariant on $\mathcal{B}$ with $c = \nu([0, 1)) < \infty$. Prove $\nu = c \cdot m$ on $\mathcal{B}$: dyadic subdivision prices ν on dyadic intervals; continuity and density extend to all intervals; then Theorem 2.13's squeeze. Conclude: honest shift-blind measurement on the line is Lebesgue's, up to the choice of yardstick.

2.12 Hints

II.1. The three exact cuts: T by E ; $T \cap E$ by F ; $T \setminus E$ by F . The four pieces reassemble to $m^*(T)$; the union's inside is three of the pieces, priced by subadditivity against their sum.

II.4. With $U \supseteq E$ open of surplus $\varepsilon/2$: U 's beads have summable lengths (total $m(U) < \infty$); choose N with tail below $\varepsilon/2$; let V be the first N beads. Then $E \setminus V \subseteq$ the tail beads together with nothing else, and $V \setminus E \subseteq U \setminus E$.

II.5. $M \cup R = [0, 1]$ and $M \cap R = \emptyset$ by construction; the courts' verdicts are Proposition 2.20's content. For "exactly one": M is not null ($m = 1$) and R is not meagre (else $[0, 1]$ would be meagre in itself, contradicting Baire — Volume One, Theorem 5.28).

II.6. $E \cap [-N, N] \uparrow E$; the crankshaft (Proposition 1.3(i)) applies to any countably additive assignment on a sigma-algebra — including ν ; that is the only use of ν 's countable additivity beyond the framed squeeze.

III.1. Well-definedness: if $A_1 \cup Z_1 = A_2 \cup Z_2$ then $A_1 \triangle A_2$ sits inside the union of the two null hulls. Universal property: any complete extension must contain every Z , hence every $A \cup Z$.

III.2. $[0, 1]$ splits into 2^n translated copies of $[0, 2^{-n}]$: additivity and invariance force $\nu([0, 2^{-n}]) = c 2^{-n}$. Dyadic endpoints are dense; half-open intervals are swelling unions of dyadic ones; continuity from below prices all intervals at $c \cdot$ length. Then run Theorem 2.13 with $c \cdot m$ as the incumbent.

2.13 Solutions

Tier I (terse). **I.1** Station left of the cake: inside slice is the whole cake, outside empty; balanced. Station inside: two subintervals, lengths adding to the whole. Station right: mirror of the first. Two-interval cake: each interval balances separately; totals add. **I.2** Beads at even n : lengths $\frac{1}{n} - \frac{1}{n+1}$; the sum telescopes within the even indices to $\sum_k (\frac{1}{2k} - \frac{1}{2k+1})$, which the desk evaluates against the complement: the odd-indexed beads price the rest of $(0, 1]$, and the two series sum to one. (Numerically: the even-bead total is $1 - \ln 2$ in the notation the course has not yet earned; the drill wants only the telescoping set-up and the complement identity.) **I.3** $[0, 1] = (\mathbb{Q} \cap [0, 1]) \sqcup (\text{irrationals})$: countable additivity (two pieces), the rationals null (Chapter 1), the interval clause: $1 = 0 + m(\text{irrationals})$. **I.4** Removals: $\sum 2^{n-1} 5^{-n} = \frac{1}{2} \cdot \frac{2/5}{1-2/5} = \frac{1}{3}$; residue $\frac{2}{3}$; nowhere density: stage pieces again shorter than 2^{-n} . **I.5** $r_n = 1 - (2/3)^n$ (each stage keeps two thirds); removed opens swell; continuity from below gives $\lim r_n = 1$. **I.6** House the k -th rational in an open interval of length $\varepsilon 2^{-k-1}$; U their union, $m(U) \leq \varepsilon/2 < \varepsilon$ (subadditivity). Complement: closed, inside the irrationals, measure $\geq 1 - \varepsilon$ (frame arithmetic). **I.7** Covers and test cakes translate and reflect with lengths intact; each invariance of m^* transfers citizenship by the shift-the-audit argument of Proposition 2.12. **I.8** $\mathbb{Q} \cap [0, 1]$: meagre and null. Irrationals of $[0, 1]$: residual and full. C : meagre (nowhere dense) and null. S : meagre and *not* null ($m = \frac{1}{2}$). M : meagre and full. R : residual and null. Warrants: the propositions of this chapter and Chapter 1's null computations.

II.2 (solution in full). *Frame-independence.* Let $[a, b] \subseteq [a', b']$ both frame E . Then $[a', b'] \setminus E = ([a, b] \setminus E) \sqcup ([a', b'] \setminus [a, b])$, the second piece a union of at most two intervals of total length $(b' - a') - (b - a)$, a citizen. Both pieces are separated by the citizen $[a, b]$, and outer measure adds across a citizen cut (the criterion applied with test set $[a', b'] \setminus E$): the two frames' inner measures agree after cancellation.

Criterion $\Rightarrow$ inner-meets-outer. The frame is one test cake: $b - a = m^*([a, b]) = m^*(E) + m^*([a, b] \setminus E)$, and rearranging gives $m_*(E) = m^*(E)$ (all quantities finite: E bounded).

Inner-meets-outer $\Rightarrow$ criterion. Let T be any test cake of finite measure and $\varepsilon > 0$. Choose open $U \supseteq T$ with $m^*(U) \leq m^*(T) + \varepsilon$ (Chapter 1's fattening; U need not sit in the frame — intersect covers as needed). From $m_*(E) = m^*(E)$ within a frame containing E : there is open $V \supseteq [a, b] \setminus E$ with $m^*(V) \leq m^*([a, b] \setminus E) + \varepsilon = (b - a) - m^*(E) + \varepsilon$. The open citizen $W = V^c \cap (a, b)$ sits inside E up to the frame's endpoints (null), with $m(W) \geq (b - a) - m^*(V) - 0 \geq m^*(E) - \varepsilon$ (beads arithmetic within the frame; endpoints null). Open sets are citizens, and citizens cut every cake exactly, so

$$m^*(T) \geq m^*(T \cap W) + m^*(T \setminus W) - 0 \quad \text{with} \quad m^*(T \cap E) \leq m^*(T \cap W) + m^*(E \setminus W),$$

and $m^*(E \setminus W) \leq \varepsilon$ by the construction of W (both E and W sit in the frame; frame arithmetic again), while $m^*(T \setminus E) \leq m^*(T \setminus W)$ by monotonicity ($W \subseteq E$ up to a null set, absorbed). Assembling: $m^*(T \cap E) + m^*(T \setminus E) \leq m^*(T) + 2\varepsilon$; ε arbitrary; the criterion holds. (The manoeuvre — replace E by an inner open citizen and let epsilons absorb the veil — is the approximation manners in miniature, and the desk flags it as such.)

II.3 (solution in full). E bounded: fix a frame $[-N, N] \supseteq E$. Apply Theorem 2.16(i) to the citizen $[-N, N] \setminus E$: open $V \supseteq [-N, N] \setminus E$ with surplus $\varepsilon/2$. Set $F = [-N, N] \setminus V$: closed, $F \subseteq E$, and

$$E \setminus F = E \cap V \subseteq V \setminus ([-N, N] \setminus E), \quad m(E \setminus F) \leq \varepsilon/2 < \varepsilon.$$

F is closed and bounded, hence compact (Volume One, Theorem 5.18). The frame bookkeeping used exactly: additivity within $[-N, N]$ (finite), and the criterion's exact cut by the citizen E . For the record, the surplus landed on the certificate's side intact — nothing was lost to the frame's endpoints, which are null and absorbed.

2.14 Notes

History. Carathéodory's criterion and extension theory appear in *Über das lineare Mass von Punktmengen* (1914) and his 1918 treatise; the four-step staircase below is his, polished by a century of textbooks. Lebesgue's inner-outer definition is the thesis (1902); de la Vallée Poussin and W. H. Young developed parallel accounts — priority in this subject is a braid, not a line. Henry Smith's fat dusts are in *On the integration of discontinuous functions* (1875), which also anticipates much of Chapter 1; Volterra's use of them (1881) closed Volume One. The category/measure duality is Oxtoby's book-length subject (*Measure and Category*); the residual-versus-null behaviour of nowhere-differentiable functions — the question Remark 2.21 files — is surveyed there and in the literature on prevalence (Hunt–Sauer–Yorke) for the measure-theoretic sense of “typical.”

The count, on credit. That $|\mathcal{B}|$ is exactly the continuum: Folland §1.6 or Kechris §22 (transfinite generation, cardinality bounds at each stage). This chapter uses the fact only in Remark 2.25's census, and nothing downstream depends on it.

Dimension. The verbatim transposition to $\mathbb{R}^d$ — box covers, the same criterion, the same staircase — is Axler chapter 5's opening or Folland §1.4–1.5 read with d general; Chapter 7 constructs the plane's measure a second way (products) and proves agreement, per the season's plan.

Reading. Axler 2B–2D (the season's companion, covering tonight's ground in the same order); Royden–Fitzpatrick chapter 2; Stein–Shakarchi chapter 1 (whose treatment of the fat dust and regularity the desk's staging follows in spirit); Tao, *Measure Theory*, §1.2 for the criterion's motivation given at even greater patience than the episode's.

Forward pointers. The catch's trial and the ghost estates: Chapter 3 (Vitali 1905; Banach–Tarski 1924 defused). The stranger return of the dust — the Cantor staircase — and the taming of the static: Chapter 4. The crown jewels: Chapter 5.

Chapter 3

Vitali and Banach–Tarski, or the Price of Choice

*Companion to Episode Three. The episode called the catch's trial: the measure problem framed and closed; Vitali's construction narrated in full (the sorting, the delegation — Choice's first flagged expenditure — the tiling, the two-sided absurdity); the ghost's own ledgers read (inner zero, outer positive) and the ghost-factory corollary (every set of positive measure harbours non-measurable subsets); the confession (Zermelo 1904; the Five Letters with Hadamard carrying the day against the course's own heroes; Zermelo's 1908 defence; the plea bargain via Gödel and Cohen); the domestic audit (this course's own quiet countable choices); Solovay's verdict, honestly fine-printed; the alphabet trick (the free group doubles); the transport to the sphere and ball; Banach–Tarski stated with full effect and defused (the pieces are ghosts; no volume was ever present to conserve; dimension's role — commutative alphabets cannot spell paradox; Banach's finitely additive extensions in dimensions one and two); the paradox-as-instrument service note; and debt six paid with the family gathered. This chapter is the paperwork. Register notes: **the bell is silent** — count holds at fifteen; the evening spent Choice, not completeness, and the two axioms run different departments. Debt thirteen (the catch) is **proved** at Theorem 3.4; debt six is **paid and closed** at Remark 3.17. Choice's expenditures are flagged at Definition 3.2 and Theorem 3.14 — the register's new column, "conjurings, declared."*

Episode Digest

The argument of the hour. The measure problem — extend Lebesgue's measure to every set — closes in the negative, and the culprit is not measure but Choice. Vitali: sort the reals into families differing by rationals (canonical, constructive); choose one delegate per family (the axiom's flagged expenditure); the delegation's rational shifts tile disjointly, blanket the unit interval, and fit inside a length-three frame — whence no measure value, zero or positive, is consistent: the contract cannot hold on all sets (debt thirteen, proved). The ghost has readable ledgers (inner zero, outer positive: Jordan's gap, incurable) and breeds (every positive-measure set contains a ghost). The confession: the Five Letters — Baire, Borel, Lebesgue objecting, each in his own key; Hadamard's diagnosis (psychology, not mathematics); Zermelo's defence; the plea bargain (consistent by Gödel, independent by Cohen, retained for its indispensability — the course's own census breakfasted on it). Solovay: without full Choice (dependent choice retained, a higher-infinite hypothesis honestly named, Shelah's necessity cited), "every set measurable" is consistent — ghosts are purchasable, never discoverable. Then the theatre: the free group's

four bins double the dictionary (the alphabet trick, pure combinatorics); two free rotations embed the alphabet in the sphere's symmetries (powers-of-three fingerprint); orbits, delegates (Choice again, flagged), the sphere doubles; exceptions absorbed by hotel manoeuvres; the ball follows; five pieces optimal. Banach–Tarski stated with effect, then defused: the pieces are non-measurable mists — the theorem *plus* measure theory proves their ghostliness; no conservation law touched; matter measurable, physics safe; dimensions one and two immune (commutative alphabets; Banach's finitely additive extensions); amenability named at the door. The word “measurable” in every hypothesis of your working library is Vitali's monument — and everything constructible is measurable: the monster's practical content is reassurance. Debt six paid: the two-population policy adopted, conjurings declared.

The episode, beat by beat. The Trial Opens / The Measure Problem. The charge on credit two weeks; the problem's name and three open years; Vitali's four pages one year after Zermelo's axiom — a dependency, not a coincidence; foundations wearing a surveyor's coat.

Sorting the Continuum. Relatives by rational difference; an equivalence (Proposition 3.1); families countable, dense, uncountably many (census in reverse); the sorting canonical — the contrast loaded for the next beat.

The Delegation. One delegate per family from the unit interval (Definition 3.2) — *Choice, flagged at the moment of expenditure*; the roll-call that cannot happen; no rule exists (most delegates are unnameable, Volume One, Chapter 3).

The Tiling. Shifts by the rationals of $[-1, 1]$: pairwise disjoint (one delegate per family); the sandwich $[0, 1] \subseteq \bigsqcup_q (V + q) \subseteq [-1, 2]$ (Lemma 3.3).

The Two-Sided Absurdity. If V were measurable at value v : countable additivity and translation invariance price the union at $\sum v$ — zero or infinite; the sandwich demands a value between one and three; both doors close (Theorem 3.4). The audit trail: only the contract's clauses were used — the catch as charged. The ghost's ledgers: $m_*(V) = 0$, $m^*(V) > 0$ (Proposition 3.5); the factory: every set of positive outer measure contains a non-measurable subset (Theorem 3.6).

The Confession. Choice stated; the aliases (well-ordering, Zorn); the Five Letters (Baire radical, Borel countably scrupulous, Lebesgue definitional, Hadamard diagnostic); Zermelo 1908; the domestic audit (the census's quiet enumerations — countable choice; dependent choice underwrites the course's sequences); the plea bargain (Gödel 1938, Cohen 1963).

Solovay's Verdict. 1970: ZF plus dependent choice plus “every set measurable” is consistent (granting an inaccessible; Shelah 1984: genuinely needed). Ghosts purchasable, not discoverable; the impossibility is Choice's bill (Remark 3.8).

The Alphabet Trick. Reduced words on two letters; four bins by leading symbol; $F = W(a) \sqcup aW(a^{-1}) = W(b) \sqcup bW(b^{-1})$: the dictionary doubles (Theorem 3.11, the guided summit); commutative alphabets collapse (the low-dimension preview).

From the Alphabet to the Sphere. Two free rotations (powers-of-three fingerprint, Remark 3.13); orbits and delegates (Choice, flagged); the four-bin carving doubles the sphere off a countable exceptional set; hotel absorptions (circle rehearsal at Problem II.6); the ball from its skin; five pieces optimal (references).

The Theatrical Summit / The Defusal. Banach–Tarski 1924 stated (Theorem 3.15, on references); pea-and-sun strong form; the pieces cannot be measurable (else one equals two); “piece”

interrogated (mists, not chunks; no roster — the demon needs the chart Solovay forbids); physics safe twice; dimensions one–two immune: Banach’s finitely additive invariant extensions (Remark 3.16); amenability at the door (Problem III.3).

The Paradox as Instrument. Why every hypothesis says “measurable”; probability’s sigma-algebras (Episode Eight seed); everything constructible is measurable — the monster’s practical content is reassurance.

The Family Gathered. Debt six paid: the unnameable numbers, the non-Borel veil, the delegation, the five pieces; the two-population policy; conjurings declared henceforth.

The toolkit acquired. The sort-choose-shift template; the two-sided squeeze; inner/outer ledger reading on arbitrary sets; the four-bin combinatorics; orbit-delegate transport; hotel absorption; the measurability-hypothesis discipline.

Dates worth keeping. Zermelo’s axiom, 1904; Vitali’s construction, 1905; the Five Letters, 1905; Zermelo’s defence, 1908; the sphere’s paradox, 1914; Banach–Tarski, 1924; Banach’s low-dimensional extensions, 1923; Gödel, 1938; Cohen, 1963; Solovay, 1970; Shelah, 1984.

3.1 Notation Ledger

Spoken in the episode	Written here	Remarks
“relatives”; “the families”	$x \sim y \iff x - y \in \mathbb{Q}$; classes $[x]$	Prop. 3.1
“the delegation”	$V \subseteq [0, 1]$: one point per class	Def. 3.2
“the shifts”	$V + q, q \in \mathbb{Q} \cap [-1, 1]$	Lem. 3.3
“the free group”; “the dictionary”	$F = F(a, b)$: reduced words	§3.3
“the bins”	$W(x)$: words with leading symbol x	Thm. 3.11
“equidecomposable”	$A \sim_G B$: finite pieces, group motions	Def. 3.9
“paradoxical”	X doubles via finitely many pieces	Def. 3.10

Conjurings, declared (the register’s new column). — Uses of the Axiom of Choice are flagged in this volume at the point of expenditure. This chapter: Definition 3.2 (the delegation) and the orbit delegates inside Theorem 3.14. The course’s standing quiet uses (countable and dependent choice beneath the census and the sequence extractions) are audited at Remark 3.7 and are Solovay-safe.

3.2 The Vitali Construction

Proposition 3.1 (The sorting). *The relation $x \sim y \iff x - y \in \mathbb{Q}$ is an equivalence on $\mathbb{R}$. Each class $[x] = x + \mathbb{Q}$ is countable and dense in $\mathbb{R}$; the classes partition $\mathbb{R}$; and there are uncountably many classes.*

Proof. Reflexive: $x - x = 0 \in \mathbb{Q}$. Symmetric: negatives of rationals are rational. Transitive: $(x - y) + (y - z) = x - z$, and $\mathbb{Q}$ is closed under addition (Volume One, Chapter 2). Each class is the bijective image $q \mapsto x + q$ of $\mathbb{Q}$: countable, and dense since $\mathbb{Q}$ is (translation preserves

density). Partition: equivalence classes always partition. Uncountably many: a countable family of countable classes would render $\mathbb{R}$ countable (the census, Volume One, Chapter 3, read in reverse) — contradicting the diagonal. $\square$

Definition 3.2 (The delegation — a conjuring, declared). Each class meets $[0, 1]$ (density). **By the Axiom of Choice**, select exactly one point of $[0, 1]$ from each class; let V be the set of delegates. No rule performs the selection: the classes are structureless, uncountably many, and almost all their members are unnameable (Volume One, Chapter 3's census); the axiom asserts the selection's existence and provides nothing further.

Lemma 3.3 (The tiling). *Let $q_1, q_2, \dots$ enumerate $\mathbb{Q} \cap [-1, 1]$. Then:*

- (i) *the sets $V + q_k$ are pairwise disjoint;*
- (ii) *$[0, 1] \subseteq \bigcup_k (V + q_k) \subseteq [-1, 2]$.*

Proof. (i) If $v + q_k = w + q_j$ with $v, w \in V$, then $v - w = q_j - q_k \in \mathbb{Q}$: $v \sim w$, so $v = w$ (one delegate per class), whence $q_k = q_j$.

(ii) Right inclusion: $V \subseteq [0, 1]$ and $q_k \in [-1, 1]$. Left: let $x \in [0, 1]$; its class has a delegate $d \in V \cap [0, 1]$; then $q = x - d \in \mathbb{Q}$ and $q \in [-1, 1]$ (both points in $[0, 1]$), so $q = q_k$ for some k and $x \in V + q_k$. $\square$

Theorem 3.4 (Vitali, 1905: the catch, proved). *V is not Lebesgue measurable. Consequently there is no assignment μ on all subsets of $\mathbb{R}$ satisfying the contract of Chapter 1 (nonnegativity, intervals to length, translation invariance, countable additivity).*

Proof. Suppose $V \in \mathcal{M}$ with $m(V) = v$. Citizenship translates (Chapter 2), so each $V + q_k \in \mathcal{M}$ with measure v ; the union $U = \bigsqcup_k (V + q_k)$ is a countable disjoint union of citizens, and countable additivity prices it:

$$m(U) = \sum_{k=1}^{\infty} m(V + q_k) = \sum_{k=1}^{\infty} v = \begin{cases} 0, & v = 0, \\ \infty, & v > 0 \end{cases}$$

(a series of identical nonnegative terms admits no third verdict; Volume One, Chapter 4). But the sandwich (Lemma 3.3) and monotonicity demand

$$1 = m([0, 1]) \leq m(U) \leq m([-1, 2]) = 3.$$

Door one ($v = 0$): $m(U) = 0 < 1$. Door two ($v > 0$): $m(U) = \infty > 3$. Both contradict. So $V \notin \mathcal{M}$. For the second statement: the argument above used only the contract's clauses — additivity, translation invariance, interval pricing, and monotonicity (additivity's child) — so any full-contract μ on all sets meets the same two doors with $\mu(V)$ in place of v . No such μ exists. *Debt thirteen: proved as charged.* $\square$

Proposition 3.5 (The ghost's ledgers). *$m^*(V) > 0$, while every closed (indeed every measurable) $F \subseteq V$ is null — the inner measure of V is zero.*

Proof. If $m^*(V) = 0$, V would be null, hence a measurable citizen (Chapter 2), contradicting Theorem 3.4. For the inner ledger, let $F \subseteq V$ be measurable. The shifts $F + q_k$ are pairwise disjoint (inherited from Lemma 3.3(i)) and lie in $[-1, 2]$; countable additivity and invariance give $\sum_k m(F) \leq 3$, forcing $m(F) = 0$. $\square$

Theorem 3.6 (The ghost factory). *Every $E \subseteq \mathbb{R}$ with $m^*(E) > 0$ contains a non-measurable subset.*

Proof. It suffices to treat $E \subseteq [0, 1]$ (E meets some unit frame in positive outer measure — countable subadditivity across frames — and subsets transfer). Write $E = \bigcup_k (E \cap (V + q_k))$, the tiling restricted to E (Lemma 3.3(ii), with $Q \cap [-1, 1]$ re-enumerated as needed). If every piece $E \cap (V + q_k)$ were measurable, each would be null — it sits inside a translate of V , and Proposition 3.5's argument applies verbatim to measurable subsets of translates — whence E , a countable union of null sets, would be null: contradicting $m^*(E) > 0$. Some piece is non-measurable. $\square$

Remark 3.7 (The confession, the audit, the verdict). **Choice and aliases.** The axiom (every family of nonempty sets admits a choice function) is equivalent to the well-ordering principle and to Zorn's lemma; Jech and the notes chart the folklore. **The Five Letters** (Hadamard–Borel–Baire–Lebesgue, 1905) are reprinted and translated in the references; the positions as narrated. **Domestic audit.** This course's quiet expenditures are countable/dependent choice: the census theorem (choosing enumerations), sequence extractions toward suprema and accumulation points, and kin. Dependent choice suffices for all of them; they are retained in Solovay's model and are hereby declared once for the whole course.

Remark 3.8 (Solovay, with the fine print). Solovay (1970): relative to the consistency of an inaccessible cardinal, $ZF +$ dependent choice + “every set of reals is Lebesgue measurable” is consistent. Shelah (1984): the inaccessible is necessary. Consequently no Choice-free construction of a non-measurable set exists; ghosts are purchased, never found. The plea bargain's other halves: Gödel (1938) — Choice is consistent; Cohen (1963) — so is its negation. References: Solovay's Annals paper; Shelah; Jech; Wagon.

3.3 The Alphabet Trick

Definition 3.9 (Equidecomposability; paradoxical sets). Let a group G act on a set X . Subsets $A, B \subseteq X$ are G -equidecomposable, $A \sim_G B$, when A and B admit partitions $A = \bigsqcup_{i=1}^n A_i$, $B = \bigsqcup_{i=1}^n B_i$ with $B_i = g_i A_i$ for some $g_i \in G$. (Transitivity requires refining pieces against pieces: Problem II.4.)

Definition 3.10 (Paradoxical). X is G -paradoxical when X contains disjoint subsets A, B with $A \sim_G X$ and $B \sim_G X$: two full copies, carved from one, with room to spare.

Theorem 3.11 (The free group doubles). *Let F be the free group on letters a, b , acting on itself by left multiplication, and for each symbol $x \in \{a, a^{-1}, b, b^{-1}\}$ let $W(x)$ be the set of reduced words with leading symbol x . Then*

$$F = \{e\} \sqcup W(a) \sqcup W(a^{-1}) \sqcup W(b) \sqcup W(b^{-1}),$$

$$F = W(a) \sqcup a W(a^{-1}), \quad F = W(b) \sqcup b W(b^{-1}),$$

whence F is F -paradoxical: the four bins assemble two copies of the dictionary, the empty word to spare.

Proof. The five-fold partition is the definition of “leading symbol.” For the first reassembly: $a W(a^{-1})$ consists of the reductions of aw with w leading with a^{-1} — write $w = a^{-1}u$ where u is reduced and does not lead with a (else w was not reduced); then $aw = u$. As w ranges over $W(a^{-1})$, u ranges over exactly the reduced words not leading with a (including e , from $w = a^{-1}$):

$aW(a^{-1}) = F \setminus W(a)$. Disjointness is the same computation read backward: a word cannot both lead with a and be the reduction of $a \cdot (a^{-1}u)$ with u not leading with a . The b -reassembly mirrors. Paradoxicality: take $A = W(a) \cup W(a^{-1})$ and $B = W(b) \cup W(b^{-1})$ — disjoint; each reassembles F with one one-letter motion. (Guided as the Tier II summit, Problem II.5, where every cancellation is tracked.) $\square$

Remark 3.12 (Commutative alphabets collapse). If $ab = ba$, the word $aba^{-1}b^{-1}$ reduces to e : bins leak into one another, the leading-symbol sorting is not even well-defined on normal forms, and no four-bin doubling exists — correctly so, since Banach’s extensions (Remark 3.16) forbid paradox in the commutative motion groups of the line and plane. Freedom of the letters is the paradox’s fuel; the desk’s amenability note (Problem 3.5) is this observation grown up.

3.4 The Sphere and the Ball

Remark 3.13 (Free rotations: the fingerprint). Two rotations by $\arccos \frac{1}{3}$ about perpendicular axes generate a free subgroup of the rotation group: in coordinates, each letter multiplies entries by thirds, and a reduced word of length n carries an uncancellable factor 3^{-n} in designated entries — a nontrivial word cannot act as the identity, whose entries are whole. Classical references (Świerczkowski; Wagon, ch. 2); the fingerprint staged for the ambitious at Problem III.1.

Theorem 3.14 (The sphere and ball double — on stated references). *The sphere’s surface, minus the countable set of poles of the free rotations’ words, is paradoxical under rotations: sort the remaining points into orbits of the embedded free group; **by Choice (conjuring, declared)** select one delegate per orbit; the four bins of Theorem 3.11, applied to the delegates’ words, carve the punctured sphere into four pieces that reassemble two punctured spheres. The countable exceptional set is absorbed (a rotation of infinite order moves it off itself; the hotel manoeuvre — circle rehearsal at Problem II.6); radial cones carry the sphere’s carving to the punctured ball; the centre is absorbed by a rotated interior circle; and with Robinson’s optimization, five pieces suffice (and four do not). Full assembly: Wagon, chs. 2–4; Stromberg’s Monthly exposition; Tomkowicz–Wagon.*

Theorem 3.15 (Banach–Tarski, 1924 — stated). *A closed ball in space is equidecomposable, under rigid motions, with the disjoint union of two copies of itself — in five pieces. Strong form: any two bounded subsets of space with nonempty interior are equidecomposable (the pea and the sun). No proof is attempted here beyond Theorem 3.14’s staging; the references carry the construction entire.*

Remark 3.16 (The defusal; dimensions one and two). If the five pieces were measurable, additivity and motion-invariance of volume (Chapter 2’s dimension license) would yield $1 = 2$; the pieces are therefore non-measurable — the theorem plus measure theory is a *proof of ghostliness*, and no measurable object ever changes size: physics is untouched. In dimensions one and two no doubling exists at all: Banach (1923) extended length and area to finitely additive, motion-invariant assignments on *all* subsets — the ghosts can be finitely priced there — because the line’s and plane’s motion groups are too commutative to spell free words (Remark 3.12); the modern dividing line is *amenability* (von Neumann 1929; Tarski’s alternative: paradoxical $\iff$ non-amenable), left ajar at Problem 3.5.

Remark 3.17 (Debt six: the family, the policy). The unnameable numbers (Volume One, Chapter 3); the non-Borel veil (Chapter 2); the delegation; the five pieces: provably existent, never exhibitable — Solovay certifying the never. Policy adopted for the course: two populations,

constructible and conjured; conjurings declared at the point of expenditure; every computable theorem lives in the first population. *Debt six: paid and closed.*

3.5 Problems

Tier I — Calisthenics

- I.1 Verify Proposition 3.1's three equivalence checks with the rational arithmetic displayed.
- I.2 Count the families: show a countable set of countable classes has countable union (cite the census and its choice footnote, Remark 3.7), and conclude the classes are uncountably many.
- I.3 Disjointness drills: show $(V + \frac{1}{3}) \cap (V + \frac{1}{2}) = \emptyset$ from one-delegate-per-family, and locate which lemma clause you used.
- I.4 The sandwich: verify both inclusions of Lemma 3.3(ii) for $x = \frac{\sqrt{2}}{2}$ — without, of course, being able to name its delegate; note where the argument dodges the roster.
- I.5 Reduce six words by hand (specified in the problem file's list, alternating letters and traps), and sort the results into the five parcels.
- I.6 One-letter prefixings: compute $a \cdot w$ for $w = a^{-1}ba, a^{-1}a^{-1}b, a^{-1}$, and place inputs and outputs in their bins.
- I.7 Prove the two-line collapse of Remark 3.12: in a commutative group, $aba^{-1}b^{-1} = e$, and the four "bins" cannot be disjoint.
- I.8 The two doors, re-audited: list every clause of the contract used in Theorem 3.4, each with its Chapter 1 or 2 citation.

Tier II — The Real Thing

- II.1 (*The absurdity, to the letter.*) Write out Theorem 3.4 with every invocation cited at its point of use, including the two monotonicity steps hidden in the sandwich.
- II.2 (*The inner ledger.*) Prove Proposition 3.5's second half in full: every measurable subset of every translate $V + q$ is null. Solved in full below, as the audio promised.
- II.3 (*The factory.*) Complete Theorem 3.6: the frame reduction, the re-enumeration, and the pigeonhole, each displayed.
- II.4 (*Equidecomposability is an equivalence.*) Prove transitivity via piece-refinement: if $A \sim B$ through pieces $\{A_i\}$ and $B \sim C$ through pieces $\{B'_j\}$, refine against the common middle and track the motions.
- II.5 (*The four-bin summit.*) Prove Theorem 3.11 with every cancellation tracked: the five-fold partition, both reassemblies, both disjointness checks, and the paradoxicality assembly. Solved in full below.

- II.6 (*The hotel on a circle.*) Let D be a countable subset of the unit circle. Produce a rotation ρ with $\rho^m D \cap \rho^n D = \emptyset$ for all $m \neq n$, and conclude the circle is equidecomposable (two pieces!) with the circle minus D . This is the absorption manoeuvre of Theorem 3.14, rehearsed where the hotel is cosiest.

Tier III — For the Ambitious ★

- III.1 ★ (*The fingerprint.*) Following Remark 3.13 and the hints: express the two rotations as matrices with entries in thirds, prove by induction the uncancellable 3^{-n} fingerprint of a reduced length- n word, and conclude freeness. References supplied.
- III.2 ★ (*The sphere, assembled.*) From the staged parts — orbits, delegates, bins, absorption — assemble the punctured-sphere paradox in full, flagging Choice’s line, and read the ball off the cones. References at every joint (Wagon chs. 2–3).
- III.3 ★ (*Amenability, at the door.*) An invariant mean on a group is a finitely additive, translation invariant “probability” on all its subsets. Show a paradoxical group admits no invariant mean; verify the integers admit one on credit (references: Følner, von Neumann); state Tarski’s alternative (paradoxical $\iff$ no invariant mean) with references, and locate Banach’s 1923 extensions (Remark 3.16) on this map.

3.6 Hints

II.1. The hidden monotonicity: $[0, 1] \subseteq U$ needs $m([0, 1]) \leq m(U)$; $U \subseteq [-1, 2]$ needs the mirror. Both are additivity’s children (Chapter 1).

II.3. Frames: $0 < m^*(E) \leq \sum_N m^*(E \cap [N, N + 1])$ forces one frame’s trace positive; translate to $[0, 1]$. Re-enumerate $\mathbb{Q} \cap [-1, 1]$ so the tiling covers the framed trace. Pigeonhole: a countable union of nulls is null (Chapter 1’s ideal).

II.4. Refine: $C_k =$ motions of overlaps $B_i \cap B'_j$; compose the two motions piecewise; count pieces honestly ($n \cdot m$ suffices).

II.6. Choose the rotation angle incommensurable with π and avoiding the countable set of angles between pairs of D (census the constraints; pick outside — dependent choice suffices and is declared). Then $\{\rho^n D\}_{n \geq 0}$ are disjoint; let $H = \bigcup_{n \geq 0} \rho^n D$ and map $H \rightarrow \rho H$ by one application of ρ , fixing the complement: two pieces, and D has been absorbed into the hotel.

III.1. Track the vector $(1, 0, 0)$; show a reduced word sends it to $\frac{1}{3^n}(x, y\sqrt{2}, z)$ with y not divisible by three (induction on the leading letter, four cases).

3.7 Solutions

Tier I (terse). **I.1** As in the proposition; the arithmetic is closure of $\mathbb{Q}$ under negation and addition. **I.2** The census enumerates a countable-by-countable array; its choice footnote is Remark 3.7; reverse: uncountable $\mathbb{R}$ refuses such an array. **I.3** A common point gives $v - w =$

$\frac{1}{2} - \frac{1}{3} = \frac{1}{6} \in \mathbb{Q}$, so $v = w$, so $\frac{1}{3} = \frac{1}{2}$ — Lemma 3.3(i)'s clause. **I.4** The delegate d exists in the class of $\frac{\sqrt{2}}{2}$ meeting $[0, 1]$; $q = \frac{\sqrt{2}}{2} - d$ is rational in $[-1, 1]$ by the two memberships; the argument names q 's *existence* only — the dodge is the wholesale quantifier. **I.5–I.6** Mechanical; the traps cancel to shorter words, and the bin of the reduction is the leading symbol *after* cancellation. **I.7** $abab^{-1}a^{-1}b^{-1}$ -style words all reduce under commutativity; $e \in aW(a^{-1})$ and $e \notin W(a)$, but with $ab = ba$, $W(a) \cap bW(b^{-1}) \ni ab \cdot b^{-1}$ -reductions: the parcels overlap. **I.8** Additivity (Carathéodory's theorem, Chapter 2), invariance (Chapter 1, II.3 + Chapter 2 citizenship), intervals (Chapter 1), monotonicity (Chapter 1) — four clauses, no more.

II.2 (solution in full). Let $F \subseteq V + q$ be measurable, and translate: $F - q \subseteq V$ is measurable (citizenship translates) with $m(F - q) = m(F)$ (invariance). The shifts $(F - q) + q_k$, $q_k \in \mathbb{Q} \cap [-1, 1]$, are pairwise disjoint — contained in the disjoint $V + q_k$ of Lemma 3.3(i) — and lie in $[-1, 2]$. Countable additivity within the citizen frame:

$$\sum_{k=1}^{\infty} m(F) = m\left(\bigsqcup_k ((F - q) + q_k)\right) \leq m([-1, 2]) = 3,$$

and a series of identical nonnegative terms bounded by three has terms zero: $m(F) = 0$. The delegation's inner measure — the supremum of measures of closed subsets (Chapter 2's certificates) — is therefore zero, while $m^*(V) > 0$ as shown; the ledgers stand apart, incurably.

II.5 (solution in full). *Partition.* Every nonempty reduced word has a unique leading symbol: the five parcels are disjoint and exhaustive by definition.

Reassembly. Fix $w \in W(a^{-1})$: $w = a^{-1}u$ with u reduced and not leading with a (an a after a^{-1} would have cancelled in w). Then $aw = aa^{-1}u = u$. The map $w \mapsto aw$ thus sends $W(a^{-1})$ into the reduced words not leading with a ; it is onto them — given u not leading with a , the word $w = a^{-1}u$ is reduced (no cancellation at the join, u not leading with a) and leads with a^{-1} , with $aw = u$; and it is injective (left multiplication in a group). So $aW(a^{-1}) = F \setminus W(a)$ exactly, including $e = a \cdot a^{-1}$. *Disjointness:* $W(a) \cap aW(a^{-1}) = W(a) \cap (F \setminus W(a)) = \emptyset$. *Union:* $W(a) \cup aW(a^{-1}) = W(a) \cup (F \setminus W(a)) = F$. The b -computation is the same with letters renamed.

Paradoxicality. $A = W(a) \sqcup W(a^{-1})$ and $B = W(b) \sqcup W(b^{-1})$ are disjoint subsets of F ; $A \sim_F F$ via pieces $W(a)$ (motion e) and $W(a^{-1})$ (motion a); $B \sim_F F$ likewise via e and b . Two copies, four bins, two one-letter motions, the empty word to spare — as the audio promised, verifiable with one's own pen.

3.8 Notes

History and texts. Vitali's *Sul problema della misura dei gruppi di punti di una retta* (1905); Zermelo's 1904 note and 1908 defence; the Five Letters (translated in Moore's *Zermelo's Axiom of Choice*, the standard history); Hausdorff's sphere paradox in *Grundzüge der Mengenlehre* (1914) — the gentleman's name stays on paper, per the course's custom; Banach–Tarski (1924); Banach's finitely additive extensions (1923); Robinson's five-piece optimum (1947). The modern bible is Wagon (now Tomkowicz–Wagon), *The Banach–Tarski Paradox*; Stromberg's 1979 Monthly article is the cleanest short assembly. Solovay (1970), Shelah (1984), Gödel (1938), Cohen (1963): Jech, *The Axiom of Choice*, collects the lot.

Amenability. Von Neumann (1929) isolated the invariant-mean property in direct response to the paradox; Tarski's alternative (paradoxical iff non-amenable) is in Wagon ch. 9; Følner's criterion supplies the integers' mean on credit as Problem III.3 states.

Forward pointers. The measurable functions and the rebuilt integral: Chapter 4, where the tribe's stability under limits — precisely what tonight proved cannot extend to all sets — becomes the working asset of the whole theory.

Chapter 4

Integration Rebuilt

*Companion to Episode Four. The episode sorted the merchant's till: measurable functions (preimage compatibility with the tribe; stability under every limit — the stability continuity never had); simple atoms and the staircase of the range; the integral in three stages, the bell ringing at the second (ring sixteen, the integral's defining supremum); the almost-everywhere dialect at full working order; the static tamed in one line (debt four, act three); the identity function re-integrated by sorting as a handshake; the old regime absorbed (with the improper-integral boundary post planted honestly); Lebesgue's criterion proved in both directions (debt ten — the vocabulary crisis dissolved); and the devil's staircase delivered with its two gifts (the composition counterexample; a measurable non-Borel specimen). This chapter is the paperwork. Register notes: the bell rings **once** — ring sixteen, ($\otimes$) at Definition 4.10; debts four, eight, ten, and eleven are retired at their boxed sites below; **one credit issued** at Remark 4.15 (general nonnegative additivity, one week's maturity, discharged with the crown jewels); one conjuring declared inside Theorem 4.22 (the ghost factory's output, inherited from Chapter 3).*

Episode Digest

The argument of the hour. Partition the range, not the domain. Measurable functions are those whose piles-above-waterlines are citizens — compatibility with the tribe, rhyming with continuity's compatibility with the topology — and the class is stable under sup, inf, limsup, and pointwise limits: the sigma was purchased for exactly this. Simple functions (discrete random variables, in the trade's dress) price by value-times-ground; every nonnegative measurable function is an increasing limit of dyadic staircases of the range; the integral is the supremum of the staircases' prices (($\otimes$) ring sixteen — one ledger, no second book, nothing refused); general functions split into gains and losses with the finiteness clause. Almost-everywhere goes to work (equal a.e. $\Rightarrow$ equal integrals; a.e.-equal to measurable $\Rightarrow$ measurable, completeness earning its keep). The static: a two-storey simple function, one storey weightless — integral zero, one line, act three complete. The bounded estate's two ledgers meet (uniform staircases), licensing bounded additivity — the general credit issued at one week. Riemann is absorbed by squeezing step staircases with null exceptions (proper integrals only; conditional impropriety refused on rearrangement grounds). Lebesgue's criterion is proved both ways — oscillation confinement forward, compact-cover assembly backward — dissolving Season One's vocabulary crisis. The devil's staircase climbs a unit while flat almost everywhere, integrates to one half by symmetry, stretches the dust to measure one, and thereby delivers a measurable function of a continuous function that is not measurable, plus a measurable non-Borel specimen.

The episode, beat by beat. The Merchant's Till. Lebesgue's own image: count coins as they come (Riemann, domain order) or sort by denomination (Lebesgue, range piles); sorting requires weighing — the season to date was the weighing-house.

Functions That Can Be Sorted. The criterion (Definition 4.1) and its equivalents (Proposition 4.2); the rhyme with continuity; extended values seated; the parade (Proposition 4.3); the algebra (Proposition 4.4); the composition caution planted (safe direction: Proposition 4.5; detonation: Theorem 4.22); **limit stability** (Theorem 4.6) — the crown of the class.

The Atoms / The Staircase of the Range. Simple functions and well-priced storeys (Definition 4.7, Proposition 4.8); the approximation theorem (Theorem 4.9): every nonnegative measurable function is an increasing limit of simple staircases, uniformly where bounded.

The Integral in Three Stages. Stage one: value times ground. Stage two: the supremum over under-staircases ((*) Definition 4.10, ring sixteen) — one ledger by design, measurability having done the vetting. The handshake: the identity re-integrated by sorting to one half (Remark 4.11). Stage three: gains minus losses, integrability as finite absolute mass (Definition 4.12); manners within tonight's honest reach (Proposition 4.13); the additivity credit (Remark 4.15).

Almost Everywhere, At Work. The two working lemmas (Proposition 4.16); modifications on null sets are clerical noise; the uniform premium was continuity's bill, never the tribe's.

The Taming of the Static. Boxed at Theorem 4.17: two storeys, one weightless; integral zero; debt four's three-act contract complete.

The Old Regime, Absorbed. The boundary post (proper integrals; conditional impropriety refused — Remark 4.18); the absorption theorem (Theorem 4.19) by squeezing step staircases, null exceptions filed.

The Prophecy, Proved. Lebesgue's criterion (Theorem 4.20), both directions, the oscillation machinery of Chapter 1 conducting; the exhibits table re-read with every verdict a theorem; debt ten retired.

The Devil's Staircase. Construction and portfolio (Theorem 4.21): continuous, nondecreasing, flat almost everywhere, climbing one unit on null ground, integral one half; the stretch $h(x) = x + c(x)$ and the two gifts (Theorem 4.22); debt eight's account closed entire.

The Season's Ledger. Ring sixteen; debts four, eight, ten, eleven retired; the additivity credit at one week; Fourier in active service.

The toolkit acquired. Threshold bookkeeping; the staircase construction; the one-ledger integral; a.e.-amnesty at the function level; squeezing-staircase absorption; oscillation confinement and compact-cover assembly; the stretch-and-pull-back manoeuvre.

Dates worth keeping. Lebesgue's integral, thesis, 1902; the criterion likewise; Vitali's ghost (fuelling tonight's gifts), 1905; the Cantor function's folklore christening as the devil's staircase, twentieth-century vintage, the desk declining to fix a date the literature has not.

4.1 Notation Ledger

Spoken in the episode	Written here	Remarks
“measurable function”	$f^{-1}((a, \infty]) \in \mathcal{M}$ for all a	Def. 4.1
“the pile above a ”	$\{f > a\}$ (and $\{f \geq a\}$, kin)	Prop. 4.2
“gains and losses”	$f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$	Def. 4.12
“a simple function”; “storeys”	$\varphi = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$	Def. 4.7
“the merchant’s multiplication”	$\int \varphi \, dm = \sum_i a_i m(A_i)$	Def. 4.7
“the integral” (nonnegative)	$\int f \, dm = \sup \{ \int \varphi : 0 \leq \varphi \leq f, \varphi \text{ simple} \}$	(*) Def. 4.10
“integrable”	$\int f \, dm < \infty$; $\int f = \int f^+ - \int f^-$	Def. 4.12
“on the ground E ”	$\int_E f \, dm = \int f \mathbf{1}_E \, dm$	Prop. 4.13
“almost everywhere”	a.e.: off a null set	Prop. 4.16
“the devil’s staircase”	$c : [0, 1] \rightarrow [0, 1]$	Thm. 4.21

Extended values. Functions take values in $[-\infty, \infty]$; $\{f = \infty\} = \bigcap_n \{f > n\}$ is a citizen; the arithmetic conventions of Chapter 1 (one prohibition) govern, with $0 \cdot \infty = 0$ in the merchant’s multiplication — the convention that makes weightless ground carry any value for free, and the taming one line.

4.2 Measurable Functions

Definition 4.1 (Measurable function). Let $E \in \mathcal{M}$. A function $f : E \rightarrow [-\infty, \infty]$ is *measurable* when $\{f > a\} = \{x \in E : f(x) > a\}$ is a citizen for every $a \in \mathbb{R}$.

Proposition 4.2 (Equivalent thresholds; Borel preimages). For $f : E \rightarrow [-\infty, \infty]$ the following are equivalent: $\{f > a\} \in \mathcal{M}$ for all a ; $\{f \geq a\} \in \mathcal{M}$ for all a ; $\{f < a\} \in \mathcal{M}$ for all a ; $\{f \leq a\} \in \mathcal{M}$ for all a . Any of them implies $f^{-1}(B) \in \mathcal{M}$ for every Borel $B \subseteq \mathbb{R}$.

Proof. $\{f \geq a\} = \bigcap_n \{f > a - \frac{1}{n}\}$ and $\{f > a\} = \bigcup_n \{f \geq a + \frac{1}{n}\}$: countable tribe operations convert the two strict/weak forms; complements within E convert to the two below-forms. For Borel preimages: the collection $\{B : f^{-1}(B) \in \mathcal{M}\}$ is a sigma-algebra (preimages respect complements and unions), and it contains the half-lines, hence the intervals, hence — the estate’s robustness, Chapter 1, Theorem 1.20 — all of $\mathcal{B}$. $\square$

Proposition 4.3 (The parade). Continuous functions on citizens are measurable ($\{f > a\}$ is the trace of an open set); monotone functions (the pile above a waterline is an interval or half-line trace); step and simple functions; indicators of citizens; the static and Thomae’s function; and any function on a null ground. Ghost indicators ($\mathbf{1}_V$, V Vitali’s delegation) are the canonical non-measurable specimens.

Proof. Each is a one-line pile audit; the desk records the static’s: $\{D > a\}$ is $\emptyset$ ($a \geq 1$), $\mathbf{Q} \cap E$ ($0 \leq a < 1$), or E ($a < 0$) — citizens all. Problem I.1 drills the rest. $\square$

Proposition 4.4 (The algebra). If f, g are measurable (finite-valued where combined) and $c \in \mathbb{R}$: cf , $f + g$, fg , $|f|$, $\max(f, g)$, $\min(f, g)$ are measurable.

Proof. The sum, the one with an idea: $f + g > a$ iff some rational q has $f > q$ and $g > a - q$ (density splits the surplus), so $\{f + g > a\} = \bigcup_{q \in \mathbb{Q}} (\{f > q\} \cap \{g > a - q\})$ — a countable union of citizens; the picket fence's contract, executed at the function level. Products via $fg = \frac{1}{4}((f + g)^2 - (f - g)^2)$ once squares are settled ($\{f^2 > a\}$ unwinds to two piles); moduli and max/min are unions and intersections of piles. Problem I.1's second half. $\square$

Proposition 4.5 (Composition: the safe direction). *If f is measurable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable (in particular continuous), then $g \circ f$ is measurable. The reverse composition — a measurable function of a continuous function — can fail: Theorem 4.22.*

Proof. $(g \circ f)^{-1}(a, \infty) = f^{-1}(g^{-1}(a, \infty))$: $g^{-1}(a, \infty)$ is Borel, and Proposition 4.2 finishes. $\square$

Theorem 4.6 (Limit stability — the crown of the class). *Let $f_1, f_2, \dots$ be measurable on E . Then $\sup_n f_n, \inf_n f_n, \limsup_n f_n, \liminf_n f_n$ are measurable; and if $f_n \rightarrow f$ pointwise on E , then f is measurable.*

Proof. $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$: some function clears the waterline — a countable union of citizens. Infima by $\inf f_n = -\sup(-f_n)$ (or the mirrored intersection). $\limsup_n f_n = \inf_n \sup_{k \geq n} f_k$: countable operations stacked twice, each stage measurable by the preceding clauses. Pointwise limits equal their limsup. No hypothesis beyond measurability was used; the sigma-algebra's countable clause is what every line spent. $\square$

4.3 Atoms, Staircases, and the Integral

Definition 4.7 (Simple functions; the merchant's multiplication). *φ is simple when it takes finitely many values, each on a citizen: $\varphi = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ with $A_i \in \mathcal{M}$ disjoint. For simple $\varphi \geq 0$,*

$$\int \varphi \, dm = \sum_{i=1}^n a_i m(A_i) \quad (0 \cdot \infty = 0).$$

Proposition 4.8 (Well-priced; simple linearity and monotonicity). *The price is independent of the representation; for simple $\varphi, \psi \geq 0$ and $c \geq 0$: $\int(\varphi + \psi) = \int \varphi + \int \psi$, $\int c\varphi = c \int \varphi$, and $\varphi \leq \psi \Rightarrow \int \varphi \leq \int \psi$.*

Proof. Two representations refine to a common one (intersect the storeys — Chapter 1's finite tribe operations), on which both prices telescope to the same sum; additivity of m on the disjoint refinement is the only fact used. Linearity and monotonicity read off the common refinement likewise. (Problem I.2 drills a two-representation specimen.) $\square$

Theorem 4.9 (The staircase of the range). *Let $f : E \rightarrow [0, \infty]$ be measurable. Define*

$$\varphi_n = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} \mathbf{1}_{A_{n,k}} + n \mathbf{1}_{\{f \geq n\}}, \quad A_{n,k} = \left\{ \frac{k-1}{2^n} \leq f < \frac{k}{2^n} \right\}.$$

Each φ_n is simple; $\varphi_n \leq \varphi_{n+1} \leq f$; and $\varphi_n \rightarrow f$ pointwise on E — uniformly on any set where f is bounded.

Proof. The bins $A_{n,k}$ are differences of piles: citizens (Proposition 4.2); finitely many storeys: simple. Monotonicity: passing $n \rightarrow n+1$ halves each denomination — a point binned at $\frac{k-1}{2^n}$ is re-binned at $\frac{2k-2}{2^{n+1}}$ or $\frac{2k-1}{2^{n+1}}$, never lower — and raises the ceiling. Convergence: if $f(x) < \infty$, then for $n > f(x)$ the assignment sits within 2^{-n} of $f(x)$; if $f(x) = \infty$, the assignments are $n \uparrow \infty$. Where $f \leq K$: for $n > K$ the error is uniformly under 2^{-n} . $\square$

Definition 4.10 ((*) The integral — ring sixteen). For measurable $f : E \rightarrow [0, \infty]$,

$$\int_E f \, dm = \sup \left\{ \int \varphi \, dm : 0 \leq \varphi \leq f, \varphi \text{ simple} \right\}.$$

The bell rings — the sixteenth: the supremum over the inventory of under-staircases exists in $[0, \infty]$ by the completeness axiom (finite case) and honest convention (infinite), the same authority that manufactured the Darboux ledgers at rings thirteen and fourteen. One ledger only, by design: measurability has done the vetting Riemann's upper book performed by hand (the bounded estate's Proposition 4.14 exhibits the second book collapsing onto the first where both exist), and Theorem 4.9 guarantees the supremum is *climbed*, not merely bounded. Nothing is refused; the value ∞ is an honest verdict.

Remark 4.11 (The handshake: the identity, re-integrated). For $f(x) = x$ on $[0, 1]$, the staircase φ_n of Theorem 4.9 has bins $A_{n,k} = [\frac{k-1}{2^n}, \frac{k}{2^n})$ of measure 2^{-n} , so

$$\int \varphi_n = \sum_{k=1}^{2^n} \frac{k-1}{2^n} \cdot \frac{1}{2^n} = \frac{1}{4^n} \cdot \frac{2^n(2^n-1)}{2} = \frac{1}{2} - \frac{1}{2^{n+1}} \uparrow \frac{1}{2},$$

and no under-staircase exceeds $\frac{1}{2}$ (each is dominated by the Riemann upper ledger, or directly: Problem I.3). The new integral returns the schoolroom's one half; the general agreement is Theorem 4.19.

Definition 4.12 (Integrable; the general integral). For measurable f : $f^\pm = \max(\pm f, 0)$ are measurable (Proposition 4.4), $f = f^+ - f^-$, $|f| = f^+ + f^-$. f is *integrable* when $\int |f| \, dm < \infty$ (equivalently both $\int f^\pm < \infty$), and then $\int f \, dm = \int f^+ - \int f^-$ — lawful subtraction. If exactly one part is infinite the integral is $\pm\infty$; if both, undefined.

Proposition 4.13 (Manners within tonight's reach). (i) *Monotonicity*: $0 \leq f \leq g$ measurable $\Rightarrow \int f \leq \int g$ (every under-staircase of f under-staircases g). (ii) $\int cf = c \int f$ for $c \geq 0$ (staircases scale). (iii) *Ground conventions*: $\int_E f = \int f \mathbf{1}_E$, the two definitions agreeing storey by storey. (iv) $f = 0$ a.e. $\Rightarrow \int f = 0$ (every under-staircase's storeys of positive value stand on null ground). (v) **Bounded additivity**: for bounded measurable f, g on finite-measure ground, $\int (f+g) = \int f + \int g$ — via Proposition 4.14's sandwiches (Problem II.2's ambit).

Proposition 4.14 (The bounded estate: two ledgers meet). Let f be measurable and bounded on E with $m(E) < \infty$. Then

$$\sup_{\substack{\varphi \leq f \\ \varphi \text{ simple}}} \int \varphi = \inf_{\substack{\psi \geq f \\ \psi \text{ simple}}} \int \psi.$$

Proof. The staircase construction at resolution n (ceiling above the bound) yields $\varphi_n \leq f \leq \psi_n + 2^{-n}$ pointwise, so $\psi_n = \varphi_n + 2^{-n}$ over-staircases f with $\int \psi_n - \int \varphi_n = 2^{-n}m(E) \rightarrow 0$; the sup and inf are pinched together. (Solved in full as Problem II.2, with the bounded-additivity dividend; the converse — ledger agreement forces measurability — is the desk's note for the ambitious, references supplied.) $\square$

Remark 4.15 (The additivity credit — one week's maturity). For arbitrary measurable $f, g \geq 0$, $\int(f+g) = \int f + \int g$: the $\geq$ direction is free (under-staircases add), and the $\leq$ direction is **stated on credit tonight**, discharged in Chapter 5 as the monotone convergence theorem's first corollary (staircases $\varphi_n \uparrow f$, $\psi_n \uparrow g$ give $\varphi_n + \psi_n \uparrow f + g$; MCT passes the limit). No argument of this chapter uses the credit: the taming, the absorption, and the criterion live in the bounded estate, where Proposition 4.14 pays cash.

Proposition 4.16 (Almost everywhere, at work). (i) If $f = g$ a.e. on E and both are measurable, then $\int_E f = \int_E g$ (whenever either side is defined). (ii) If f is measurable and $g = f$ a.e., then g is measurable — completeness of $\mathcal{M}$ earning its keep.

Proof. (ii) $\{g > a\}$ differs from the citizen $\{f > a\}$ within the null disagreement set; the amnesty (Chapter 2, completeness) makes the difference a citizen, and the tribe's algebra reassembles. (i) For $f, g \geq 0$: any under-staircase of f converts to one of g by zeroing its storeys on the null disagreement — prices unchanged, $0 \cdot \infty = 0$ and Proposition 4.13(iv) covering the leftovers; the suprema agree. General case: positive and negative parts disagree only on the null set. $\square$

Theorem 4.17 (The taming of the static — debt four, act three). $\int_{[0,1]} D \, dm = 0$.

Proof. $D = 1 \cdot \mathbf{1}_{\mathbb{Q} \cap [0,1]} + 0 \cdot \mathbf{1}_{[0,1] \setminus \mathbb{Q}}$ is simple; $\int D = 1 \cdot m(\mathbb{Q} \cap [0,1]) = 1 \cdot 0 = 0$. One line, as promised in Volume One, Chapter 8. The three-act contract — born at Volume One's Definition 6.24, victorious over Riemann in Volume One, Chapter 8, tamed here — is complete. $\square$

4.4 The Old Regime, Absorbed

Remark 4.18 (The boundary post: improper integrals). The absorption below covers the *proper* Riemann integral. A conditionally convergent improper integral — finite by cancellation between infinite gains and infinite losses — has $\int |f| = \infty$ and is refused by Definition 4.12, on the same grounds Volume One's rearrangement theorem convicted conditional series: totals that depend on the order of summation have no order-free value, and the sorted count is order-free by construction. Standard specimen: the sign-alternating bump train with masses $1, \frac{1}{2}, \frac{1}{3}, \dots$ — improperly convergent to the alternating harmonic total, Lebesgue-undefined. Absolutely convergent improper integrals are absorbed (Problem III.2's ambit, via next week's tools). Lebesgue honours everything Riemann *proved*; some things Riemann merely got away with.

Theorem 4.19 (Riemann absorbed). Let f be (properly) Riemann integrable on $[a, b]$. Then f is measurable, integrable, and $\int_{[a,b]} f \, dm = (R) \int_a^b f$.

Proof. f is bounded (Riemann's ticket, Volume One). Take partitions P_n , each refining the last, mesh $\rightarrow 0$, with Darboux ledgers $L(f, P_n) \uparrow$ and $U(f, P_n) \downarrow$ toward the common Riemann value R (integrability). Let $\varphi_n \leq f \leq \psi_n$ be the step functions taking the slice-infima and slice-suprema on P_n 's slices: simple functions with $\int \varphi_n = L(f, P_n)$ and $\int \psi_n = U(f, P_n)$ — the merchant's multiplication reading Darboux's books aloud. Refinement makes $\varphi_n \uparrow \varphi$ and $\psi_n \downarrow \psi$ pointwise (monotone bounded sequences; Volume One's license), with $\varphi \leq f \leq \psi$; both limits are measurable (Theorem 4.6). On the finite-measure frame, the bounded estate prices

$$\int(\psi - \varphi) \leq \int(\psi_n - \varphi_n) = U(f, P_n) - L(f, P_n) \rightarrow 0$$

(monotonicity; bounded additivity for the differences, Proposition 4.13(v)); so $\int(\psi - \varphi) = 0$ with $\psi - \varphi \geq 0$, forcing $\psi = \varphi$ a.e. (Problem I.6's little lemma: a nonnegative function of zero integral vanishes a.e.). Then f , trapped between a.e.-equal measurable functions, equals φ a.e.: measurable by amnesty (Proposition 4.16(ii)), and

$$\int f = \int \varphi = \lim_n \int \varphi_n = \lim_n L(f, P_n) = R$$

(the middle equality by the bounded two-ledger pinch against ψ_n ; partition endpoints, countably many, are null and filed). $\square$

4.5 The Prophecy, Proved

Theorem 4.20 (Lebesgue's criterion — debt ten retired). *Let f be bounded on $[a, b]$, Δ_f its set of discontinuity points. Then f is Riemann integrable if and only if $m(\Delta_f) = 0$.*

Proof. Write $D_\sigma = \{x : \omega_f(x) \geq \sigma\}$ (the loud sets; oscillation as in Chapter 1), so $\Delta_f = \bigcup_n D_{1/n}$ (Chapter 1's Lemma 1.24). Let K bound $|f|$.

($\Rightarrow$) Fix $\sigma > 0$ and $\varepsilon > 0$. Choose a partition with $U - L < \sigma\varepsilon$. On any slice whose *interior* meets D_σ , the slice's swing is at least σ (the gauge reads a swing of at least σ in every window of an interior loud point); letting w be those slices' total width,

$$\sigma\varepsilon > U - L = \sum_i (\text{swing}_i) \Delta x_i \geq \sigma w,$$

so $w < \varepsilon$: the interior-loud slices cover D_σ save the finitely many partition points, null. Hence $m^*(D_\sigma) < \varepsilon$ for every ε : D_σ is null, and Δ_f is a countable union of nulls — null (Chapter 1's ideal).

($\Leftarrow$) Suppose $m(\Delta_f) = 0$; fix $\varepsilon > 0$ and set $\sigma = \varepsilon / (2(b - a))$. The loud set D_σ is closed (Chapter 1, Problem II.5's grain — proved there), bounded: compact; null (inside Δ_f); so finitely many open intervals cover it with total length below $\varepsilon / (8K)$ (null + compact: a countable ε -cover admits a finite subcover). Off those intervals, the leftover F is compact with every point quiet below σ : each $x \in F$ owns $\delta_x > 0$ with the window $(x - \delta_x, x + \delta_x)$ swinging under σ (the gauge's approximation property); the half-radius windows cover F ; extract a finite subcover (Heine–Borel, Volume One, Theorem 5.18 — the same manoeuvre as Heine–Cantor's proof, Volume One 6.22, and the kinship is the point); let δ be the least half-radius. Now partition $[a, b]$ by: the cover intervals' endpoints, and a mesh finer than δ elsewhere. Slices inside the loud cover: total width below $\varepsilon / (8K)$, swing at most $2K$ — contribution below $\varepsilon / 4$. Slices meeting F : each lies inside some full window (mesh under the half-radius), swing at most σ — contribution at most $\sigma(b - a) = \varepsilon / 2$. Hence $U - L < \varepsilon$; the auditor's criterion (Volume One, Chapter 8) stamps. *The Volume One finale's prophecy stands proved; the vocabulary crisis is dissolved; the exhibits table reads as theorems, line by line.* $\square$

4.6 The Devil's Staircase

Theorem 4.21 (The Cantor function — debt eight's account closes). *There is a function $c : [0, 1] \rightarrow [0, 1]$, continuous and nondecreasing, with $c(0) = 0$, $c(1) = 1$, constant on each interval removed in the dust's construction — hence differentiable with $c' = 0$ at almost every point — and $\int_{[0,1]} c \, dm = \frac{1}{2}$. Its entire unit of ascent is accomplished on the null dust.*

Proof. Construction. Assign to the stage- n removed middles, read left to right, the values $\frac{1}{2^n}, \frac{3}{2^n}, \dots, \frac{2^n-1}{2^n}$ (each middle the dyadic mean of its neighbours' values); the assignments are consistent under staging and monotone across the complement. For $x \in C$ set $c(x) = \sup\{c(t) : t \text{ in removed middles, } t \leq x\}$ (sup over a nonempty bounded set — small change), with $c(0) = 0$; monotonicity extends.

Continuity. c is monotone with range dense in $[0, 1]$ — every dyadic value is attained on a middle. A monotone function's only discontinuities are jumps, and a jump omits an interval of values, contradicting dense range. So c is continuous (and $c(1) = 1$ by the sup over values climbing to one).

Flatness a.e. On each removed middle c is constant, so $c' = 0$ there; the middles carry measure one (Chapter 2); the dust where ascent happens is null.

The integral. The construction is symmetric under $x \mapsto 1 - x$ with values reflecting through $\frac{1}{2}$: $c(x) + c(1 - x) = 1$ for all x (verified on middles, extended by continuity). Reflection invariance of m (Chapter 3, Problem I.7) gives $\int c(1 - x) dm = \int c dm$, so $2 \int c = \int (c(x) + c(1 - x)) dm = m([0, 1]) = 1$ (bounded additivity), and $\int c = \frac{1}{2}$. $\square$

Theorem 4.22 (The two gifts). *Let $h(x) = x + c(x)$ on $[0, 1]$. Then h is a homeomorphism onto $[0, 2]$ with $m(h(C)) = 1$; and there exist*

- (i) *a measurable set $A \subseteq C$ (null, hence a citizen) that is not Borel;*
- (ii) *a measurable function g and a continuous function F with $g \circ F$ non-measurable — the composition caution, detonated.*

(One conjuring, declared: the ghost inside $h(C)$ is Chapter 3's.)

Proof. h is continuous and strictly increasing (identity strictly, c nondecreasing): a continuous bijection $[0, 1] \rightarrow [0, 2]$, hence a homeomorphism (compact domain; Volume One's machinery). On each removed middle, c is constant, so h is a translation there: lengths preserved; the complement's beads carry total measure one into $[0, 2]$, and additivity leaves $m(h(C)) = 2 - 1 = 1$. The ghost factory (Chapter 3, Theorem 3.6) supplies non-measurable $N \subseteq h(C)$. Set $A = h^{-1}(N) \subseteq C$: null by heredity, hence a citizen by amnesty. *Not Borel:* were A Borel, then $N = (h^{-1})^{-1}(A)$ would be a Borel set's preimage under the continuous h^{-1} — Borel (Proposition 4.2) — but ghosts are not even measurable. *The detonation:* $g = \mathbf{1}_A$ is measurable (citizen indicator); $F = h^{-1} : [0, 2] \rightarrow [0, 1]$ is continuous; and $g \circ F = \mathbf{1}_N$, whose unit pile is the ghost: non-measurable. *The veil made specimen:* Chapter 2's count argued that non-Borel citizens exist; here is one, a single conjuring back. $\square$

4.7 Problems

Tier I — Calisthenics

- I.1 Threshold drills: verify measurability from Definition 4.1 for the static, Thomae's function, the step function, and a monotone staircase; then complete Proposition 4.4's squares-and-products unwinding.
- I.2 Price $\varphi = 2 \cdot \mathbf{1}_{[0, 1/2]} + 3 \cdot \mathbf{1}_{(1/2, 1]}$ under two representations (split $[0, \frac{1}{2}]$ at $\frac{1}{3}$) and confirm Proposition 4.8.

- I.3 Compute the staircase φ_2 of Theorem 4.9 for $f(x) = x^2$ on $[0, 1]$: bins, measures, price; compare with the true integral one third.
- I.4 Decompose into gains and losses, and audit integrability on the stated grounds: $x \mapsto x$ on $[-1, 1]$; $1/x$ on $(0, 1]$; $1/x^2$ on $[1, \infty)$; the static minus one half on $[0, 1]$; $(-1)^{\lfloor x \rfloor} / \lceil x \rceil$ -style bump trains on $[0, \infty)$ (state the verdict Remark 4.18 predicts).
- I.5 Almost-everywhere drills: modify x^2 on $\mathbb{Q}$; redefine the cornered ramp at its corner; verify integrals unmoved, citing Proposition 4.16.
- I.6 The little lemma: if $f \geq 0$ is measurable with $\int f = 0$, then $f = 0$ a.e. (Consider $\{f \geq 1/n\}$ and the under-staircases $\frac{1}{n} \mathbf{1}_{\{f \geq 1/n\}}$.)
- I.7 Reproduce the taming with every citation (Theorem 4.17), then tame Thomae the same way.
- I.8 Spot-check the absorption on the step function of Volume One's cornered ramp: Darboux value and merchant's value, shown equal.

Tier II — The Real Thing

- II.1 (*Limsup, to the letter.*) Write out Theorem 4.6's limsup clause with every countable operation displayed, and exhibit the theorem failing for *continuity* in its place (the moving spike's limit).
- II.2 (*The two ledgers meet.*) Prove Proposition 4.14 in full, and derive bounded additivity (Proposition 4.13(v)) from the sandwiches. Solved in full below, as the audio promised.
- II.3 (*Absorption's bookkeeping.*) Complete Theorem 4.19: the refinement monotonicity of φ_n, ψ_n , the null filing of partition endpoints, and the final pinch. Solved in full below.
- II.4 (*The forward confinement.*) Re-derive the ($\Rightarrow$) direction of Theorem 4.20 as Volume One, Chapter 8, Problem III.2 intended: state Riemann's oscillation criterion and show tonight's two-line confinement is its cash value.
- II.5 (*The staircase's continuity.*) Certify the monotone-plus-dense-range argument of Theorem 4.21 with the jump-omits-values lemma proved, not waved.
- II.6 (*The staircase's integral, twice.*) Evaluate $\int c$ by the reflection symmetry (as in the text), then a second way: sum the merchant's prices over the removed middles (value times bead length, a series the desk's hint arranges) and squeeze the dust's null contribution.

Tier III — For the Ambitious ★

- III.1 ★ (*The gifts, end to end.*) Assemble Theorem 4.22 with every step displayed and the one conjuring flagged at its line; then show no such example can be *Borel*-flavoured: if g is Borel measurable and F continuous, $g \circ F$ is measurable — the caution is precisely about the completed tribe.

III.2 ★ (*The converse assembly, audited.*) Rebuild the ($\Leftarrow$) direction of Theorem 4.20 with every compactness step displayed and each Volume One citation verified at its address; then, using next week's dominated tools *on credit*, absorb absolutely convergent improper integrals (Remark 4.18's open half).

4.8 Hints

II.1. $\{\limsup f_n > a\} = \bigcup_{q>a, q \in \mathbb{Q}} \bigcap_n \bigcup_{k \geq n} \{f_k > q\}$ is one lawful unwinding; justify each layer. The spike: continuous functions with pointwise limit the zero function — continuity survives, but exhibit a variant (the sliding indicator train) whose limit is the static's cousin.

II.4. Riemann's criterion (Volume One, Chapter 8, Problem III.2): integrable iff for every σ, ε some partition has loud-slice width below ε . Tonight's display $\sigma w \leq U - L$ is its forward half; conclude $m^*(D_\sigma) \leq w + 0$.

II.5. If a monotone c jumps at x_0 , the interval $(\lim_{x \uparrow x_0} c, \lim_{x \downarrow x_0} c)$ (minus possibly $c(x_0)$) is omitted from the range; a dense subset of $[0, 1]$ cannot avoid an interval.

II.6. Stage- n middles carry values $\frac{2j-1}{2^n}$ on lengths 3^{-n} ; the double series $\sum_n 3^{-n} \sum_{j=1}^{2^{n-1}} \frac{2j-1}{2^n}$ evaluates by the arithmetic-series formula to $\frac{1}{2}$ (the desk expects the $\frac{4}{6}$ -ratio geometric assembly); the dust contributes nought by Proposition 4.13(iv).

III.1. For the Borel half: chase Proposition 4.5 and locate exactly which preimage fails when g is merely Lebesgue measurable (the Borel preimage lemma needs $g^{-1}(a, \infty)$ *Borel*, not just a citizen).

III.2. Dominate the truncations $f \mathbf{1}_{[a, b_n]}$ by $|f|$; next week's theorem passes the limit; tonight's credit is repaid at the same counter.

4.9 Solutions

Tier I (terse). I.1–I.5 As drilled; I.4's verdicts: integrable; not integrable ($\int_0^1 1/x = \infty$: staircases $\sum 2^k \mathbf{1}$ -flavoured climb unboundedly); integrable (mass one); integrable (bounded on finite ground); the bump train with masses $\frac{1}{n}$: refused, per the boundary post. I.6 $\frac{1}{n} m(\{f \geq \frac{1}{n}\}) \leq \int f = 0$ forces each pile null; their union is $\{f > 0\}$. I.7 Thomae: simple it is not, but $t = 0$ a.e. (defects on $\mathbb{Q}$), so $\int t = 0$ by Proposition 4.16(i) against the zero function. I.8 Ramp's step: Darboux $\frac{1}{2}$ (Volume One); merchant: $0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$.

II.2 (solution in full). *The pinch.* With K a bound and resolution $n > K$: Theorem 4.9's φ_n has $f - 2^{-n} < \varphi_n \leq f$ pointwise (the bin width), so $\psi_n = \varphi_n + 2^{-n} \mathbf{1}_E$ is simple with $f \leq \psi_n$ and $\int \psi_n - \int \varphi_n = 2^{-n} m(E)$. Hence $\inf_{\psi \geq f} \int \psi - \sup_{\varphi \leq f} \int \varphi \leq 2^{-n} m(E) \rightarrow 0$; the reverse inequality is monotonicity (any $\varphi \leq f \leq \psi$ has $\int \varphi \leq \int \psi$ via Proposition 4.8). *Bounded additivity.* For bounded $f, g \geq 0$ on finite ground: take under-staircases $\varphi \leq f, \chi \leq g$; $\varphi + \chi \leq f + g$ is simple, so $\int f + \int g \leq \int (f + g)$ (sup over sums). For $\leq$: over-staircases $\psi \geq f, \omega \geq g$ give

the simple $\psi + \omega \geq f + g$, whence every under-staircase of $f + g$ prices at most $\int \psi + \int \omega$ (Proposition 4.8); infimize using the pinch on each: $\int (f + g) \leq \int f + \int g$. Signed bounded cases split into parts, all finite on finite ground. *Dividend banked: every use in Theorems 4.19 and 4.20 cites this problem.*

II.3 (solution in full). *Monotonicity of the step sandwiches.* P_{n+1} refines P_n : each P_{n+1} -slice sits inside a P_n -slice, so slice-infima rise ($\varphi_{n+1} \geq \varphi_n$) and slice-suprema fall, pointwise off the (countable, null) set of all partition points — where values may be redefined freely under the a.e. dialect (Proposition 4.16). *Limits.* Monotone bounded sequences converge pointwise (Volume One, MCT for sequences); φ, ψ measurable by Theorem 4.6. *The pinch.* $\int (\psi - \varphi) \leq \int (\psi_n - \varphi_n) = U - L \rightarrow 0$ uses monotonicity of the integral on the nonnegative differences (Proposition 4.13(i)) and bounded additivity (II.2) to split $\int \psi_n - \int \varphi_n$; I.6 then forces $\psi = \varphi$ a.e.; amnesty passes measurability to f ; and $\int f = \int \varphi = \lim L(f, P_n) = R$, the middle step because $\varphi_n \leq \varphi \leq \psi_n$ pinches $\int \varphi$ between $L(f, P_n)$ and $U(f, P_n)$.

4.10 Notes

History. The integral, the criterion, and the staircase-of-the-range construction are all in Lebesgue's thesis (1902) and his *Leçons sur l'intégration* (1904), the merchant's metaphor included; the one-ledger design's contrast with Darboux is Lebesgue's own advertisement. The Cantor function is Cantor's (1884, correspondence) and Scheeffer's; "devil's staircase" is folklore vintage. The composition counterexample and the measurable-non-Borel specimen via $x + c(x)$ are textbook classics (Royden; Stein–Shakarchi ch. 1, exercise circle).

Scope notes. General nonnegative additivity: Chapter 5, first corollary of monotone convergence, per Remark 4.15 — the desk keeps the credit visible rather than quietly circular. The bounded-ledger converse (agreement forces measurability): Royden's bounded-function chapter, for the ambitious.

Reading. Axler 2E–3A (the season's companion); Stein–Shakarchi ch. 2; Royden–Fitzpatrick ch. 4; Tao §1.3–1.4. Lebesgue's own *Leçons*, for the pleasure of the source.

Forward pointers. The convergence theorems and the additivity discharge: Chapter 5. The modes of convergence and Littlewood's principles: Chapter 6. Fubini: Chapter 7, where the staircase construction returns as the workhorse of product integration.

Chapter 5

The Convergence Theorems

*Companion to Episode Five. The episode restated the course's oldest question with two seasons' interest, exhibited the three escapes (bump, spike, spread — the complete list), proved the climbing license (monotone convergence, by discount-swell-exhaust), discharged the additivity credit on its one-week schedule (D14), freed nonnegative series-integral interchange, installed Fatou's one-way valve (with the spike as mnemonic), and crowned the season: dominated convergence by the double valve, with the mean-convergence rider, the bounded corollary (Season One's finale interchange re-proved in two lines), differentiation under the integral sign (Feynman's trick licensed), the counterexample parade, and the ceremonial closure of debt five — the swap account, outrage to law in two seasons. This chapter is the paperwork. Register notes: **the bell is silent** — count holds at sixteen; the jewels run entirely on ring sixteen's supremum, the crankshaft, and the staircases ("when a season's crown jewels ring nothing new, the architecture is finished"). **D14 discharged** at Corollary 5.3; **D5 closed** at Remark 5.16.*

Episode Digest

The argument of the hour. When does the integral of a limit equal the limit of the integrals? Mass escapes in exactly three ways — off to the horizon (the bump), up the spike, spread thin — and each license bars exits: monotonicity forbids all descent (MCT, proved by discounting an under-staircase, swelling sufficiency sets, exhausting storeys by the crankshaft, and letting the discount expire); nonnegativity makes the valve one-way (Fatou: the integrals remember mass the limit forgot; via climbing tail-infima and MCT); one integrable lid bricks every exit (DCT: Fatou on lid-plus and lid-minus, the lid's finite integral cancelling twice — with the mean-convergence rider on the dominated discrepancies). Additivity discharges as MCT's first corollary (staircases add; the limit passes); nonnegative series swap freely (partial sums climb); the bounded corollary retires Season One's interchange theorem in two lines; continuity in a parameter and differentiation under the integral sign follow the lid; the parade violates each hypothesis once. Debt five closes: uniformity's premium against one lid — "what those theorems ask is nearly nothing."

The episode, beat by beat. The Question, Restated. The account's history (S1E4 opening quoted back; instalments S1E7, S1E8 with the stinginess codicil); the trade's stakes (limits past expectations).

The Three Escapes. The bump ($\mathbf{1}_{[n,n+1]}$), the spike ($n\mathbf{1}_{(0,1/n)}$), the spread ($\frac{1}{n}\mathbf{1}_{[0,n]}$); conservation

along the sequence, destitution in the limit; the spread's uniform convergence — Season One's license fails on infinite ground (Remark 5.12).

The Climbing License. MCT (Theorem 5.1): free half by monotonicity; substance by the discount $c < 1$, the swelling sufficiency sets E_n , the crankshaft's exhaustion of storeys, the discount's expiry. A.e. version; the sign clause load-bearing. Application: the improper absolute integrals absorbed by truncate-climb-license (Remark 5.2 — Chapter 4's boundary post, half retired).

The Credit, Discharged. Additivity (Corollary 5.3, boxed; D14 retired on schedule); nonnegative series-integral interchange (Corollary 5.4); the standard operating procedure seeded for Chapter 7.

The One-Way Valve. Fatou (Theorem 5.5) by climbing tail-infima; the spike certifies strictness; the cautions (no limsup mirror without hypotheses; no unsigned valve — Remark 5.6).

The Crown Jewel. DCT (Theorem 5.8): the double valve, both cancellations of the lid's finite integral marked; the escapes priced against the lid (no integrable lid over bump, spike, or spread — Proposition 5.7); the riders — mean convergence (Corollary 5.9) and inherited integrability. Wonder beat: no new machinery — staircases, crankshaft, climb, valve, jewel.

The Bounded Corollary. BCT (Corollary 5.10): constant lids on finite ground; probability's workhorse (Episode Eight seed); Season One's finale interchange theorem re-proved in two lines (Remark 5.11).

What the Lid Buys. Signed series under absolute check; continuity in a parameter (Theorem 5.13); differentiation under the integral sign (Theorem 5.14) with the mean-value bridge; Feynman licensed; the sport integral (Problem II.6) with the logarithm's credit cited.

The Parade. Each hypothesis dropped once (Table at Remark 5.15): bump vs monotonicity; descending tower vs finite first floor; negative spike vs the valve's sign; the three escapes vs the lid; the spread vs uniformity-as-substitute.

The Account, Closed. D5's full history and closing comparison; the bell silent at sixteen on finished architecture.

The toolkit acquired. Discount-swell-exhaust; truncate-climb-license; the valve and its double; lid-hunting; the SOP schema (nonnegative free; signed after absolute check).

Dates worth keeping. Levi's theorem, 1906; Fatou's lemma, 1906; Lebesgue's dominated theorem, in his *Leçons*' orbit, 1908–1910.

5.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"climbs to"	$f_n \uparrow f: f_n \leq f_{n+1}, f_n \rightarrow f$	Thm. 5.1
"the tail-infima"	$g_n = \inf_{k \geq n} f_k$	Thm. 5.5
"the lid"	$ f_n \leq g, g \text{ integrable}$	Thm. 5.8
"the discrepancies"	$\int f_n - f dm \rightarrow 0$	Cor. 5.9
"the bump / spike / spread"	$\mathbf{1}_{[n, n+1]}; n\mathbf{1}_{(0, 1/n)}; \frac{1}{n}\mathbf{1}_{[0, n]}$	Rem. 5.15

5.2 The Climbing License

Theorem 5.1 (Monotone convergence — Levi, 1906). *Let $f_n : E \rightarrow [0, \infty]$ be measurable with $f_n \leq f_{n+1}$ pointwise and $f_n \rightarrow f$ pointwise. Then f is measurable and*

$$\int_E f_n dm \uparrow \int_E f dm.$$

(A.e. monotonicity and convergence suffice: patch the null set, Proposition 4.16.)

Proof. f is measurable as a pointwise limit (Theorem 4.6). *Free half:* $f_n \leq f$ gives $\int f_n \leq \int f$ (manners); the integrals climb (monotonicity again), so $L = \lim_n \int f_n$ exists in $[0, \infty]$ with $L \leq \int f$.

Substance. Fix a simple φ with $0 \leq \varphi \leq f$ and a discount $c \in (0, 1)$; write $\varphi = \sum_{i=1}^N a_i \mathbf{1}_{A_i}$. The sufficiency sets

$$E_n = \{x \in E : f_n(x) \geq c \varphi(x)\}$$

are citizens ($f_n - c\varphi$ measurable), swell with n (the climb), and exhaust E : where $\varphi(x) = 0$ membership is immediate; where $\varphi(x) > 0$, $c\varphi(x) < \varphi(x) \leq f(x)$, so some $f_n(x)$ has cleared $c\varphi(x)$. Then

$$\int f_n \geq \int_{E_n} f_n \geq c \int_{E_n} \varphi = c \sum_{i=1}^N a_i m(A_i \cap E_n) \xrightarrow{n \rightarrow \infty} c \sum_{i=1}^N a_i m(A_i) = c \int \varphi,$$

the arrow by the crankshaft (continuity along swelling chains, Proposition 1.3(i)) applied to each of the finitely many storeys $A_i \cap E_n \uparrow A_i$. Hence $L \geq c \int \varphi$ for every $c < 1$: $L \geq \int \varphi$ (let the discount expire); and taking the supremum over under-staircases, $L \geq \int f$ by ring sixteen's definition. With the free half, $L = \int f$. $\square$

Remark 5.2 (The sign clause; the improper absorption). Signed climbing fails: $f_n = -\frac{1}{x} \mathbf{1}_{(0, 1/n)}$ -style descending starts have $\int f_n = -\infty$ throughout while the limit vanishes; a finite first floor restores the mirror (Problem I.8, the descending tower). *Absorption:* for measurable $f \geq 0$ on $(0, 1]$ with improper Riemann integral finite, the truncations $f \mathbf{1}_{[1/n, 1]}$ climb to f ; each truncated integral is Riemann's (Chapter 4's absorption on the frame); MCT passes the limit — e.g. $\int_{(0, 1]} x^{-1/2} dm = \lim_n (2 - 2/\sqrt{n}) = 2$. Chapter 4's boundary post, absolutely convergent half: retired.

Corollary 5.3 (Additivity — debt fourteen discharged). *For measurable $f, g \geq 0$: $\int (f + g) = \int f + \int g$; hence for integrable f, g and scalars, the integral is linear.*

Proof. Staircases $\varphi_n \uparrow f$, $\psi_n \uparrow g$ (Theorem 4.9); $\varphi_n + \psi_n \uparrow f + g$; simple additivity (Proposition 4.8) and Theorem 5.1 on all three sequences pass the identity through the limit. The signed case splits into parts, every term finite by integrability. *The one-week credit* (Remark 4.15) is retired on schedule. $\square$

Corollary 5.4 (Nonnegative series swap freely). *For measurable $f_k \geq 0$: $\int \sum_{k=1}^{\infty} f_k dm = \sum_{k=1}^{\infty} \int f_k dm$ (both sides possibly ∞ , together).*

Proof. Partial sums climb; Corollary 5.3 prices each; Theorem 5.1 passes the limit. $\square$

5.3 The One-Way Valve

Theorem 5.5 (Fatou, 1906). *For measurable $f_n \geq 0$:*

$$\int \liminf_n f_n \, dm \leq \liminf_n \int f_n \, dm.$$

Proof. $g_n = \inf_{k \geq n} f_k$ is measurable (Theorem 4.6), climbs, and $g_n \uparrow \liminf f_n$ by definition. From $g_n \leq f_n$: $\int g_n \leq \int f_n$, so $\lim \int g_n \leq \liminf \int f_n$ (the left limit exists: MCT's free half). MCT prices the left limit as $\int \liminf f_n$. $\square$

Remark 5.6 (Cautions). Strictness: the spike ($\liminf = 0$; integrals = 1). No unsigned valve: the negative spike $-n\mathbf{1}_{(0,1/n)}$ has $\int \liminf = 0 > -1 = \liminf \int$ read against the claimed direction — the sign gasket is structural. No limsup mirror without a lid: the bump has $\int \limsup = 0 < 1 = \limsup \int$; with an integrable lid g , the reverse valve holds ($\limsup \int f_n \leq \int \limsup f_n$, by Theorem 5.5 on $g - f_n$ — Problem II.3's ambit).

5.4 The Crown Jewel

Proposition 5.7 (No lids over the escapes). *Any measurable g dominating all bumps has $g \geq \mathbf{1}_{[0,\infty)}$ a.e.-eventually in the obvious sense and $\int g = \infty$; dominating all spikes forces $g(x) \geq \sup_n n\mathbf{1}_{(0,1/n)}(x)$, which exceeds $\frac{1}{2x}$ near nought; dominating all spreads forces $g \geq \frac{1}{n}$ on $[0, n]$ for all n , whence $\int g \geq \sum_n \frac{1}{n} \cdot 1$ -flavoured lower bounds diverge (the desk's one-line versions: Problem I.4). No integrable lid exists in any of the three galleries.*

Proof. Problem I.4, with the harmonic comparison for the spread and the $1/(2x)$ comparison for the spike (between $1/(k+1)$ and $1/k$ the sup of spikes is at least $k \geq 1/(2x)$). $\square$

Theorem 5.8 (Dominated convergence — Lebesgue). *Let f_n be measurable, $f_n \rightarrow f$ a.e., and $|f_n| \leq g$ a.e. for every n , with g integrable. Then f is integrable and $\int f_n \, dm \rightarrow \int f \, dm$.*

Proof. Patch the null exceptions (the amnesty); assume the hypotheses everywhere. $|f| \leq g$ in the limit: f integrable with room ($\int |f| \leq \int g < \infty$). The sequences $g + f_n \geq 0$ and $g - f_n \geq 0$ are lawful valve inputs. Fatou on the first:

$$\int g + \int f = \int (g + f) \leq \liminf \int (g + f_n) = \int g + \liminf \int f_n,$$

additivity (Corollary 5.3) expanding both ends; cancel $\int g$ — finite, the lid's integrability spent, first time: $\int f \leq \liminf \int f_n$. Fatou on the second, the same bookkeeping (the second expenditure): $\int g - \int f \leq \int g - \limsup \int f_n$, i.e. $\limsup \int f_n \leq \int f$. The pinch closes: $\lim \int f_n = \int f$. $\square$

Corollary 5.9 (The mean-convergence rider). *Under Theorem 5.8's hypotheses, $\int |f_n - f| \, dm \rightarrow 0$.*

Proof. $|f_n - f| \rightarrow 0$ a.e., dominated by $2g$; Theorem 5.8 applied to the discrepancies. $\square$

Corollary 5.10 (Bounded convergence — probability's workhorse). *On E with $m(E) < \infty$: if $f_n \rightarrow f$ a.e. and $|f_n| \leq M$ (one constant, all n), then $\int f_n \rightarrow \int f$.*

Proof. $g = M\mathbf{1}_E$ is integrable on finite ground; the jewel applies. $\square$

Remark 5.11 (Season One's finale theorem, retired with honours). Volume One, Chapter 8's interchange theorem (uniform convergence on $[a, b]$ licenses the swap) is now two lines: uniform convergence eventually pins $|f_n| \leq |f| + 1 \leq M + 1$ — f bounded as a uniform limit of bounded functions — a constant lid on finite ground; Corollary 5.10. The spread (Remark 5.12) marks where the retirement's finite-ground clause was always working.

Remark 5.12 (The spread). $\frac{1}{n}\mathbf{1}_{[0,n]} \rightrightarrows 0$ uniformly, yet $\int = 1 \not\rightarrow 0$: on infinite ground, uniformity neither implies domination nor substitutes for it (the failing line of the old proof: the tube's cross-section times the ground's width, $\frac{1}{n} \cdot n$, refuses to vanish). Problem II.5 dissects.

5.5 What the Lid Buys

Theorem 5.13 (Continuity in a parameter). *Let $f(x, t)$ be measurable in x for each t in an interval, continuous in t for a.e. x , and dominated: $|f(x, t)| \leq g(x)$, g integrable, for all t . Then $F(t) = \int f(x, t) dm(x)$ is continuous.*

Proof. Sequential continuity suffices (Volume One): $t_k \rightarrow t$ gives $f(x, t_k) \rightarrow f(x, t)$ a.e., dominated by g ; Theorem 5.8. $\square$

Theorem 5.14 (Differentiation under the integral sign). *Let $f(x, t)$ be integrable in x for each t in an open interval, differentiable in t for a.e. x , with the derivative dominated: $|\partial_t f(x, t)| \leq g(x)$, g integrable, for all t in the interval. Then $F(t) = \int f(x, t) dm(x)$ is differentiable, and*

$$F'(t) = \int \partial_t f(x, t) dm(x).$$

Proof. Fix t ; take any $h_k \rightarrow 0$. The quotients $q_k(x) = (f(x, t + h_k) - f(x, t))/h_k$ converge a.e. to $\partial_t f(x, t)$; the mean-value bridge: for each x and k , Volume One's mean value theorem plants a witness s between t and $t + h_k$ with $q_k(x) = \partial_t f(x, s)$, so $|q_k(x)| \leq g(x)$ — the derivative's lid covers every quotient. Theorem 5.8 passes the limit through the integral; linearity identifies the quotient of F with $\int q_k$. Every sequence h_k agreeing, $F'(t)$ exists and equals the display. $\square$

Remark 5.15 (The parade, tabulated).

Specimen	Violated clause	Failure
bump	MCT's climb; DCT's lid	$\int \rightarrow 1 \neq 0$
spike	the lid ($\geq 1/2x$)	$\int = 1$, limit 0; Fatou strict
spread	the lid (harmonic); uniformity no substitute	$\int = 1 \neq 0$ despite $\rightrightarrows$
negative spike	Fatou's sign	valve leaks backward
descending tower	finite first floor	signed climb, $\int \equiv -\infty$

Each hypothesis of each license, violated exactly once; no theorem wounded.

Remark 5.16 (Debt five: the account, closed). Opened S1E4 (the rearrangement); instalment one S1E7 (limits past limits, uniformly); instalment two S1E8 (limit past integral, uniformly, stinginess codicil filed); closed tonight: climbing, nonnegative, or dominated mass swaps freely on any measurable ground. The codicil is discharged with prejudice — Remark 5.11 prices the old license as a pocket corollary. *Converted into standing law.*

5.6 Problems

Tier I — Calisthenics

- I.1 Price the three escapes at $n = 3, 10, 100$: integrals, pointwise limits, and (for the spread) the uniform sag.
- I.2 MCT warm-ups: the dyadic staircases of x^2 on $[0, 1]$ climb; verify $\int \varphi_n \rightarrow \frac{1}{3}$ against Chapter 4's Problem I.3.
- I.3 Valve drills: compute both sides of Fatou for the bump, the spike, and the constant sequence $\mathbf{1}_{[0,1]}$.
- I.4 Lid-hunting (Proposition 5.7): complete the three one-line divergences; then exhibit honest lids for sin-free bounded sequences on $[0, 1]$ of your construction, and for $x^{-1/2}\mathbf{1}_{[1/n,1]}$.
- I.5 Replay Corollary 5.3 on $f = \mathbf{1}_{[0,2]}$, $g = x\mathbf{1}_{[0,1]}$ with explicit staircases at resolution two.
- I.6 Series swap: $\sum_k 2^{-k}\mathbf{1}_{[k,k+1]}$ — integrate twice, once per side of Corollary 5.4.
- I.7 A.e. patching: rerun Theorem 5.8 where the hypotheses fail on $\mathbb{Q}$, writing the patch once.
- I.8 The descending tower: $f_n = -\mathbf{1}_{[n,\infty)}$ climbs (!) to 0 with $\int f_n \equiv -\infty$; reconcile with Theorem 5.1's sign clause and state the finite-floor mirror.

Tier II — The Real Thing

- II.1 (*The discount, to the letter.*) Write Theorem 5.1's substance half with every epsilon and the crankshaft citation at its line. Solved in full below.
- II.2 (*The double valve.*) Prove Theorem 5.8 with both cancellations marked and the exact point where $\int g < \infty$ is spent, twice. Solved in full below.
- II.3 (*The reverse valve.*) With an integrable lid, prove $\limsup \int f_n \leq \int \limsup f_n$; drop the lid and convict with the bump.
- II.4 (*The bridge.*) Assemble Theorem 5.14 from the jewel with the mean-value bridge displayed; state precisely why the lid must cover an *interval* of parameters.
- II.5 (*The spread, dissected.*) Identify the failing line of Volume One's tube proof on infinite ground, and prove the finite-ground clause was necessary by any counterexample of your own manufacture distinct from the spread.
- II.6 (*For sport.*) Evaluate $F(t) = \int_{(0,1)} \frac{x^t-1}{\log x} dm(x)$ for $t \geq 0$: differentiate under the sign (domination: $x^t \leq 1$), integrate $F'(t) = \frac{1}{t+1}$, and conclude $F(t) = \log(1+t)$ — the logarithm riding on Volume One's standing credit, cited.

Tier III — For the Ambitious ★

- III.1 ★ (*Varying lids.*) Prove the generalized dominated theorem: if $g_n \rightarrow g$ a.e., all integrable, with $\int g_n \rightarrow \int g < \infty$ and $|f_n| \leq g_n$, then the conclusion of Theorem 5.8 holds. (Fatou on $g_n \pm f_n$; references: Royden.)
- III.2 ★ (*The valve on measures.*) For citizens A_n , prove $m(\liminf A_n) \leq \liminf m(A_n)$ via indicator sequences, identify the swelling-chain crankshaft as the special case $A_n \uparrow$, and state the matching limsup clause with its finite-mass proviso (the shrinking caveat's return).

5.7 Hints

- II.3. Apply Theorem 5.5 to $g - f_n \geq 0$ and unfold $\liminf(-a_n) = -\limsup a_n$.
- II.4. The witness s ranges over the interval as x and k vary; a lid pinned only at t itself covers no quotient.
- II.5. The tube bound is $|\int f_n - \int f| \leq d(f_n, f) \cdot m(E)$; on infinite ground the right factor is $\infty \cdot 0$ -flavoured. Manufacture: any mass-one plateau train of vanishing height.
- III.2. $\mathbf{1}_{\liminf A_n} = \liminf \mathbf{1}_{A_n}$ pointwise; Fatou. For limsup: complements within a finite-mass frame.

5.8 Solutions

Tier I (terse). I.1–I.8 As drilled; I.8's mirror: if $f_n \downarrow f \geq 0$ pointwise with $\int f_1 < \infty$, then $\int f_n \downarrow \int f$ (apply Theorem 5.8 with lid f_1 , or the crankshaft on complements).

II.1 (solution in full). As Theorem 5.1's substance, expanded: fix φ, c ; the sets E_n swell and exhaust (two cases displayed); for each storey, $m(A_i \cap E_n) \uparrow m(A_i)$ is Proposition 1.3(i) applied to the swelling chain $A_i \cap E_n$ (citizens, intersections); summing finitely many storeys preserves the limit (Volume One's algebra of limits); whence $L \geq c \int \varphi$; the map $c \mapsto c \int \varphi$ is continuous on $(0, 1)$ with supremum $\int \varphi$ (or: given ε , pick $c > 1 - \varepsilon/(1 + \int \varphi)$ when $\int \varphi < \infty$, and any c with $c \int \varphi > M$ arbitrary when $\int \varphi = \infty$); the under-staircase supremum closes onto ring sixteen's definition.

II.2 (solution in full). As Theorem 5.8, with the ledger: integrability of f from the limit lid; nonnegativity of $g \pm f_n$ from the domination; additivity's two expansions (Corollary 5.3, tonight's discharge — the proof is not circular: the corollary rests on MCT alone); the two cancellations, each a subtraction of the *finite* $\int g$ (the season's one prohibition respected — this is where integrability of the lid is spent, and the proof collapses without it: take $g = \infty$ -flavoured lids and the inequalities read $\infty \leq \infty$, empty); the final pinch $\limsup \leq \int f \leq \liminf$.

5.9 Notes

History. Levi (1906); Fatou (1906), in a paper on Fourier series — the account's creditor collecting interest even here; Lebesgue's dominated theorem in the *Leçons* tradition (1908–1910). Feynman's account of the differentiation trick is in *Surely You're Joking*; the calculus text was Woods' *Advanced Calculus*.

Reading. Axler 3A–3B (the season's companion); Bartle, *The Elements of Integration*, chs. 5–6, still the cleanest short treatment; Stein–Shakarchi ch. 2; Tao §1.4–1.5.

Forward pointers. The modes of convergence and the typewriter: Chapter 6 (mean convergence, tonight's rider, becomes a geometry there). The SOP canonized for double integrals: Chapter 7 (Tonelli and Fubini). Expectation swaps on mass-one ground: Chapter 8, where Corollary 5.10 runs the shop.

Chapter 6

The Zoo of Convergence, or Littlewood's Three Principles

*Companion to Episode Six. The episode read Littlewood's three principles as field guide, paraded the five modes with their working currencies, walked the implication map compass in hand, staged the typewriter (wonder beat: "you can hear the carriage return"), extracted the subsequence consolation with its Borel–Cantelli skeleton flagged for the finale, proved Egorov aloud in strategy (shrinking crankshaft + the gambit), assembled Lusin from three nearlies, built the mean distance (amnesty as geometry), rang **ring seventeen** at Riesz–Fischer on ring twelve's precedent, and exiled Hölder and Minkowski by name — "the honest division of labour is not a compromise; it is the design." This chapter is the paperwork; its core is the diagram with a proof or a counterexample on every arrow, per the plan's mandate. Register notes: the bell rings once — (⊗) at Theorem 6.10, flagged for veto, precedent Volume One's ring twelve.*

Episode Digest

The argument of the hour. Five modes, one map, three principles. Uniform, pointwise, almost-everywhere, in-the-mean, in-measure — each the currency of a working question (guarantees, trajectories, expected loss, consistency). The map: uniform at the headwaters; mean $\rightarrow$ measure by Chebyshev's embryo (mass bounds acreage); a.e. $\rightarrow$ measure on finite ground (the bump patrols the frontier); mean and a.e. incomparable (the spike one way, the typewriter the other). The typewriter — blocks sweeping the interval in halving generations — converges in measure and mean, pointwise nowhere: smallness of offence and persistence of offence are different failures. In-measure extracts a.e. along a fast subsequence (the gambit + tail-trap; Borel–Cantelli in embryo). Egorov: on finite ground, a.e. convergence is uniform off epsilon acreage (shrinking crankshaft, finite floor cited; the bump witnesses the clause). Lusin: measurable functions are continuous off epsilon acreage (restriction, not pointwise — the static on the irrationals pins the distinction; the property characterizes measurability); assembled from staircases + the first principle + Egorov + the three-legged stool. The mean distance makes the integrable functions a space (points = a.e.-classes); Riesz–Fischer: it is complete — fast subsequence, stacked series, climbing license, dominated closure — ring seventeen on ring twelve's precedent. Hölder and Minkowski: named, motivated, exiled; proved at the desk via Young.

The episode, beat by beat. **The Field Guide / The Modes in the Wild.** The three principles; "nearly" as exact currency; the five modes' working questions.

The Five Modes / The Geography. Definitions (Definition 6.1); the walked map; Chebyshev's embryo (Lemma 6.2); the full diagram (Theorem 6.3).

The Typewriter. Construction and both verdicts (Proposition 6.4); the wonder beat.

The Consolation Prize. The subsequence lemma (Theorem 6.5); the skeleton flagged (Remark 6.6).

Egorov's Theorem. Theorem 6.7; the finite floor at its line; the bump as witness.

Lusin's Theorem. Statement and footnotes (Theorem 6.8, sketch with references per plan); the characterization remark.

The Mean as a Distance / Riesz–Fischer. The space L^1 (Definition 6.9); $(*)$ Theorem 6.10, the guided centrepiece solved in full.

The Exiles. Young, Hölder, Minkowski (Theorems 6.11–6.13); the L^p glimpse and nesting.

The toolkit acquired. The exchange-rate map; the fast-subsequence harvest; the shrinking-tails manoeuvre; amnesty-as-geometry; the exile discipline.

Dates worth keeping. Riesz and Fischer, independently, 1907, both via Fourier series; Egorov, 1911; Lusin, 1912; Littlewood's principles as printed maxims, 1944 (*Lectures on the Theory of Functions*).

6.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"uniformly"	$f_n \Rightarrow f$	Vol. One, Ch. 7
"almost everywhere"	$f_n \rightarrow f$ a.e.	Ch. 4
"in the mean"	$\ f_n - f\ _1 = \int f_n - f dm \rightarrow 0$	Def. 6.1
"in measure"	$\forall \varepsilon : m\{ f_n - f \geq \varepsilon\} \rightarrow 0$	Def. 6.1
"the mean distance"	$d_1(f, g) = \ f - g\ _1$ on L^1	Def. 6.9
"conjugate exponents"	$\frac{1}{p} + \frac{1}{q} = 1, 1 < p, q < \infty$	Thm. 6.12

6.2 The Map

Definition 6.1 (The two integral modes). On measurable ground E : $f_n \rightarrow f$ *in the mean* when $\int_E |f_n - f| dm \rightarrow 0$; *in measure* when for every $\varepsilon > 0$, $m\{x \in E : |f_n(x) - f(x)| \geq \varepsilon\} \rightarrow 0$. (Uniform, pointwise, and a.e. are of long standing.)

Lemma 6.2 (Chebyshev's embryo: mass bounds acreage). For measurable $g \geq 0$ and $\varepsilon > 0$: $m\{g \geq \varepsilon\} \leq \frac{1}{\varepsilon} \int g dm$.

Proof. $\varepsilon \mathbf{1}_{\{g \geq \varepsilon\}} \leq g$ pointwise; integrate (manners, Chapter 4). The finale christens the general form. $\square$

Theorem 6.3 (The diagram, every arrow warranted). *On ground E :*

- (i) *uniform $\Rightarrow$ pointwise $\Rightarrow$ a.e.; uniform $\Rightarrow$ mean when $m(E) < \infty$ ($\|f_n - f\|_1 \leq d(f_n, f) m(E)$); the spread kills the finiteness clause.*
- (ii) *mean $\Rightarrow$ measure, always (Lemma 6.2 on $|f_n - f|$).*
- (iii) *a.e. $\Rightarrow$ measure when $m(E) < \infty$ (via Theorem 6.7: off acreage ε the discrepancy is eventually uniform, hence eventually below ε everywhere there; the offending ground is eventually inside the sacrificed acreage). The bump kills the clause on infinite ground.*
- (iv) *mean $\nRightarrow$ a.e. (the typewriter, Proposition 6.4); a.e. $\nRightarrow$ mean (the spike); measure $\nRightarrow$ a.e. (typewriter); measure $\nRightarrow$ mean (spike, which converges in measure); pointwise $\nRightarrow$ uniform (Volume One's ramps); a.e. $\nRightarrow$ pointwise (redefine on $\mathbb{Q}$).*

The five-by-five table with one-line warrants is Problem I.1's completed grid.

Proof. All displayed; (iii)'s derivation from Egorov is Problem II.1, solved in full, with the bump's frontier patrol. $\square$

Proposition 6.4 (The typewriter). *For $n = 2^k + j$, $0 \leq j < 2^k$, let $T_n = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k}]}$ on $[0, 1]$. Then $T_n \rightarrow 0$ in measure and in the mean ($m(\text{supp } T_n) = \int T_n = 2^{-k} \rightarrow 0$), and $(T_n(x))$ converges for no $x \in [0, 1]$: each generation's carriage passes every point once, so $T_n(x) = 1$ infinitely often and $T_n(x) = 0$ infinitely often.*

Proof. The two computations are displayed; the pointwise verdict is Problem II.2, solved in full (for each k there is exactly one j with $x \in [j2^{-k}, (j+1)2^{-k}]$ — two at a dyadic seam, which only helps — so ones recur eternally; zeros recur since $2^k > 1$ blocks per generation miss x). $\square$

Theorem 6.5 (The consolation: a.e. along a subsequence). *If $f_n \rightarrow f$ in measure on E , there are indices $n_1 < n_2 < \dots$ with $f_{n_k} \rightarrow f$ almost everywhere.*

Proof. Harvest: choose n_k with $m(A_k) \leq 2^{-k}$ for $A_k = \{|f_{n_k} - f| \geq 2^{-k}\}$ (in-measure convergence supplies each). Tail-trap: let $B_K = \bigcup_{k \geq K} A_k$, so $m(B_K) \leq 2^{1-K}$ (the gambit's series), and $N = \bigcap_K B_K$ is null. Off N : x eventually avoids every A_k beyond some K , so $|f_{n_k}(x) - f(x)| < 2^{-k}$ for all $k \geq K$: convergence. $\square$

Remark 6.6 (The skeleton, flagged). The step “summable acreages $\Rightarrow$ the visited-infinitely-often set is null” is the first Borel–Cantelli lemma, proved here in embryo; the finale states it under its name and spends it on almost-sure convergence. A converse service: if every subsequence of (f_n) has a further subsequence converging a.e. to f , then $f_n \rightarrow f$ in measure (Problem II.4 — the trade's standard trick for upgrading subsequential verdicts).

Theorem 6.7 (Egorov, 1911). *Let $m(E) < \infty$ and $f_n \rightarrow f$ a.e. on E , all measurable and finite a.e. For every $\varepsilon > 0$ there is a citizen $A \subseteq E$ with $m(E \setminus A) < \varepsilon$ and $f_n \rightrightarrows f$ on A .*

Proof. For $k, n \in \mathbb{N}$ let $E_{n,k} = \bigcup_{m \geq n} \{|f_m - f| > \frac{1}{k}\}$: citizens, shrinking in n . Their intersection over n is contained in the null non-convergence set (a point in every $E_{n,k}$ suffers $\frac{1}{k}$ -discrepancies beyond every stage), so, the finite floor being available ($m(E) < \infty$ — the crankshaft's shrinking clause, Proposition 1.3(ii), spent here and nowhere else), $m(E_{n,k}) \rightarrow 0$ as $n \rightarrow \infty$. Choose n_k with $m(E_{n_k,k}) < \varepsilon 2^{-k}$ (the gambit); set $A = E \setminus \bigcup_k E_{n_k,k}$, so $m(E \setminus A) < \varepsilon$. On A : for every k and every

$m \geq n_k$, $|f_m - f| \leq \frac{1}{k}$ — one threshold per tolerance, valid across all of A : uniform convergence. The bump on $[0, \infty)$ converges everywhere yet is uniform on no co-finite-acreage set (its unit block always escapes any fixed bounded remainder): the finiteness clause is real. $\square$

Theorem 6.8 (Lusin, 1912 — sketched, with references). *Let f be measurable and finite a.e. on $[a, b]$. For every $\varepsilon > 0$ there is a closed $F \subseteq [a, b]$ with $m([a, b] \setminus F) < \varepsilon$ such that the restriction $f|_F$ is continuous. Moreover the property characterizes measurability.*

Assembly (the three nearlies, epsilons split by the gambit). Staircases $\varphi_j \rightarrow f$ a.e. (Theorem 4.9); each φ_j is continuous off open acreage $\varepsilon 2^{-j-1}$ — its finitely many storeys are nearly finite unions of intervals (Littlewood's first principle, Chapter 2's manners, Problem II.4 there), and off the boundary slivers the storey structure is locally constant; Egorov (Theorem 6.7) makes $\varphi_j \rightarrow f$ uniform off acreage $\varepsilon/4$; a uniform limit of continuous restrictions is continuous (Volume One's three-legged stool, on the restricted set); the closed certificate comes from inner regularity (Chapter 2). Full details: Axler 2D, Royden ch. 3, Stein–Shakarchi ch. 1; the characterization converse: Problem III.1's references. *Footnote pinned in the episode: continuity of the restriction, not continuity at points — the static restricted to the irrationals is identically 0.*

6.3 The Mean Distance and Riesz–Fischer

Definition 6.9 (L^1 ; the mean distance). $L^1(E)$: the integrable functions on E , identified when equal a.e. (the amnesty as geometry: distance-zero pairs are one point); $d_1(f, g) = \int_E |f - g| dm$. Metric axioms: symmetry and nonnegativity are manifest; $d_1(f, g) = 0$ iff $f = g$ a.e. (Chapter 4's little lemma, I.6); the triangle inequality is additivity on the pointwise triangle $|f - h| \leq |f - g| + |g - h|$.

Theorem 6.10 ((*) Riesz–Fischer, 1907 — ring seventeen). $L^1(E)$ is complete: every Cauchy sequence in the mean distance converges in the mean to a member of $L^1(E)$.

Proof. (The guided centrepiece, solved in full; each jewel cited at its expenditure.) Let (f_n) be d_1 -Cauchy. *Harvest* n_k with $d_1(f_{n_{k+1}}, f_{n_k}) \leq 2^{-k}$ (Cauchy supplies; the gambit's rates). *Stack*: let $g = |f_{n_1}| + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|$, a nonnegative measurable function; the climbing license through the series corollary (Corollary 5.4) prices $\int g \leq \int |f_{n_1}| + \sum_k 2^{-k} < \infty$: g is integrable, hence finite a.e. *Limit*: where g is finite, the telescoping series $f_{n_1} + \sum_k (f_{n_{k+1}} - f_{n_k})$ converges absolutely (Volume One's comparison against the stack); call the a.e. limit f — measurable (Theorem 4.6), and $|f| \leq g$: integrable. *Closure*: $|f - f_{n_k}| \leq \sum_{j \geq k} |f_{n_{j+1}} - f_{n_j}| \leq g$ with a.e. convergence to 0; the dominated jewel's rider (Corollary 5.9, via Theorem 5.8 with lid g) gives $d_1(f, f_{n_k}) \rightarrow 0$; and a Cauchy sequence with a convergent subsequence converges (Volume One's huddle lemma, verbatim in any metric): $d_1(f, f_n) \rightarrow 0$. *The bell rings — the seventeenth*: the completeness of the space in which twentieth-century analysis lives, on the precedent of ring twelve (Volume One's C under the uniform distance); flagged for veto, per standing practice. $\square$

6.4 The Exiles

Theorem 6.11 (Young's inequality). *Let $p, q \in (1, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. For $a, b \geq 0$: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$, with equality iff $a^p = b^q$.*

Proof. Fix $b \geq 0$ and maximize $h(a) = ab - a^p/p$ over $a \geq 0$: $h'(a) = b - a^{p-1}$ vanishes at $a_* = b^{1/(p-1)} = b^{q-1}$ — a maximum, $h'' < 0$ (Volume One's harvest). Then $h(a_*) = b^{q-1}b - b^{(q-1)p}/p = b^q(1 - \frac{1}{p}) = b^q/q$, using $(q-1)p = q$. So $ab - a^p/p \leq b^q/q$ for all a , which is the claim; equality exactly at $a = a_*$, i.e. $a^p = b^q$. (No logarithms; the credit stands unspent.) $\square$

Theorem 6.12 (Hölder's inequality). *With p, q conjugate and f, g measurable:*

$$\int |fg| dm \leq \left(\int |f|^p \right)^{1/p} \left(\int |g|^q \right)^{1/q} = \|f\|_p \|g\|_q.$$

Proof. If either norm is 0 or ∞ , the desk's two-line cases settle it (a null factor annihilates; an infinite bound is vacuous). Otherwise normalize: $u = |f|/\|f\|_p$, $v = |g|/\|g\|_q$; Young pointwise: $uv \leq u^p/p + v^q/q$; integrate (additivity): $\int uv \leq \frac{1}{p} + \frac{1}{q} = 1$; unnormalize. $\square$

Theorem 6.13 (Minkowski's inequality). $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ for $1 \leq p < \infty$.

Proof. $p = 1$ is additivity on the pointwise triangle. For $p > 1$:

$$\int |f + g|^p \leq \int |f| |f + g|^{p-1} + \int |g| |f + g|^{p-1} \leq (\|f\|_p + \|g\|_p) \| |f + g|^{p-1} \|_q$$

by the pointwise triangle then Hölder twice; and $\| |f + g|^{p-1} \|_q = (\int |f + g|^p)^{1/q}$ since $(p-1)q = p$. Divide (finiteness first: $|f + g|^p \leq 2^p(|f|^p + |g|^p)$ keeps the divisor finite; zero divisor: nothing to prove). $\square$

Remark 6.14 (The L^p glimpse). On finite ground, $L^p \subseteq L^r$ for $p > r$ (Hölder with exponent p/r against the constant one: Problem I.8); the family's completeness generalizes Theorem 6.10 (Problem III.1). The course's boundary: duality, Hilbert space, and Fourier's settlement belong to a further season; the Notes point onward.

6.5 Problems

Tier I — Calisthenics

- I.1 Complete the five-by-four verdict grid: the bump, spike, spread, typewriter against the five modes, one-line warrants.
- I.2 Typewriter arithmetic at generations three, five, ten: index ranges, block widths, both integral verdicts.
- I.3 Chebyshev-in-embryo drills: acreage bounds for the spike and the identity's staircases at stated tolerances.
- I.4 Harvest a fast subsequence from a given in-measure rate $(m\{|f_n| \geq \varepsilon\} \leq 1/\log n$ -flavoured), exhibiting the indices.
- I.5 Egorov warm-up: for the spike on $[0, 1]$, exhibit the set A explicitly at $\varepsilon = \frac{1}{10}$ (cut a neighbourhood of nought) and the uniform threshold on it.
- I.6 Conjugate arithmetic: the (p, q) table for $p = \frac{3}{2}, 2, 3, 4$; verify $(p-1)q = p$ throughout.

- I.7 Mean distances between residents: d_1 of the static and the zero function; of Thomae and zero; of two typewriter blocks at one generation.
- I.8 Nesting on $[0, 1]$: prove $\|f\|_1 \leq \|f\|_2$ via Hölder against **1**; exhibit failure on $[0, \infty)$.

Tier II — The Real Thing

- II.1 (*The subtle arrow.*) Prove a.e. $\Rightarrow$ in-measure on finite ground via Egorov, and convict the infinite case with the bump. Solved in full below.
- II.2 (*The typewriter's verdict.*) Prove pointwise convergence fails at every point of $[0, 1]$, dyadic seams included. Solved in full below.
- II.3 (*The finite floor.*) Locate the exact line of Theorem 6.7 that fails for the bump on $[0, \infty)$, and prove no co-finite-acreage uniform set exists.
- II.4 (*The upgrade trick.*) If every subsequence of (f_n) has a further subsequence $\rightarrow f$ a.e., prove $f_n \rightarrow f$ in measure. (Contrapositive: a stubborn ε -acreage along a subsequence survives every further extraction, contradicting Theorem 6.5.)
- II.5 (*Lusin, assembled.*) Write the three-nearly assembly with each epsilon's ledger line and each cited instrument's address; state precisely why the certificate may be taken closed.
- II.6 (*The exile's paperwork.*) Young to Hölder to Minkowski with every normalization case handled, including the zero-and-infinite-norm edges.

Tier III — For the Ambitious ★

- III.1 ★ L^p completeness for $1 \leq p < \infty$: run Theorem 6.10's argument with p -th powers (Minkowski for the stack; the climbing license on g^p); references: Axler 3B, Royden ch. 7. Include Lusin's converse characterization from the same shelves.
- III.2 ★ Metrizability of the modes: exhibit a metric for convergence in measure on finite ground ($\int \min(|f - g|, 1)$ -flavoured) and prove pointwise a.e. convergence is induced by *no* metric — the typewriter plus the subsequence trick (a metric mode in which every subsequence has an a.e.-convergent sub-subsequence would force a.e. convergence of the typewriter itself). References: Royden, exercises; Folland §2.4.

6.6 Hints

- II.3. The crankshaft's shrinking clause demands a finite first floor; $E_{n,1}$ for the bump has infinite measure for every n .
- II.4. If $f_n \not\rightarrow f$ in measure: some ε_0, δ_0 and indices n_j with $m\{|f_{n_j} - f| \geq \varepsilon_0\} \geq \delta_0$; no further subsequence can converge a.e. (it would converge in measure on the finite frame trapping δ_0 of the acreage — II.1).
- II.5. Closed: inner regularity (Chapter 2) inside the Egorov set; the three sacrifices: boundary slivers ($\varepsilon/2$ summed), Egorov ($\varepsilon/4$), regularity ($\varepsilon/4$).

III.2. In a metric space, $x_n \rightarrow x$ iff every subsequence has a sub-subsequence $\rightarrow x$; the typewriter violates the conclusion while satisfying the premise for a.e.-mode.

6.7 Solutions

Tier I (terse). **I.1** Grid as in Theorem 6.3; the spread's line: uniform yes, pointwise yes, a.e. yes, mean no, measure yes (acreage $\{\frac{1}{n} \geq \varepsilon\}$ is empty for large n — note it converges in measure though not in mean). **I.5** $A = [\frac{1}{10}, 1]$: beyond $n = 10$ the spike's support misses A ; threshold $N = 10$ for every tolerance. **I.7** $d_1(D, 0) = 0$ — the static IS the zero of L^1 : one point of the space, the amnesty as geometry made vivid; $d_1(t, 0) = 0$ likewise; two generation- k blocks: $2 \cdot 2^{-k}$ when disjoint. **I.8** $\|f\|_1 = \int |f| \cdot 1 \leq \|f\|_2 \|1\|_2 = \|f\|_2$; on $[0, \infty)$: $f = \frac{1}{x} \mathbf{1}_{[1, \infty)}$ has $\|f\|_2 < \infty = \|f\|_1$.

II.1 (solution in full). Let $m(E) < \infty$, $f_n \rightarrow f$ a.e., $\varepsilon > 0$, $\delta > 0$. Egorov supplies A with $m(E \setminus A) < \delta$ and uniform convergence on A : beyond some N , $|f_n - f| < \varepsilon$ on all of A , so $\{|f_n - f| \geq \varepsilon\} \subseteq E \setminus A$ for $n \geq N$: acreage below δ . As δ was arbitrary, the acreage tends to 0: in-measure convergence. The bump: $\{|\mathbf{1}_{[n, n+1]}| \geq \frac{1}{2}\}$ has measure 1 for every n — the arrow dies without the finite floor, and Egorov's own hypothesis was the casualty (Problem II.3).

II.2 (solution in full). Fix $x \in [0, 1]$ and a generation k . The blocks $[j2^{-k}, (j+1)2^{-k}]$, $j = 0, \dots, 2^k - 1$, cover $[0, 1]$, so some j has x in its block: $T_{2^k+j}(x) = 1$. Ones occur in every generation: infinitely often. For zeros: $2^k \geq 2$ blocks exist from generation one onward and at most two contain x (a seam), so some block misses x : $T_n(x) = 0$ infinitely often. A sequence with both values infinitely often has no limit; no x is spared, seams included. (The subsequence of first blocks T_{2^k} converges to 0 at every $x > 0$: the consolation prize's illustration.)

6.8 Notes

History. Egorov (1911, *Comptes Rendus*); Lusin (1912) — teacher and student, Moscow's founding pair; Riesz and Fischer (1907), both via Fourier series (the account's creditor again); Littlewood's maxims in *Lectures on the Theory of Functions* (1944). The typewriter is folklore of uncertain paternity; Stein and Shakarchi print it as the standard witness.

Reading. Axler 2D, 3B; Bartle ch. 7 (the cleanest modes chapter in print); Royden ch. 3, 7; Folland §2.4 (the modes' zoo with the full metrizable discussion).

Forward pointers. Product measure and the two-dimensional merchant: Chapter 7. Borel–Cantelli under its own name, Chebyshev christened, convergence in probability: Chapter 8, on mass-one ground, where tonight's map becomes the working atlas of your trade.

Chapter 7

Fubini, or the Crime of Swapping Integrals

*Companion to Episode Seven. The episode built the plane from two lines (rectangles, the product tribe, Carathéodory rerun; the two routes agreeing), sharpened both knives (sections, with the monotone-class beam flagged honestly as the season's one labelled borrowing), proved Tonelli by the four-stage reel (rectangles, tribe, staircases, climb), derived Fubini by gains and losses, canonized the standard operating procedure (check with Tonelli, spend with Fubini), convicted the staircase-and-shadows function at the crime scene (both itineraries computed; diagnosis: Season One's rearrangement crime in planar dress), witnessed the sigma-finite fine print at the counting-measure diagonal, and banked the layer cake (wonder beat: the two seasons' pictures of an integral meeting in the subgraph). This chapter is the paperwork. Register notes: **the bell is silent** — count holds at seventeen; the product construction reruns architecture already rung. The SOP promise of Chapter 5 is discharged at Remark 7.9.*

Episode Digest

The argument of the hour. When may a double integral be computed in either order? Build the plane honestly: rectangle atoms priced by the area rule; the product tribe generated from above; the product measure by Carathéodory on rectangle covers, unique by the frame squeeze, agreeing with the direct planar construction. Sections are measurable (the monotone-class beam, labelled). Tonelli: nonnegative functions swap freely, always — rectangles agree by inspection, the beam carries agreement to the tribe, staircases and the climbing license carry it to everything. Fubini: integrable functions swap after the absolute check, bad slices null and amnestied. The SOP canonized. The crime scene: the diagonal staircase with rightward shadows — columns total one then nought (the ground floor casts no shadow into itself), rows total nought without exception; iterated integrals one and nought, both finite, both correct; absolute mass infinite; the rearrangement theorem in planar dress. Fine print: sigma-finiteness (the counting-measure diagonal: nonnegative, measurable, itineraries nought and one). The layer cake: integrate the subgraph's indicator both ways — vertical slices give the integral, horizontal slices give the acreage-above-height — Tonelli's somersault; the bridge to tail probabilities pre-signed for the finale.

The episode, beat by beat. **The Manoeuvre / The Plane, Built from the Line.** Two routes to planar measure (direct: the dimension license; product: tonight's, the one that generalizes); rectangles (Definition 7.1); the product tribe and measure (Theorem 7.2); the two routes agree (Corollary 7.3); sections and the two knives (Definition 7.4); the section lemmas with the beam flagged (Lemma 7.5, Remark 7.6).

Tonelli's Theorem. Statement and the four-stage reel (Theorem 7.7).

Fubini's Theorem. Statement with the almost-every-slice clause; gains and losses (Theorem 7.8); the SOP (Remark 7.9).

The Crime Scene. The staircase-and-shadows function computed to the digit (Proposition 7.10); the diagnosis.

The Fine Print. Sigma-finiteness; the counting-measure diagonal (Remark 7.11).

The Layer Cake. The identity by the subgraph somersault (Theorem 7.12); the finale's tail-probability form pre-signed.

The toolkit acquired. Rectangle-first proofs; the beam (stated, half-proved, shelved); the four-stage reel; the SOP; subgraph slicing.

Dates worth keeping. Fubini, 1907; Tonelli, 1909; the counting-measure caution, folklore of the same vintage.

7.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"a rectangle"	$A \times B, A, B \in \mathcal{M}$	Def. 7.1
"the product tribe"	$\mathcal{M} \otimes \mathcal{M} = \sigma(\text{rectangles})$	Thm. 7.2
"the product measure"	$(m \times m)(A \times B) = m(A)m(B)$, extended	Thm. 7.2
"the x-section at y"	$E^y = \{x : (x, y) \in E\}; f^y(x) = f(x, y)$	Def. 7.4
"iterated integrals"	$\int \left(\int f(x, y) dm(x) \right) dm(y)$, and the reverse	Thm. 7.7
"the subgraph"	$S_f = \{(x, t) : 0 \leq t < f(x)\}$	Thm. 7.12

7.2 The Product Construction

Definition 7.1 (Rectangles). A *measurable rectangle* is $A \times B$ with $A, B \in \mathcal{M}$ (on the two factor lines), priced $\rho(A \times B) = m(A)m(B)$ (extended arithmetic; $0 \cdot \infty = 0$).

Theorem 7.2 (The product measure). *On $\mathcal{M} \otimes \mathcal{M} = \sigma(\{\text{rectangles}\})$ there is a measure $m \times m$ agreeing with ρ on rectangles; on sigma-finite factors (our case) it is the unique such.*

Proof. Existence: run Carathéodory on countable rectangle covers — define the planar outer measure as the infimum of $\sum_k \rho(R_k)$ over rectangle covers; the 1914 staircase (Chapter 2, quoted, not re-proved: symmetry, composition, proportion, limit) yields a sigma-algebra of splitters carrying a countably additive restriction; rectangles split cleanly (the half-plane turnstiles $A \times \mathbb{R}$

and $\mathbb{R} \times B$ mirror Chapter 2's half-lines — Problem II.1 performs the audit), so the splitters include the rectangles, hence the generated tribe. That ρ is itself countably additive on rectangles — the covering bound needs it — is Problem II.1's second half, a Tonelli-in- miniature for indicator slices (or: the desk's direct route via the line's MCT on sections of a disjoint rectangle union). Uniqueness: two agreeing measures on rectangles agree on finite frames $[-N, N]^2$ by the frame squeeze of Chapter 2's uniqueness theorem transplanted verbatim (rectangles play intervals; the estate's robustness plays itself), and sigma-finiteness patches the frames by the crankshaft. $\square$

Corollary 7.3 (The two routes agree). *The direct planar Lebesgue measure (Chapter 2's dimension license) and $m \times m$ coincide on the product tribe.*

Proof. Both keep the rectangle clause (boxes are rectangles; the direct construction prices boxes by area — its own interval theorem run twice); uniqueness closes. (The product tribe sits inside the direct construction's full commonwealth; on the common jurisdiction they agree, and the completion remark of the Notes handles the veil.) $\square$

Definition 7.4 (Sections). For E in the plane and y fixed: $E^y = \{x : (x, y) \in E\}$; dually E_x . For functions: $f^y(x) = f(x, y)$, $f_x(y) = f(x, y)$.

Lemma 7.5 (Sections are measurable). *If $E \in \mathcal{M} \otimes \mathcal{M}$, every section E^y, E_x is a citizen of its line; if f is product-measurable, every section function is measurable.*

Remark 7.6 (The monotone-class beam — the labelled borrowing). The collection of sets with all sections measurable contains the rectangles ($(A \times B)^y$ is A or $\emptyset$) and is a sigma-algebra (sections commute with complements and countable unions — the accessible half, proved here in four lines: $(\bigcup_k E_k)^y = \bigcup_k E_k^y$ and $(E^c)^y = (E^y)^c$); hence it contains the generated tribe, and Lemma 7.5 follows — *for sets*. For the *measure-theoretic* upgrades Tonelli's stage two needs — that $y \mapsto m(E^y)$ is measurable with $\int m(E^y) dm(y) = (m \times m)(E)$ — the standard induction over the tribe wants the *monotone class theorem*: stated here (a monotone class containing an algebra contains its generated sigma-algebra), its proof cited to the shelves (Folland §2.1; Axler 5A; Royden ch. 20) per the plan's instruction to flag the machinery honestly. The sigma-finite bookkeeping of the induction is Problem III.1's guided skeleton. *One beam, borrowed, labelled; everything else in the chapter is house timber.*

7.3 The Matched Pair

Theorem 7.7 (Tonelli, 1909). *Let $f \geq 0$ be product-measurable (sigma-finite factors). Then $y \mapsto \int f^y dm$ and $x \mapsto \int f_x dm$ are measurable, and*

$$\iint f d(m \times m) = \int \left(\int f(x, y) dm(x) \right) dm(y) = \int \left(\int f(x, y) dm(y) \right) dm(x),$$

all three together, ∞ included.

Proof. The four-stage reel. *Rectangles*: for $f = \mathbf{1}_{A \times B}$ all three read $m(A)m(B)$ by inspection. *Tribe*: the class of sets whose indicators satisfy the display contains the rectangles and is a monotone class closed under the algebra's operations — the beam (Remark 7.6) carries it to $\mathcal{M} \otimes \mathcal{M}$; sigma-finite frames guard the complementation step (the guided skeleton). *Staircases*: linearity (Chapter 5's discharged additivity, on the plane and the line alike) extends the display to simple functions.

Climb: the staircase theorem writes f as an increasing simple limit; monotone convergence (Chapter 5) passes the limit through all three integrals at once — the inner integrals climb by MCT on each section, and their integrals climb by MCT again. Rectangles, tribe, staircases, climb. $\square$

Theorem 7.8 (Fubini, 1907). *Let f be product-measurable with $\int \int |f| d(m \times m) < \infty$. Then for almost every y the section f^y is integrable (and dually), the a.e.-defined iterated integrals exist, and both equal $\int \int f d(m \times m)$.*

Proof. Tonelli on $|f|$ prices $\int (\int |f^y|) dm(y) < \infty$, so the inner integral is finite for a.e. y (Chapter 4's little lemma, I.6, in its finite-a.e. form): bad slices are null, amnestied. Split $f = f^+ - f^-$; Tonelli prices each part's display; subtract — lawful, both totals finite by the hypothesis — section by section on the good slices and integral by integral overall (linearity). $\square$

Remark 7.9 (The standard operating procedure — canonized). To swap: if $f \geq 0$, swap (Tonelli). Else compute $\int \int |f|$ by either itinerary (Tonelli again — the check itself is swappable); if finite, swap (Fubini); if infinite, do not — you stand at the crime scene. Chapter 5's small promise, discharged; the constitution is Season One's series doctrine (absolute licenses rearrangement) in planar dress.

7.4 The Crime Scene and the Fine Print

Proposition 7.10 (The staircase and its shadows). *On $[0, \infty)^2$ let*

$$f = \sum_{k=0}^{\infty} \left(\mathbf{1}_{[k, k+1) \times [k, k+1)} - \mathbf{1}_{[k+1, k+2) \times [k, k+1)} \right).$$

Then both iterated integrals exist and

$$\int \left(\int f dm(y) \right) dm(x) = 1, \quad \int \left(\int f dm(x) \right) dm(y) = 0,$$

while $\int \int |f| = \infty$: the integrability check fails, and no license was issued.

Proof. Columns (fix $x \in [j, j+1)$, integrate $dm(y)$): for $j = 0$ the column meets only the $k=0$ positive block: inner integral 1. For $j \geq 1$: the positive block $k = j$ contributes +1 on $y \in [j, j+1)$ and the shadow with $k+1 = j$, i.e. $k = j-1$, contributes -1 on $y \in [j-1, j)$: inner integral 0. Outer: $\int = 1 \cdot 1 = 1$. Rows (fix $y \in [k, k+1)$, integrate $dm(x)$): the storey's positive block on $x \in [k, k+1)$ and its own shadow on $x \in [k+1, k+2)$: inner integral 0, every $k \geq 0$ — shadows are cast sideways, so no storey lacks one. Outer: 0. Absolute mass: each of infinitely many blocks carries 1: $\int \int |f| = \infty$ (Tonelli's own count, either order). Diagnosis: the two itineraries are two orderings of the conditionally convergent block-series $1 - 1 + 1 - 1 + \dots$ -flavoured mass; Volume One's rearrangement theorem, staged on the quarter-plane. Solved in full as Problem II.2, block by block. $\square$

Remark 7.11 (Sigma-finiteness's witness). Pair $([0, 1], m)$ with $([0, 1], \#)$ — counting measure, not sigma-finite — and take $f = \mathbf{1}_{\Delta}$, the diagonal's indicator. Then $\int (\int f d\#(y)) dm(x) = \int 1 dm = 1$ (each x -section is one point, counted once), while $\int (\int f dm(x)) d\#(y) = \int 0 d\# = 0$ (each y -section is one point, length nought). Nonnegative, measurable in the appropriate joint sense, disagreeing: the sigma-finite clause in Theorems 7.7–7.8 is real. (Three lines, as promised; the measurability fine print of the diagonal in the product tribe is the Notes' small-print reference.)

7.5 The Layer Cake

Theorem 7.12 (The layer-cake representation). *For measurable $f \geq 0$ on sigma-finite ground E :*

$$\int_E f \, dm = \int_0^\infty m\{x \in E : f(x) > t\} \, dm(t).$$

Proof. Tonelli's somersault on the subgraph indicator $\mathbf{1}_S$, $S = \{(x, t) : 0 \leq t < f(x)\}$ (product-measurable: $S = \bigcup_{q \in \mathbb{Q}^+} \{f > q\} \times [0, q)$ -flavoured assembly, Problem II.4). Heights first: $\int_0^\infty \mathbf{1}_S(x, t) \, dm(t) = f(x)$; then ground: the left side. Ground first: $\int_E \mathbf{1}_S(x, t) \, dm(x) = m\{f > t\}$; then heights: the right side. Tonelli licenses the equality unconditionally. — The finale's uniform: expectation equals the integral of tail probabilities. $\square$

7.6 Problems

Tier I — Calisthenics

- I.1 Rectangle pricing drills, extended arithmetic included ($m(A) = 0$, $m(B) = \infty$).
- I.2 Sections of named sets: the diagonal, the subgraph of x^2 , the staircase-and-shadows support; both knives.
- I.3 Iterated integrals both ways: $f(x, y) = \mathbf{1}$ on the unit square; $x + y$ via staircases; a two-block signed specimen inside the license.
- I.4 SOP audits on four candidates (two licensed, the crime scene, the diagonal-under-counting): verdicts with warrants.
- I.5 Replay Remark 7.11 with every set displayed.
- I.6 Layer-cake evaluations: the identity on $[0, 1]$ (acreage $1 - t$); a two-storey simple function.
- I.7 The crime scene's absolute mass: Tonelli's count both ways, shown infinite twice.
- I.8 Convolution warm-up: for indicators of $[0, 1]$, compute $(f * g)(x) = \int f(x - y)g(y) \, dm(y)$ — the tent, by direct slicing.

Tier II — The Real Thing

- II.1 (*The turnstiles.*) Show the half-planes $A \times \mathbb{R}$, $\mathbb{R} \times B$ split every rectangle cover cleanly (Chapter 2's half-line audit, run twice); and prove ρ countably additive on rectangles via the line's MCT on sections.
- II.2 (*The crime scene, to the digit.*) Both itineraries block by block, the absolute count, and the rearrangement diagnosis drawn in form. Solved in full below.
- II.3 (*The a.e.-slice clause.*) In Theorem 7.8, exhibit an integrable f with a genuinely non-integrable section, and verify the bad set is null.
- II.4 (*The subgraph's papers.*) Prove S_f is product-measurable for measurable $f \geq 0$, by the rational assembly.

- II.5 (*The cake, weighted.*) Prove $\int f^p = \int_0^\infty p t^{p-1} m\{f > t\} dm(t)$ for $p \geq 1$ — the trade's moment formula (Tonelli on the weighted subgraph).
- II.6 (*Convolution exists.*) For integrable $f, g \geq 0$: $(x, y) \mapsto f(x - y)g(y)$ is product-measurable (translation machinery + composition's safe direction), Tonelli prices $\iint = \|f\|_1 \|g\|_1$, and hence $(f * g)(x)$ is finite for a.e. x with $\|f * g\|_1 = \|f\|_1 \|g\|_1$; extend to signed integrable f, g with $\leq$. Solved in full below.

Tier III — For the Ambitious ★

- III.1 ★ (*The beam.*) State and prove the monotone class theorem from the shelves' skeleton, then run the guarded induction that completes Lemma 7.5's measure-theoretic upgrade, sigma-finite frames displayed. (Folland §2.1; Axler 5A.)
- III.2 ★ (*Three factors.*) Construct $m \times m \times m$ and prove associativity of the product construction — $(m \times m) \times m = m \times (m \times m)$ on the common tribe — the bookkeeping guarding higher-dimensional joints.

7.7 Hints

II.3. One bad slice suffices and one is null: take any integrable $f_0(x, y)$ and redefine it on the single line $y = y_0$ to be a non-integrable section there (the spike family's supremum, say, $1/x$ on $(0, 1]$). The line is planar-null, so the redefinition changes neither product-measurability (amnesty) nor the double integral; yet the y_0 -section is honestly non-integrable. The theorem's "almost every slice" is exactly calibrated to tolerate it.

II.6. Measurability: $(x, y) \mapsto x - y$ is continuous; f measurable on the line; the safe direction (Chapter 4) needs f Borel — upgrade by amnesty: replace f by a Borel a.e.-equal version (Chapter 2's structure theorem), the null discrepancy sweeping into planar null by Tonelli on the indicator.

III.2. Rectangles with three sides generate both parenthesizations; uniqueness on sigma-finite frames closes.

7.8 Solutions

Tier I (terse). I.3's middle: $\int_0^1 \int_0^1 (x + y) = \frac{1}{2} + \frac{1}{2}$ by staircase symmetry, either order. I.6: $\int_0^1 (1 - t) dt$ computed by the identity's own layer cake — the season eating its own cooking. I.8: the tent on $[0, 2]$, peak one at $x = 1$: $\min(x, 2 - x)^+$.

II.2 (solution in full). As Proposition 7.10, expanded: the column table for $j = 0, 1, 2$ and the general row; the absolute count $\sum_k (1 + 1) = \infty$; and the diagnosis displayed as a rearrangement: enumerate the blocks column-first and the partial sums visit 1 eternally; row-first and they visit 0 eternally — two orderings of ± 1 masses, Volume One, Chapter 4's theorem quoted with its chapter number.

II.6 (solution in full). *Measurability.* Take $\tilde{f}$ Borel with $\tilde{f} = f$ a.e. (Chapter 2, Theorem 2.17 — an F_σ -built version); then $(x, y) \mapsto \tilde{f}(x - y)$ is Borel-continuous: product-Borel-measurable (Chapter 4's safe direction on the plane); $\tilde{f}(x - y)g(y)$ product-measurable (algebra). The exceptional set $\{\tilde{f}(x - y) \neq f(x - y)\}$ is planar-null: its y -sections are translates of a line-null set, and Tonelli on its indicator prices it at $\int 0 = 0$. *Pricing.* Tonelli (nonnegative case), inner over x first: $\int f(x - y) dm(x) = \|f\|_1$ for every y (translation invariance), so $\iint f = \|f\|_1 \|g\|_1 < \infty$. *Existence a.e.* The x -sections' finiteness for a.e. x is the little lemma on the finite double integral (via the reversed itinerary — Tonelli again). Signed case: run the above on $|f|, |g|$ for the lid, then Fubini's bookkeeping; $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$ by the triangle inequality under the integral. *The algebra of convolution — commutativity, the smoothing folklore — is shelved (Notes); tonight only its existence papers were promised.*

7.9 Notes

History. Fubini (1907), Tonelli (1909); the modern matched- pair presentation is the textbooks' distillation. The staircase-and-shadows crime scene is the standard discrete-flavoured witness (Folland's exercise circle); the classical smooth witness (the trade knows it with arctangents) is avoided here deliberately — its evaluation spends trigonometry, and Volume One's loan L1 remains unspent by this course's proofs.

The diagonal's small print. In Remark 7.11, joint measurability of Δ holds in the product of Borel tribes; the pathology is additivity's, not measurability's — Folland §2.3 for the careful version.

Reading. Axler 5A–5B; Folland §2.1–2.3 (whose monotone-class staging this chapter's labelled borrowing follows); Stein–Shakarchi ch. 2; Royden ch. 20. Convolution's onward algebra: Folland §8.1.

Forward pointers. Independence as product measure, the dictionary, Borel–Cantelli under its name, Chebyshev christened, the laws of large numbers, Radon–Nikodym and conditional expectation: Chapter 8 — the finale.

Chapter 8

Probability Is Measure Theory

*Companion to Episode Eight, the series finale. The episode read Kolmogorov's dictionary item by item (the wonder beat: "you have been using this course your entire career"), made independence a product measure and Episode Seven load-bearing, named Borel–Cantelli (the first lemma being Episode Six's tail-trap, cross-reference stamped), christened Chebyshev and proved the weak law in four sentences, honoured the strong law with a statement and Etemadi's sketch, stated Radon–Nikodym with the devil's staircase as its singular witness (the course's fondest monster retiring as fine print), framed conditional expectation as best guess and projection, set the Kullback–Leibler wink as the course's final star, retraced the arc Hippasus-to-Kolmogorov, audited every account of two seasons aloud (Debt Twelve settled on the spot — the desk executes the payment in Section 8.5 — and one loan gracefully carried beyond, declared), read the seventeen-ring peal one final time, and closed the door. This chapter is the paperwork, and the last. Register notes: **the bell is silent and retires at seventeen**; the full audit is the episode's text, mirrored in the LEDGER; the further-reading essay closes the volume per the plan's final mandate.*

Episode Digest

The argument of the hour. Probability is this course. Mass-one measure spaces; events are citizens; random variables are measurable functions; distributions are pushforwards; expectation is the integral with all of Chapter 5 attached; almost surely is the amnesty; the convergence zoo transfers with its map. Independence is product measure (factored expectations by the SOP). Borel–Cantelli: finite total risk, finitely many disasters (the tail-trap under its name); independent infinite risk, infinitely many (the complement bound) — a zero-one dichotomy. Chebyshev from the embryo; the weak law by linearity-independence-Chebyshev; the strong law stated with respect. Radon–Nikodym: densities exist exactly under absolute continuity (necessity proved; sufficiency stated with references); the staircase's distribution is singular continuous — mass on the dust, no atoms, no density: the clause's standing witness. Conditional expectation: the R–N dividend; cell averages in the finite case; the projection reading in the seventeenth ring's geometry; the KL divergence as an expected log-derivative — the final star. The audit executes its last payment on the spot: every bounded derivative integrates to the net change of its primitive (Theorem 8.12, dominated convergence presiding) — Debt Twelve's bounded estate recovered as promised. The audit balances; the peal is read; the door closes.

The episode, beat by beat. The Pamphlet / The Dictionary. Kolmogorov 1933; the table (Section 8.1) read as identities with addresses; the wonder beat.

Independence. Product measure; factored expectations (Proposition 8.2).

Borel and Cantelli. First lemma = Chapter 6's tail-trap, stamped (Theorem 8.3); second with independence (Theorem 8.4); the dichotomy and the zero-one door (Problem III.1).

The Laws. Chebyshev at rank (Proposition 8.5); WLLN in full (Theorem 8.6); SLLN stated, Etemadi sketched (Theorem 8.7).

Radon and Nikodym. Statement, necessity, worked densities, the singular staircase (Theorem 8.8, Proposition 8.9).

Conditional Expectation. Definition via R–N; the finite case solved; properties; projection (Definition 8.10 and following).

The Arc / The Audit / The Door. The episode's text; Debt Twelve's settlement executed at the desk (Section 8.5); the LEDGER mirrors; the peal at seventeen; the loan carried, declared.

Dates worth keeping. Kolmogorov's *Grundbegriffe*, 1933; Borel–Cantelli, 1909/1917; Chebyshev, 1867; Radon, 1913; Nikodym, 1930; Etemadi's proof, 1981.

8.1 The Dictionary

The trade says	This course proved	Address
probability space $(\Omega, \mathcal{F}, P)$	measure space, $P(\Omega) = 1$	Chs. 1–2
event	citizen of the tribe	Ch. 2 (and Ch. 3's why)
random variable X	measurable function	Ch. 4
distribution of X	pushforward $P_X(B) = P(X^{-1}B)$	Lem. 8.1
expectation $\mathbb{E}X$	$\int_{\Omega} X dP$	Chs. 4–5
almost surely	almost everywhere, mass one	Chs. 1, 4
in probability	in measure	Ch. 6
independence	product measure	Ch. 7, Prop. 8.2
density / likelihood ratio	Radon–Nikodym derivative	Thm. 8.8
conditional expectation	the R–N dividend; projection	Def. 8.10

This table is the volume's last notation ledger and, by design, the course's one-page index.

Lemma 8.1 (Pushforward). *For a random variable X on $(\Omega, \mathcal{F}, P)$, $P_X(B) = P(X^{-1}(B))$ defines a measure on the Borel line with total mass one, and for $g \geq 0$ Borel, $\mathbb{E} g(X) = \int g dP_X$ (the change-of-variables identity).*

Proof. Preimages respect complements and countable disjoint unions (Chapter 4's preimage bookkeeping), so countable additivity transfers; mass one is $X^{-1}(\mathbb{R}) = \Omega$. The identity: true for indicators by definition, lifts by staircases and the climb (Chapter 5's reel, one dimension). $\square$

Proposition 8.2 (Independence; factored expectations). *Random variables X, Y are independent iff the joint pushforward on the plane is the product of the marginals: $P_{(X,Y)} = P_X \times P_Y$. In that case, for integrable X, Y : $\mathbb{E}[XY] = \mathbb{E}X \cdot \mathbb{E}Y$.*

Proof. The iff is definitional bookkeeping on rectangles plus uniqueness (Chapter 7). Factoring: for $X, Y \geq 0$, Tonelli on the product (via Lemma 8.1 to the plane): $\mathbb{E}[XY] = \iint xy d(P_X \times P_Y) = \int x dP_X \int y dP_Y$; signed case by the SOP — the absolute check factors first, licensing Fubini. $\square$

8.2 Borel–Cantelli

Theorem 8.3 (First lemma: finite risk). *If $\sum_n P(A_n) < \infty$, then $P(A_n \text{ infinitely often}) = 0$: almost surely only finitely many A_n occur.*

Proof. This is Chapter 6’s tail-trap under its proper name, cross-reference stamped (Theorem 6.5’s engine): with $B_K = \bigcup_{n \geq K} A_n$, $P(B_K) \leq \sum_{n \geq K} P(A_n) \rightarrow 0$ (subadditivity; the convergent tail), and $\{A_n \text{ i.o.}\} = \bigcap_K B_K$ has mass at most every tail: zero. The gambit’s skeleton, redeemed as promised. $\square$

Theorem 8.4 (Second lemma: independent infinite risk). *If the A_n are independent and $\sum_n P(A_n) = \infty$, then $P(A_n \text{ i.o.}) = 1$.*

Proof. Complements: for $K \leq M$, $P(\bigcap_{n=K}^M A_n^c) = \prod_{n=K}^M (1 - P(A_n))$ (independence passes to complements: Problem I.5) $\leq \prod \exp(-P(A_n)) = \exp(-\sum_K^M P(A_n)) \rightarrow 0$ as $M \rightarrow \infty$ (divergence; the inequality $1 - t \leq e^{-t}$ is the desk’s one-line convexity-free calculus: the difference vanishes at $t = 0$ with nonpositive derivative — Volume One’s harvest; the exponential rides Volume One’s e -manufacture). So $P(\bigcup_{n \geq K} A_n) = 1$ for every K ; shrinking intersection over K (mass one throughout, the finite floor trivial) keeps mass one. $\square$

8.3 The Laws

Proposition 8.5 (Chebyshev, christened). *For X with $\mathbb{E}X = \mu$ and $\text{Var } X = \mathbb{E}(X - \mu)^2 = \sigma^2 < \infty$, and any $\varepsilon > 0$: $P(|X - \mu| \geq \varepsilon) \leq \sigma^2 / \varepsilon^2$.*

Proof. Chapter 6’s embryo (mass bounds acreage, Lemma 6.2) applied to $g = (X - \mu)^2$ at level ε^2 . $\square$

Theorem 8.6 (The weak law of large numbers). *Let $X_1, X_2, \dots$ be independent with a common distribution, $\mathbb{E}X_i = \mu$, $\text{Var } X_i = \sigma^2 < \infty$; let $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. Then $\bar{X}_n \rightarrow \mu$ in probability.*

Proof. $\mathbb{E}\bar{X}_n = \mu$ (linearity, Chapter 5’s discharged credit). Variance: $\text{Var } \bar{X}_n = \frac{1}{n^2} \sum_i \text{Var } X_i = \sigma^2 / n$, the cross-terms $\mathbb{E}(X_i - \mu)(X_j - \mu)$ vanishing for $i \neq j$ by factored expectations (Proposition 8.2) — the two-line computation displayed at Problem II.2. Chebyshev: $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2 / (n\varepsilon^2) \rightarrow 0$. Four sentences, as the episode promised. $\square$

Theorem 8.7 (The strong law — stated with respect). *With only $\mathbb{E}|X_i| < \infty$ (independent, common distribution): $\bar{X}_n \rightarrow \mu$ almost surely.*

Sketch and references. Etemadi’s elementary route (1981): reduce to nonnegative variables; truncate $Y_i = X_i \mathbf{1}_{X_i \leq i}$ (Borel–Cantelli reconciling X_i with Y_i via $\sum P(X_i > i) \leq \mathbb{E}X_1 < \infty$ — the layer cake pricing the sum); control variances of truncations along geometric subsequences $n_k = \lfloor \alpha^k \rfloor$ by Chebyshev and a summable bound; sandwich general n between geometric marks by monotonicity. Every instrument is the course’s; the bookkeeping is a further evening’s, and the honesty clause files it: Durrett §2.4, Williams ch. 12, Etemadi’s original two pages.

8.4 Radon–Nikodym and Conditional Expectation

Theorem 8.8 (Radon–Nikodym — stated; necessity proved). *Let μ, ν be sigma-finite on a common tribe. There exists measurable $w \geq 0$ (the density, $w = \frac{d\nu}{d\mu}$, unique up to μ -amnesty) with*

$$\nu(A) = \int_A w d\mu \quad \text{for all citizens } A$$

if and only if $\nu \ll \mu$ (absolute continuity: $\mu(A) = 0 \Rightarrow \nu(A) = 0$). Necessity is one line: an integral over a μ -null set is zero (Chapter 4, manners (iv)). Sufficiency is stated on references (von Neumann’s Hilbert- space route rides the seventeenth ring’s geometry; the classical route rides a supremum-of-candidates exhaustion): Axler 9A, Folland §3.2, Royden ch. 18. Worked densities: Problems I.6–I.7 (uniform, reweighted, two-model likelihood ratio); the derivative calculus (chain rule, change of measure under integrals): Problem II.4.

Proposition 8.9 (The staircase’s distribution: the singular witness). *Let c be the Cantor function (Chapter 4) read as a cumulative distribution: the measure μ_c with $\mu_c([0, x]) = c(x)$. Then μ_c is a mass-one measure carried entirely by the dust ($\mu_c(C) = 1$, though $m(C) = 0$), with no atoms ($\mu_c\{x\} = 0$ for every x), and consequently no density with respect to length exists: $\mu_c \not\ll m$. Singular continuous — the standing witness that Theorem 8.8’s clause is real.*

Proof. Existence and countable additivity of μ_c : c is continuous and nondecreasing; the Stieltjes-flavoured construction runs Carathéodory on intervals priced by increments $c(b) - c(a)$ — Chapter 2’s staircase verbatim (Problem II.5 audits the rectangle clauses). Mass on the dust: each removed middle carries increment zero (c constant there), so the complement’s countably many beads carry μ_c -mass zero; additivity leaves everything on C . No atoms: $\mu_c\{x\} = c(x) - c(x^-) = 0$ by continuity. No density: $m(C) = 0$ with $\mu_c(C) = 1$ violates absolute continuity; necessity (proved above) forbids the density. The course’s fondest monster retires as the fine print of its final theorem. $\square$

Definition 8.10 (Conditional expectation). Let X be integrable on $(\Omega, \mathcal{F}, P)$ and $\mathcal{G} \subseteq \mathcal{F}$ a sub-tribe. The *conditional expectation* $\mathbb{E}[X | \mathcal{G}]$ is the (P -a.s. unique) $\mathcal{G}$ -measurable, integrable random variable Z with

$$\int_G Z dP = \int_G X dP \quad \text{for every } G \in \mathcal{G}.$$

Existence and uniqueness: apply Theorem 8.8 on $(\Omega, \mathcal{G})$ to $\nu(G) = \int_G X dP$ against $P \upharpoonright_{\mathcal{G}}$ ($X \geq 0$ first; parts for signs) — $\nu \ll P$ by manners (iv), and the density is Z . (The finale states R–N; conditional expectation inherits the same stated-with-references standing for existence, with everything else below proved outright.)

Proposition 8.11 (The finite case; first properties). *(i) If $\mathcal{G}$ is generated by a finite partition $\{G_1, \dots, G_m\}$ with $P(G_j) > 0$, then on each cell $\mathbb{E}[X | \mathcal{G}] = \frac{1}{P(G_j)} \int_{G_j} X dP$: the cell average, the obvious best guess. (ii) Linearity; monotonicity. (iii) Tower rule: for $\mathcal{H} \subseteq \mathcal{G}$, $\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}]$; in particular $\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}X$. (iv) Taking out what is known: for bounded $\mathcal{G}$ -measurable Y : $\mathbb{E}[XY | \mathcal{G}] = Y \mathbb{E}[X | \mathcal{G}]$.*

Proof. (i): the displayed Z is $\mathcal{G}$ -measurable (constant on cells) and matches integrals on each generator, hence on the generated tribe (finite unions); uniqueness by the defining property. (ii): integrate the defining identity. (iii): both sides are $\mathcal{H}$ -measurable and match integrals over every

$H \in \mathcal{H} \subseteq \mathcal{G}$ — two applications of the definition. (iv): for indicators $Y = \mathbf{1}_G$ the identity is the definition restricted to $G \cap G'$; staircases and bounded convergence (Chapter 5's BCT, mass-one ground) lift to bounded Y . Worked two-block portfolio: Problem II.3. Projection characterization (square-integrable X : the a.s.-unique $\mathcal{G}$ -measurable minimizer of $\mathbb{E}(X - Z)^2$, orthogonal projection in the seventeenth ring's geometry): stated, with the L^2 fine print, references Axler 9B, Williams ch. 9 — Problem III.2 stages it. $\square$

8.5 The Last Settlement: the Bounded Estate Recovered

One account in the audit deserves its payment executed rather than merely announced. Debt Twelve, issued at Season One's summit: Volterra's function is differentiable at every point with a bounded derivative, and the Riemann integral refuses to read that derivative — the fundamental theorem's crack. The terms fixed at issue promised that Volume Two would recover the *bounded estate*: every bounded derivative integrates, in the new theory, to the net change of its primitive. The season's machine is now complete, and the payment takes four steps. (The name does double duty, and not by accident: a bounded derivative on finite land lives precisely in Chapter 4's bounded estate.)

Theorem 8.12 (The bounded estate recovered — D12, bounded case). *Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable at every point of $[a, b]$ (one-sidedly at the endpoints), with $|f'(x)| \leq M$ for all x . Then f' is Lebesgue integrable on $[a, b]$ and*

$$\int_{[a,b]} f' dm = f(b) - f(a).$$

Proof. Step 1: extend. Prolong f to $[a, b + 1]$ along the tangent line at b : set $f(x) := f(b) + f'(b)(x - b)$ for $x \in (b, b + 1]$. The extension is differentiable at every point of $[a, b + 1]$ (at b the one-sided derivatives agree at $f'(b)$), its derivative is bounded by M throughout, and f is continuous.

Step 2: quotients under a constant lid. For $n \geq 1$ define on $[a, b]$

$$q_n(x) := n \left(f\left(x + \frac{1}{n}\right) - f(x) \right).$$

Each q_n is continuous, hence measurable; differentiability at each point gives $q_n \rightarrow f'$ everywhere on $[a, b]$ (at b , exactly: the linear extension makes $q_n(b) = f'(b)$ for every n). So f' is a pointwise limit of measurable functions — measurable by limit stability (Theorem 4.6) — and bounded by M , hence integrable on finite land (Proposition 4.14). The mean value theorem — Volume One's Chapter Seven workhorse, proved there in full — supplies the lid: $q_n(x) = f'(\xi)$ for some $\xi \in (x, x + \frac{1}{n})$, so $|q_n| \leq M$: a constant, integrable on $[a, b]$.

Step 3: the licensed swap. Dominated convergence (Theorem 5.8) gives

$$\int_{[a,b]} f' dm = \lim_n \int_{[a,b]} q_n dm.$$

Step 4: compute the right-hand side with Volume One's own instruments. Each q_n is continuous, so its Lebesgue integral is its Riemann integral (Theorem 4.19); we may compute in the old regime. Let $F(x) := \int_a^x f$ (Riemann) on $[a, b + 1]$; the fundamental theorem's first part (Volume

One, Chapter Eight; f continuous) gives $F' = f$, the chain rule (Volume One, Chapter Seven) gives $\frac{d}{dx}F(x + \frac{1}{n}) = f(x + \frac{1}{n})$, and the evaluation part then yields

$$\int_a^b f(x + \frac{1}{n}) dx = F(b + \frac{1}{n}) - F(a + \frac{1}{n}),$$

whence, subtracting $\int_a^b f = F(b) - F(a)$ and scaling,

$$\int_{[a,b]} q_n dm = n \int_b^{b+1/n} f - n \int_a^{a+1/n} f.$$

For any c , continuity of f at c prices the average:

$$\left| n \int_c^{c+1/n} f - f(c) \right| = \left| n \int_c^{c+1/n} (f - f(c)) \right| \leq \sup_{[c, c+1/n]} |f - f(c)| \rightarrow 0.$$

The two averages converge to $f(b)$ and $f(a)$, and the theorem follows. $\square$

Remark 8.13 (The terms, honoured; the crack's witness reads). Volterra's function — differentiable everywhere, derivative bounded, Riemann helpless before it (Volume One, Chapter Eight) — is a citizen of Theorem 8.12: the Lebesgue integral reads its derivative and returns the net change, and the fundamental theorem's bounded estate is recovered exactly as Debt Twelve's terms promised. Every instrument in the payment is on the premises and paid for: limit stability and the two ledgers from Chapter 4, the constant lid from Volume One's mean value theorem, the swap from Chapter 5's dominated convergence, the computation by the absorption theorem and Volume One's fundamental theorem — no new completeness deployment anywhere (the bell's business tonight is retirement, not new rings). The *unbounded* estate — derivatives existing everywhere but unbounded — remains honestly beyond the course, in the gauge integral's jurisdiction (Denjoy, Perron, Henstock, Kurzweil; named at Volume One's close), precisely as declared at issue.

8.6 Problems — the Final Set

Tier I — Calisthenics

- I.1 Dictionary drills: translate five textbook statements into course language and back, citing chapter addresses.
- I.2 Pushforwards computed: a two-valued variable; the identity on $([0, 1], m)$; the staircase draw's CDF read from Proposition 8.9.
- I.3 Independence audits: three joint tables, verdicts with rectangle warrants.
- I.4 Borel–Cantelli warm-ups: dyadic-risk events; the typewriter's blocks as events on $([0, 1], m)$ (risk 2^{-k} per generation-block: what does the first lemma conclude, and how does it reconcile with pointwise-nowhere convergence of the full sequence?).
- I.5 Independence passes to complements: prove it from the product identity.

- I.6 Worked densities: the uniform draw reweighted by $w(x) = 2x$ — verify mass one and compute three event prices both ways.
- I.7 A likelihood ratio: two dyadic-block models on $[0, 1]$; compute $d\nu/d\mu$ cell by cell.
- I.8 Chebyshev drills: bound stray probabilities for the uniform draw's mean at three tolerances; compare with exact computations.

Tier II — The Real Thing

- II.1 (*The second lemma, to the letter.*) Write Theorem 8.4 in full: complement independence, the $1 - t \leq e^{-t}$ calculus by Volume One's harvest, the shrinking bookkeeping. Solved in full below.
- II.2 (*The variance computation.*) The cross-term annihilation and the σ^2/n identity, displayed; then the WLLN rate at $\varepsilon = \sigma$.
- II.3 (*The two-block portfolio.*) On four equal cells with stated X -values: compute $\mathbb{E}[X \mid \mathcal{G}]$ for two nested coarsenings and verify the tower rule numerically. Solved in full below.
- II.4 (*Derivative calculus.*) Prove: $\nu \ll \mu \ll \lambda$ gives $\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda}$ a.e., and $\int f d\nu = \int f \frac{d\nu}{d\mu} d\mu$ for $f \geq 0$ (staircases and the climb).
- II.5 (*The Stieltjes audit.*) Verify the increment-pricing of Proposition 8.9 satisfies Carathéodory's rectangle clauses on intervals, so Chapter 2's machinery constructs μ_c .
- II.6 (*An almost-sure verdict.*) Independent draws uniform on $[0, 1]$: prove that almost surely only finitely many X_n fall below 2^{-n} , and that almost surely infinitely many fall below $1/n$ — both lemmas exercised on one sequence.

Tier III — For the Ambitious ★

- III.1 ★ (*Zero-one.*) Kolmogorov's zero-one law: tail events of independent sequences have probability zero or one. Guided skeleton from the shelves (independence of the tail from every finite prefix; the self-independence trick $P(A) = P(A)^2$); references: Williams ch. 4, Durrett §2.5.
- III.2 ★ (*The projection.*) For square-integrable X : prove $\mathbb{E}[X \mid \mathcal{G}]$ minimizes $\mathbb{E}(X - Z)^2$ over $\mathcal{G}$ -measurable Z , via the orthogonality identity; state the L^2 -completeness fine print and its shelf.
- III.3 ★ (*The final star: the closing wink.*) For $P \ll Q$ mass-one measures, define $\text{KL}(P \parallel Q) = \int \log \frac{dP}{dQ} dP$ — the expectation, under P , of the log of the Radon–Nikodym derivative: your trade's daily loss, in the course's last vocabulary. Prove $\text{KL} \geq 0$ with equality iff $P = Q$, via $\log t \leq t - 1$ — the logarithm and its inequality taken, per the register's standing annotation, from the credit Volume One extended (references there); every other symbol is a citizen of this course. The final problem of the course, sir; set with the wink it deserves.

8.7 Hints

II.4. Indicators first, both identities; staircases; the climb. Uniqueness of densities (a.e.) closes the chain rule.

II.6. $\sum 2^{-n} < \infty$: first lemma. $\sum 1/n = \infty$ and the events are independent: second lemma.

III.3. $-\text{KL} = \int \log \frac{dQ}{dP} dP \leq \int (\frac{dQ}{dP} - 1) dP = 1 - 1 = 0$, the inequality pointwise from the credit, the pricing by II.4's change of measure. Equality: $\log t = t - 1$ only at $t = 1$; the derivative is 1 P -a.s.; conclude $P = Q$ on every citizen.

8.8 Solutions

Tier I (terse). **I.4** Per-block risks within generation k sum to 1 (there are 2^k blocks of risk 2^{-k}): the full sequence's total risk diverges and the first lemma says nothing — consistent with pointwise-nowhere convergence; the *first-block* subsequence has risks 2^{-k} , summable: almost surely finitely many hits, matching Chapter 6's consolation. **I.5** $P(A^c \cap B) = P(B) - P(A \cap B) = P(B)(1 - P(A))$. **I.6** $\int_0^1 2x dm = 1$; prices both ways per II.4's identity. **I.8** Uniform: $\mu = \frac{1}{2}$, $\sigma^2 = \frac{1}{12}$ (the desk's staircase computation, Chapter 5's drill).

II.1 (solution in full). As Theorem 8.4, expanded: complement independence for finite families by I.5 and induction; the product bound $\prod_K^M (1 - p_n) \leq \exp(-\sum_K^M p_n)$ from $1 - t \leq e^{-t}$ on $[0, 1]$ — proved by Volume One's harvest: $h(t) = e^{-t} - 1 + t$ has $h(0) = 0$ and $h'(t) = 1 - e^{-t} \geq 0$ (the exponential's monotonicity, manufactured in Volume One, Chapter 4's desk); divergence sends the exponent to $-\infty$, the product to 0; hence $P(\bigcap_{n \geq K} A_n^c) = 0$ by shrinking chains from the finite stages (mass-one ground: the finite floor is free), so $P(\bigcup_{n \geq K} A_n) = 1$ for every K ; the i.o. set is the shrinking intersection of these mass-one tails: mass one.

II.3 (solution in full). Cells $C_1, \dots, C_4$, each mass $\frac{1}{4}$, with $X = 1, 3, 5, 7$. Fine coarsening $\mathcal{G}$: pairs $\{C_1 C_2\}, \{C_3 C_4\}$ — $\mathbb{E}[X \mid \mathcal{G}] = 2$ on the first pair, 6 on the second (cell averages). Coarse $\mathcal{H}$: everything — $\mathbb{E}[X \mid \mathcal{H}] = 4$. Tower, numerically: the $\mathcal{H}$ -average of the $\mathcal{G}$ -guess is $\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 6 = 4$: matches. Each defining integral displayed cell by cell; uniqueness by the finite case of Proposition 8.11(i).

8.9 Notes, and the Consolidated Further Reading

History. Kolmogorov's *Grundbegriffe der Wahrscheinlichkeitsrechnung* (1933) — the sixty pages; Borel (1909) and Cantelli (1917); Chebyshev (1867), in Markov's telling; Radon (1913), Nikodym (1930); Etemadi (1981). The dictionary's philosophical reception — probability as measure, full stop — took a generation; Doob's advocacy settled it.

The further-reading essay, per the plan's final mandate. The course ends at the doorstep, and three doors stand labelled. **The martingale door:** Williams, *Probability with Martingales* — conditional expectation, tonight's crowning dividend, becomes the subject itself; the strong law, the convergence theorems' probabilistic descendants, optional stopping; a Season Three's

natural spine, and the repayment venue for loan L1 (power series arrive with characteristic functions) and the logarithm's credit alike. **The functional-analytic door:** Axler chs. 6–10 and Folland chs. 5–6 — the seventeenth ring's geometry taken seriously; Hilbert space, duality (Hölder's exile repaid with interest), the spectral theorem; Fourier's account, this course's oldest, settles in these rooms — the mean-distance arch was built for exactly that traffic, as Episode Six promised. **The stochastic door:** Brownian motion and the Itô calculus (Durrett; Øksendal) — measure theory on path space, where the typewriter's lessons about modes of convergence become daily bread. Each door is described, none opened: a possible Season Three, as the plan has said from Session Zero, and not scope creep for this one.

Reading for this chapter. Williams chs. 1–9 (the finale's spiritual companion); Durrett §1–2; Axler 9A–9B; Folland §3.2.

The course closes. Sixteen chapters, two volumes, seventeen rings, one loan gracefully carried. The register balances; the desk is yours, sir.

The Season's Notation, Consolidated

Every chapter opened with a notation ledger bridging the episode's spoken phrases to their written symbols. This table gathers the season's standing instruments — the notation that outlived its introducing episode — grouped by the chapter that coined it. Episode-local devices remain in their own chapters' ledgers. The probability instruments are deliberately absent: Chapter 8's Dictionary is the season's final ledger page — it doubles, by design, as a one-page index of the entire course — and this table defers to it rather than duplicate it.

Spoken	Written	Coined
"the outer measure"	$m^*(A) = \inf\{\sum_k I_k : A \subseteq \bigcup_k I_k\}$	Ch. 1 (⊗)
"null; almost everywhere"	$m^*(Z) = 0$; a.e.	Ch. 1
"a sigma-algebra; the tribe"	$\mathcal{A}$	Ch. 1
"the ladder's first rungs"	F_σ, G_δ	Ch. 1
"the oscillation gauge"	$\omega_f(x)$	Ch. 1
"the measurable citizens"	$\mathcal{M}$ (Carathéodory's tribe)	Ch. 2
"Lebesgue measure"	$m = m^*$ restricted to $\mathcal{M}$	Ch. 2
"the Borel tribe"	$\mathcal{B} = \sigma(\text{open sets})$	Ch. 2
"the fat dust"	the Smith–Volterra–Cantor sets	Ch. 2
"the ghost"	V (a Vitali set)	Ch. 3
"equidecomposable"	finitely many congruent pieces	Ch. 3
"a measurable function"	preimages of open sets in $\mathcal{M}$	Ch. 4
"a staircase" (simple function)	$\varphi = \sum_i c_i \mathbf{1}_{A_i}$	Ch. 4
"the integral"	$\int f dm = \sup\{\int \varphi : 0 \leq \varphi \leq f\}$	Ch. 4 (⊗)
"the devil's staircase"	$c(x)$; the stretch $h(x) = x + c(x)$	Ch. 4
"the lid"	dominating g : $ f_n \leq g$, $\int g < \infty$	Ch. 5
"converges a.e. / in measure / in mean"	the three working modes, mapped	Ch. 6
"L-one; the mean distance"	L^1 ; $d_1(f, g) = \ f - g\ _1$	Ch. 6 (⊗)
"the typewriter"	the sliding-block sequence	Ch. 6
"the product tribe; the product measure"	$\mathcal{M} \otimes \mathcal{M}$; $m \times m$	Ch. 7
"the sections"	E_x, E^y	Ch. 7
"the layer cake"	$\int f dm = \int_0^\infty m(\{f > t\}) dt$	Ch. 7

Pronunciation Guide

The master lexicon’s table in full — the complete spoken cast of both seasons, fifty-two names, superseding the first volume’s edition of the same guide — for the reader who intends to say these names aloud in company. The machine-readable forms (exact IPA, both apostrophe variants of every possessive) live in the course’s pronunciation-lexicon files, from which this table is generated; the IPA column is omitted here rather than set approximately in a face that lacks the glyphs — the honesty clause extends to typography.

Name	Respelling	Remark
Abel	Ah-bel	Norwegian; not “AY-bel”
Ampère	Am-pair	
Arago	Ah-rah-goh	
Baire	Bair	one syllable; not “BAY-er”
Banach	Bah-nahk	the bishop is BARK-lee
Berkeley	Barkley	
Bolzano	Bolt-sah-no	
Borel	Bo-rell	
Carathéodory	Kara-tay-oh-doh-ree	
Cauchy	Koh-shee	
Chebyshev	Cheb-ish-off	
Darboux	Dar-boo	
Dedekind	Day-duh-kint	
Dirichlet	Dee-ree-shlay	not English “kind” anglicized convention; the German hard-K variant also circulates
Egorov	Yeh-gor-off	spoken in the S2E8 strong-law sketch
Etemadi	Eh-teh-MAH-dee	
Euler	Oiler	
Eudoxus	You-dock-sus	

Name	Respelling	Remark
Fatou	Fah-too	
Fourier	Foor-ee-ay	usually fine unaided; belt and braces
Gödel	Gur-del	
Grandi	Grahn-dee	
Hadamard	Ah-dah-mar	French; initial H silent
Heine	Hy-nuh	
Hermite	Air-meet	not “HER-might”
Hippasus	Hip-pah-sus	
Hölder	Herl-der	
Kolmogorov	Kol-muh-gor-off	
Kronecker	Kroh-neck-er	
L'Hôpital	Loh-pee-tahl	both apostrophe forms in PLS
Lebesgue	Luh-beg	the heaviest-traffic entry in the course
Leibler	Libe-ler	Kullback–Leibler
Leibniz	Libe-nits	
Liouville	Lee-oo-veel	
Lusin	Loo-zin	
Méray	May-ray	Charles Méray; Cantor's rival-firm precursor

Name	Respelling	Remark
Minkowski	Min-koff-ski	
Nikodym	Nik-oh-deem	
Oresme	Or-em	
Radon	Rah-don	the mathematician, not the gas
Poincaré	pwan-kar-AY	French; added Session 7 (spoken E7)
Riemann	Ree-mahn	
Rolle	rohl	rhymes with “hole”; added Session 7 (spoken E7)
Riesz	Reess	Riesz–Fischer
Solovay	Sol-uh-vay	
Stieltjes	Steel-chess	
Takagi	Tah-kah-ghee	hard G
Thomae	Toe-mah-uh	German final vowel sounded; anglicized “TOE-may” also circulates
Vitali	Vih-tah-lee	
Volterra	Vol-tair-ah	
Weierstrass	Vye-er-shtrahss	
Zermelo	Tsair-may-lo	

Colophon

This volume is the desk's half of *The Measure of All Things*, the second and final season of the audio course begun in *The Anatomy of the Continuum* — eight episodes narrated by Jeeves for and with P. Milovanov, Esq., produced one episode and one chapter per session on the twelfth of July, twenty twenty-six: the eight sessions of the season dispatched in a single day, a fact the register records without further comment. The season ran from the question the first volume left on its table — how long is a set? — to Kolmogorov's dictionary: length made honest by countable covers, the tribe selected, Choice's bill presented and paid knowingly, the integral rebuilt on measured ground, the convergence theorems converting two seasons of outrage into standing law, the zoo mapped, the plane licensed and policed, and probability revealed at the last as measure theory in evening dress. Seventeen marquee deployments of a single axiom — one sentence about least upper bounds — carry both volumes: fourteen rung in the first season, three in this one (the outer measure's birth, the integral's defining supremum, and the completeness of L^1); the full peal is read one final time in Chapter 8, and the bell is retired with the course. The debt register closes with every account paid or honestly declared beyond the course; one loan — schoolroom trigonometry, opened for an oscillating specimen in the first season — is carried forward, clearly marked, interest gracefully waived, against a further season that would repay it in its first evening. The text is set in Palatino (MATHPAZO), the face adopted at the first season's assembly and kept for the second.

The establishment closes its doors with the books balanced, the bell at seventeen, and its gratitude entirely unfeigned. Season Three is not commissioned; the management, as ever, stands ready.